Complete Build Kit

69 feature groups • 28 foundations • sequenced implementation prompts

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Bilingual spare-parts enquiries with deterministic compatibility, durable specialist workflows and human approval.

Next.js / Vercel · Supabase · FastAPI / LangGraph · OpenAI · AWS EC2 / Docker Compose

Read AGENTS.md first. Follow the execution index. This guide specifies the application build; implementation and hosted acceptance run when you execute the prompts.

Contents

Chapter numbers follow reading order. Implementation order is governed by prompts/README.md; covered features introduce no duplicate implementation.

001 README.md 4
002 AGENTS.md 5
003 CLAUDE.md 6
004 MASTER_PROMPT.md 7
005 docs/REQUIREMENTS.md 8
006 docs/DISCREPANCIES.md 12
007 docs/ARCHITECTURE.md 13
008 docs/CONSTRAINTS.md 15
009 docs/DECISIONS.md 16
010 docs/FLOW.md 17
011 docs/CONTRACTS.md 21
012 docs/TECH_SPEC.md 22
013 docs/DEPLOYMENT.md 51
014 docs/TEST_CHECKLIST.md 53
015 docs/ROLLBACK.md 54
016 docs/FIELD_GUIDE_MAP.md 55
017 docs/HANDOVER.md 56
018 docs/SESSION_LOG.md 57
019 docs/SELF_AUDIT.md 58
020 brand/BRAND.md 59
021 prompts/README.md 60
022 docs/templates/BUG_TEMPLATE.md 64
023 docs/templates/FEATURE_TEMPLATE.md 65
024 docs/templates/PROMPT_TEMPLATE.md 66
025 docs/templates/SESSION_HANDOFF.md 69
026 prompts/00-bootstrap-and-preserve-legacy.md 70
027 prompts/01-shared-contracts-and-clients.md 73
028 prompts/02-catalogue-migration.md 76
029 prompts/03-workflow-schema-and-database-permissions.md 79
030 prompts/04-private-storage.md 82
031 prompts/05-synthetic-catalogue-and-demo-fixtures.md 85
032 prompts/06-extend-legacy-search.md 88
033 prompts/F20-machine-identification.md 93
034 prompts/F21-machine-mismatch-detection.md 95
035 prompts/F22-compatibility-gate.md 97
036 prompts/F23-aliases-and-typo-matching.md 99
037 prompts/F24-semantic-retrieval.md 101
038 prompts/F25-ranked-candidates.md 103
039 prompts/F30-old-part-numbers.md 105
040 prompts/F33-fulfilment-planning.md 107
041 prompts/F34-transfer-estimates.md 109
042 prompts/07-authentication-and-demo-sessions.md 111
043 prompts/08-enquiry-intake-and-limits.md 114
044 prompts/09-deterministic-durable-workflow.md 117
045 prompts/F12-conversation-memory.md 121
046 prompts/F50-durable-workflows.md 123
047 prompts/10-intake-specialist.md 125
048 prompts/F10-structured-extraction.md 128
049 prompts/11-parts-specialist-and-tools.md 130
050 prompts/F26-ambiguity-handling.md 133
051 prompts/12-deterministic-verification.md 135
052 prompts/F28-compatibility-references.md 138
053 prompts/F46-verification-gate.md 140
054 prompts/13-reply-specialist-and-renderer.md 142
055 prompts/F45-reply-drafting.md 145
056 prompts/14-approval-and-revision-transactions.md 147
057 prompts/F40-approval-invalidation.md 150
058 prompts/F47-human-publication-gate.md 152
059 prompts/F49-audit-trail.md 154
060 prompts/15-worker-leases-and-sweeper.md 156

061 prompts/F51-background-jobs.md 159
062 prompts/F52-retry-and-deduplication.md 161
063 prompts/16-nameplate-vision-job.md 163
064 prompts/F14-nameplate-ocr.md 166
065 prompts/17-approved-enquiry-pdf.md 168
066 prompts/18-frontend-shell-and-authentication.md 171
067 prompts/F01-branded-landing.md 176
068 prompts/19-customer-chat-and-demo.md 178
069 prompts/F02-demo-scenarios-and-assets.md 182
070 prompts/F05-demo-reset.md 184
071 prompts/F09-bilingual-chat.md 186
072 prompts/F11-clarifying-questions.md 188
073 prompts/F13-private-uploads.md 190
074 prompts/F15-ocr-confirmation.md 192
075 prompts/20-staff-workspace-and-timeline.md 194
076 prompts/F03-demo-role-switching.md 198
077 prompts/F18-progress-timeline.md 200
078 prompts/F27-match-explanations.md 202
079 prompts/F32-warehouse-stock.md 204
080 prompts/F37-review-workspace.md 206
081 prompts/F39-review-actions.md 208
082 prompts/F61-enquiry-pdf.md 210
083 prompts/F16-photo-quality-feedback.md 212
084 prompts/F17-multi-part-enquiries.md 216
085 prompts/F19-enquiry-history.md 220
086 prompts/F55-saved-machines.md 223
087 prompts/F56-repeat-enquiry.md 226
088 prompts/F59-catalogue-versioning.md 229
089 prompts/F57-catalogue-browser.md 233
090 prompts/F58-catalogue-imports.md 236
091 prompts/F60-missing-parts-queue.md 240
092 prompts/F35-stock-freshness.md 243
093 prompts/F29-evidence-page-viewer.md 247
094 prompts/F31-candidate-comparison.md 250
095 prompts/F38-correct-extracted-details.md 253
096 prompts/F42-assignments.md 257
097 prompts/F41-internal-notes.md 260
098 prompts/F43-overdue-indicators.md 263
099 prompts/F44-conversation-summary.md 266
100 prompts/F36-staff-inbox.md 270
101 prompts/F07-arabic-and-english-interface.md 273
102 prompts/F08-light-and-dark-themes.md 277
103 prompts/F06-responsive-layouts.md 280
104 prompts/F48-input-isolation.md 283
105 prompts/21-labelled-evaluations.md 286
106 prompts/F62-labelled-evaluations.md 289
107 prompts/23-observability-and-credit-limits.md 291
108 prompts/F64-latency-and-ai-cost.md 294
109 prompts/F67-traces.md 296
110 prompts/F63-metrics-dashboard.md 298
111 prompts/F65-correction-feedback.md 302
112 prompts/22-security-and-failure-recovery-gate.md 305
113 prompts/F04-demo-isolation.md 308
114 prompts/F53-graceful-failures.md 310
115 prompts/F54-access-and-permissions.md 312
116 prompts/24-production-containers-and-caddy.md 314
117 prompts/25-ci-and-controlled-deployment.md 317
118 prompts/F66-tests-and-ci.md 320
119 prompts/26-aws-free-plan-deployment.md 322
120 prompts/27-vercel-and-hosted-acceptance.md 325
121 prompts/F68-recruiter-documentation.md 328
122 prompts/F69-demo-walkthrough.md 331
123 features/README.md 334

001 README.md

PartsIQ complete build kit

Start with BUILD_GUIDE.docx for the consolidated reading edition, then execute prompts in prompts/README.md order. This ZIP contains an implementation kit, exact schema/contract specifications, original legacy backend and brand sources. It is not a claim that the full application has already been built or deployed.

Included

• 28 corrected foundation prompts, 00 through 27.

• 69 feature prompt files and 69 feature trace files. Foundation-covered features have verification prompts rather than duplicate implementation.

• 54 implementation steps in dependency order, plus coverage checks.

• Exact additive SQL, per-table RLS, private Storage policies and typed contracts.

• Verified direct dependency release receipts and resolved dependency references.

• Logo exports, semantic tokens and licensed bilingual fonts.

• Deployment, rollback, handover, evaluation, testing and recruiter guidance.

• Original backend ZIP and AI Collaboration Field Guide.

Begin

1. Extract this kit into a new work folder.

2. Give the coding agent AGENTS.md,docs/HANDOVER.md and prompts/00-bootstrap-and-preserve-legacy.md.

3. Execute one prompt at a time in the README order. Review actual tests and commits.

4. Run the associated covered-feature checks when their foundation is complete.

5. Configure real credentials only in local environment files/provider secret stores. No key belongs in this ZIP.

The owner approved 69 feature groups,assigned-only staff review,synthetic public demo,client-configurable retention,OpenAI using their existing free-credit keys,and no paid fallback. AWS Free Plan is conditional and expires; local Compose is the fallback. Do not promise always-on free hosting.

Kit validation: `python tools/kit/verify_kit.py`. This validates the documents/manifests; it does not run the future application's acceptance tests.

002 AGENTS.md

PartsIQ coding-agent operating manual

Session start

Read AGENTS.md,docs/HANDOVER.md,docs/CONSTRAINTS.md,docs/ARCHITECTURE.md,the current prompt,and its READ FIRST files. Summarize current state and intended change in simple English. The source-of-truth order is owner decisions,approved kit contracts,legacy characterization,then implementation convenience.

Work rules

1. Execute one implementation prompt at a time in prompts/README.md order.

2. State files,packages,tests and risks before code. The owner has approved the sequence and listed changes; continue without asking for repeated permission. Ask when a new decision exceeds that approval.

3. A source file has one creator in spec/file-ownership.json. Later prompts modify it. Keep schema,contracts,routes and environment names synchronized.

4. Preserve legacy domain code and numeric catalogue IDs. Characterize first,then make the approved corrections for successor fitment and ambiguity margins.

5. Never edit an applied migration. The candidate-key migration is explicitly approved; other destructive changes need a concrete migration and rollback proposal.

6. Secrets remain server-side. No real customer data or credentials in fixtures,logs,traces or screenshots.

7. Report actual commands and outcomes. Blocked tests are blocked,not passed. A written prompt is not an implemented feature.

8. Comments explain why,who calls the code,and assumptions that affect safety. Do not comment obvious syntax.

9. Use branch prompt/00-bootstrap for foundation00 and the corresponding ID/title for subsequent tasks. Commit each task independently.

10. Review the diff,update handover and explain the result before moving to the next task.

Stop conditions

Stop for a new cross-service boundary,unapproved package or algorithm,reduced compatibility/verification/approval control,public demo access to real data,paid fallback,production deletion,or a contradictory requirement. Show the concrete change and smallest compliant alternative. Approved schema additions and named migrations do not require a new permission loop.

Definition of done

Named files implemented; exact package pins retained; success/failure tests run; security and revision cases covered; documentation updated; actual implementation commit recorded; unresolved external deployment checks explicitly identified. Human approval in the product is mandatory independently of coding-agent authorization.

Documentation habits

Apply all 15 habits mapped in docs/FIELD_GUIDE_MAP.md. Record actual coding-model identifier when available; write 'not exposed' when it is unavailable. Never invent a model version or test result.

003 CLAUDE.md

PartsIQ agent entry point

Read AGENTS.md first. It is the shared operating manual for all coding agents. Use docs/HANDOVER.md and prompts/README.md for current state and execution order. No agent-specific alternative architecture is authorized.

004 MASTER_PROMPT.md

Complete and execute PartsIQ

The planning approval is recorded in docs/DECISIONS.md. Implement this kit one prompt at a time. Do not regenerate scope from memory. Read AGENTS.md and follow prompts/README.md. Foundation-covered F prompts verify coverage without a second implementation. Every feature keeps its trace. Package and schema receipts are specifications until runtime checks execute; do not confuse kit validation with application readiness.

Before deployment,inspect account credit eligibility,actual model access,host sizing,domain binding and security checks. No paid fallback is authorized. Produce a concrete preflight result; if external inputs are unavailable,finish all local work and report the exact blocked activation step.

005 docs/REQUIREMENTS.md

Approved requirements

Recruiter-facing bilingual spare-parts enquiry system with deterministic compatibility and human approval. All original selected behaviours retained by grouping scenario/photos asF02 and access/roles asF54. No voice or quotation builder.

ID	Feature	Behaviour
F01	Branded landing	Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching.
F02	Demo scenarios and assets	Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records.
F03	Demo role switching	Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session.
F04	Demo isolation	Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default.
F05	Demo reset	Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection.
F06	Responsive layouts	Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content.
F07	Arabic and English interface	Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi.
F08	Light and dark themes	Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme.
F09	Bilingual chat	Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence.
F10	Structured extraction	Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity.
F11	Clarifying questions	Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only.
F12	Conversation memory	Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/ revision; never mix customer sessions.
F13	Private uploads	Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR.
F14	Nameplate OCR	Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision.
F15	OCR confirmation	Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text.
F16	Photo quality feedback	Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning.
F17	Multi-part enquiries	Track up to 20 items with independent

ID	Feature	Behaviour
		quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests.
F18	Progress timeline	Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff.
F19	Enquiry history	Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.
F20	Machine identification	Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input.
F21	Machine mismatch detection	Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved.
F22	Compatibility gate	Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate.
F23	Aliases and typo matching	Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context.
F24	Semantic retrieval	Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path.
F25	Ranked candidates	Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit.
F26	Ambiguity handling	Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates.
F27	Match explanations	Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose.
F28	Compatibility references	Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot.
F29	Evidence page viewer	Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document.
F30	Old part numbers	Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration.
F31	Candidate comparison	Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts.
F32	Warehouse stock	Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot.
F33	Fulfilment planning	Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders.
F34	Transfer estimates	Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days.
F35	Stock freshness	Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible.
F36	Staff inbox	List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope.
F37	Review workspace	Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages.

ID	Feature	Behaviour
F38	Correct extracted details	Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after.
F39	Review actions	Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key.
F40	Approval invalidation	Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval.
F41	Internal notes	Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content.
F42	Assignments	Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped.
F43	Overdue indicators	Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC.
F44	Conversation summary	Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change.
F45	Reply drafting	Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference.
F46	Verification gate	Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results.
F47	Human publication gate	Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates.
F48	Input isolation	Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates.
F49	Audit trail	Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads.
F50	Durable workflows	Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart.
F51	Background jobs	Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup.
F52	Retry and deduplication	Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token.
F53	Graceful failures	Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input.
F54	Access and permissions	Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup.
F55	Saved machines	Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue.
F56	Repeat enquiry	Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again.
F57	Catalogue browser	Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect

ID	Feature	Behaviour
		fitment and sources.
F58	Catalogue imports	Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish.
F59	Catalogue versioning	Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action.
F60	Missing-parts queue	Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation.
F61	Enquiry PDF	Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download.
F62	Labelled evaluations	Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls.
F63	Metrics dashboard	Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments.
F64	Latency and AI cost	Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance.
F65	Correction feedback	List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits.
F66	Tests and CI	Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs.
F67	Traces	Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable.
F68	Recruiter documentation	Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools.
F69	Demo walkthrough	Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions.

006 docs/DISCREPANCIES.md

Corrections applied

Original conflict	Correction
71 selected entries called69	Two approved groupings preserve all behaviours
New package/table names silently replace legacy	Original numeric schema and modules preserved
RRF replaces weighted ranking	Weighted legacy fusion retained
Catalogue worker loads BGE	Search-only encoder and ingestion API
Embedding before search exists	Base loading05,embedding06
Gateway cannot write enquiry/jobs	Intake/session/staff write ownership explicitly granted
Worker resume HTTP forbidden but implied	Durable workflow event consumption
All replies require review	Clarification/status exception;final recommendation gated
Single item state	Per-item identities and aggregate stock verification
Demo reviewer equals global staff	Own-session capability separated from membership
Model confidence used as accuracy	Mandatory identifier confirmation
Arbitrary model Jinja template	Structured slots with code-owned rendering
Usage counted only after spend	Atomic reservation and conservative timeout accounting
Redis is durable source	DB jobs/events plus sweeper
Runtime checkpointer DDL	Migration-only setup and explicit library schema receipt
Audit impossible for anyone to change	Append-only application roles;owner privilege acknowledged
Four language demos replace business cases	Mixed photo,missing serial,incompatible part,local shortage
Every feature implemented twice	Foundation coverage prompts plus unique implementation owner
Missing package pins	Official release receipts and dependency resolution references
New SSR helper conflicts with client	SSR0.8.0 chosen for client2.99.1 compatibility
Free key assumed free forever	Confirmed allowance/expiry,default disabled,no paid fallback
Free host assumed enough RAM	Measured preflight and local fallback

007 docs/ARCHITECTURE.md

Architecture

1. What is in the system?

Next.js customer and staff interfaces on Vercel call a FastAPI gateway behind Caddy on EC2. Supabase Auth establishes identity; PostgreSQL stores authoritative data; Storage holds private photos,evidence and exports. The same EC2 Docker network hosts search,orchestrator,worker,sweeper,orchestrator event consumer and Redis. OpenAI is outbound language/vision; BGE-M3 runs only inside search. Specialists are modules,not independent servers.

2. Who owns each responsibility?

Responsibility	Owner	Exact home
Auth,demo,customer/staff CRUD	Gateway	services/gateway/app/ auth.py,permissions.py,demo.py,enquiries.p y,features
Identification	Search	services/search/app/identification.py
Lexical ranking	Search	services/search/app/lexical.py
Retrieval coordination	Search	services/search/app/ service.py,repository.py
Allocation	Search	services/search/app/fulfilment.py
Embedding generation	Search	services/search/app/embedding_ ingestion.py
Routing and durable state	Orchestrator	services/orchestrator/app/ controller.py,graph.py,events.py
Intake/parts/reply	Orchestrator specialists	services/orchestrator/app/specialists
Verification	Orchestrator	services/orchestrator/app/verification.py
Approval/revisions	Orchestrator	services/orchestrator/app/ approval.py,revisions.py
Nameplate specialist	Worker	services/worker/app/jobs/nameplate.py
PDF/signing/cleanup	Worker	services/worker/app/jobs/export_ pdf.py,storage.py
Imports	Worker and catalogue domain	services/worker/app/features/f58_ catalogue_imports.py,tools/catalog/catalog
Catalogue publication	Catalogue application	tools/catalog/catalog/versioning.py
Shared contracts,audit,credit	Common	libs/partsiq_common/partsiq_common
Screens and accessibility	Web	apps/web
Schema,RLS and Storage policies	Migrations	supabase/migrations
Metrics and corrections API	Gateway	services/gateway/app/features
Tests,evaluations and release	Tooling	tests,tools/evals,infra,.github/workflows

3. Why this design?

The controller uses deterministic transitions because missing serials,verification failures and approval gates do not need model routing. Separate specialists give bounded inputs and testable contracts. Compatibility is a database relation,not a similarity judgement. Redis loss is recoverable because job state lives in PostgreSQL. A single host limits operational overhead; it also creates a documented single-host availability limitation. No high-availability claim is made.

4. What may touch what?

Web calls gateway and Supabase Auth; it may transfer an authorized private Storage object. Gateway verifies identity,writes intake/session/staff records and calls internal search/orchestrator/worker signing routes. Search reads catalogue and inventory and writes only embeddings through its administrative ingestion entry point. Orchestrator reads/writes workflow data,uses search and OpenAI,and owns final publication. Worker consumes jobs,uses Storage/OpenAI vision,and writes durable workflow events; it never calls orchestrator HTTP. Catalogue publication runs under the restricted catalogue role after validation and admin authorization. Runtime services never use migration credentials.

Within a service: route -> application use case -> domain function -> repository/adapter. Preserve existing legacy modules initially; an abstract layering preference is not permission to rewrite them.

5. How does data move?

Gateway commits enquiry,message,revision and start event. Event consumer starts LangGraph. Optional OCR job returns a durable completion event. Customer confirms identifiers. Controller invokes identification and compatible retrieval. Customer/staff resolves ambiguity. Planner computes allocations. Reply specialist fills prose slots; renderer inserts structured facts. Verification records rule results. Staff approves current revision;

approval transaction creates one customer recommendation. Export worker renders the approved snapshot. FLOW.md binds these steps to concrete prompt-created functions.

6. What cannot break?

Compatible-set membership,explicit evidence,current revision,quantity/stock bounds,assigned/session access,no model writes,no unapproved publication,no stale worker commits,and no paid fallback. CONSTRAINTS.md is the complete rule list.

7. Where does new code belong?

Use spec/file-ownership.json as the exhaustive planned application file inventory. A new file must have one creator. Feature backend modules live under the relevant service app/features directory; existing legacy modules retain their paths. UI feature components live in apps/web/components/features. Tests are colocated by service or live in root acceptance/integration directories. Do not introduce another orchestrator,API or catalogue implementation.

8. When must the agent ask?

Ask for an unapproved change to locked architecture,permissions,paid services,destructive production migration or safety gates. The specified additions and migrations are already approved. Account IDs,domains and secrets are deployment inputs,not values to invent. Finish local reversible work before reporting an external activation blocker.

008 docs/CONSTRAINTS.md

Constraints

1. Next.js runs on Vercel; Supabase owns data,Auth and private files in separate dev/prod projects.

2. Production backend is one EC2 host with Docker Compose,Caddy,gateway,search,orchestrator,worker,sweeper and Redis. Render is an alternative appendix only.

3. OpenAI language and vision only. Use the owner's key after explicit credit allowance configuration. No Gemini switch or paid fallback.

4. BGE-M3 loads only in search. Catalogue tools and workers submit embedding work to search.

5. Compatibility,quantity validation,stock planning,verification,workflow transitions and approval live in code. Models can extract customer statements and propose prose,not authorize facts.

6. Specialist tools are read-only. A deterministic application layer persists validated results. An LLM never receives a write,approval or catalogue-publication tool.

7. Final recommendations require current-revision human approval. Customer messages,code-owned clarification and safe progress messages do not require recommendation approval.

8. Only orchestrator approval code publishes recommendation messages; database trigger and unique publication index backstop it.

9. Assigned-only staff access for real enquiries; own active demo session for demo review. Admin assignment is audited.

10. No secret in frontend,NEXT_PUBLIC variables,URLs,logs,fixtures,trace payloads or commits. Publishable Supabase key is not a secret.

11. Runtime SQL uses separate restricted service login roles. Gateway sets transaction-local verified actor context. Browser does not query application tables directly.

12. Private Storage transfer is the narrow browser exception. Gateway authorizes; signed links expire; worker storage credential is never used for application-table REST access.

13. Public containers: Caddy80/443 only. Internal service token and request ID required; do not expose search,Redis,worker or orchestrator.

14. Redis is dispatch/cache,not durable truth. Jobs,events and workflow checkpoints survive Redis loss.

15. A stale revision,session generation or lease token cannot commit derived results. Approval is transactional and idempotent.

16. Preserve legacy modules,numeric IDs,weighted lexical/semantic ranking and transfer_days unless a separately approved ADR changes them.

17. Published evidence/revision snapshots are immutable. Audit is append-only for application roles; database-owner access is not magically prevented.

18. No voice,quotation builder,prices,payments,external messaging,actual stock reservations,ERP sync or order creation.

19. No unapproved dependency,folder pattern or service. Exact direct pins and transitive lock references are supplied; verify again when executing on the target runtime.

20. Do not edit applied migrations or reset production. Preview/local uses dev or disposable Supabase only.

21. Production client retention must be configured before real-client onboarding. Public V1 uses synthetic demo data.

22. No fake accuracy,latency,uptime,client affiliation,test result or deployed-status claim. Measured evidence must name dataset and environment.

009 docs/DECISIONS.md

Decision record

All entries below reflect the owner's approval in the planning conversation on 2026-10-11. Authoring model identifier was not exposed; no invented model version is recorded.

ADR	Decision	Reason and consequence
ADR-001	Extend legacy backend	Preserve existing modules,tables,numeric IDs and characterizations; additive migrations only except explicitly approved candidate key upgrade
ADR-002	EC2 Compose behind Caddy	One host,low operational complexity; conditional AWS Free Plan,local fallback; Render alternative only
ADR-003	Deterministic LangGraph controller	No LLM supervisor for rule-based transitions
ADR-004	Supabase dev/prod plus Vercel	Separate projects and secrets; preview never points at prod
ADR-005	OpenAI language/vision	Owner has keys/free credits; retain provider; no Gemini change
ADR-006	Search-only BGE-M3	One encoder instance; tools submit embedding work
ADR-007	Structured facts and code renderer	LLM prose cannot authorize part numbers,quantities or delivery claims
ADR-008	Human approval bound to revision	Edits/catalogue changes invalidate pending derived results
ADR-009	Durable jobs/events and lease fencing	Redis dispatch is recoverable; no exactly-once external-call guarantee
ADR-010	69 grouped features	Combine scenarios/photos and access/roles; preserve all selected behaviours
ADR-011	Foundation ownership plus F coverage prompts	69 trace files; no duplicate implementation for covered features
ADR-012	Public synthetic demo; real signup disabled	Recruiter-first scope; assigned-only operational review
ADR-013	Mandatory OCR confirmation	Do not treat model confidence as calibrated correctness
ADR-014	Worker Storage capability	Private worker credential only for authorized object operations; gateway uses end-user upload policy
ADR-015	Application free-credit allowance	Default disabled; manually confirm allowance/expiry; hard local reserve; no automatic paid fallback
ADR-016	Client-defined retention	Demo default24h; real-client onboarding blocked until policy configured
ADR-017	Preserve weighted ranking	No unapproved RRF/trigram rewrite; characterize then evaluate changes
ADR-018	Supersession integrity correction	Terminal current part needs explicit fitment; calculate confidence margin before truncation
ADR-019	Supabase SSR0.8.0	Replaces proposed0.12.7 because it required supabase-js2.114.0; matched pair resolves with client2.99.1
ADR-020	Brand source preserved	Navy/gold tokens; supplied PNG artwork; licensed Noto fonts; no fabricated SVG

010 docs/FLOW.md

Execution flow and function ownership

The primary enquiry flow is 08 intake persistence ->09 event/controller ->10 intake ->16 optional OCR ->19 confirmation ->11 search tools backed by06 ->13 draft ->12 verification ->20 review ->14 publication ->17 export. The build order differs: a function may be created before its caller is wired. Production readiness remains false until required handlers are registered.

Function registry

Every entry below is created or explicitly preserved by its named prompt. Detailed file paths,request/result types and validation live in that prompt and CONTRACTS.md.

Prompt	Owner	Entry contract	Build prerequisites
00	Repository	def verify_environment(env: Mapping[str,str]) -> list[str]: return missing or unsafe environment names without values; reject prod database URLs when APP_ENV is local.	start
01	Common	async def call_ structured(request: AIRequest, budget: BudgetReservation) -> AIResult; async def append_ audit(conn: AsyncConnection, event: AuditEvent) -> int; async def actor_transaction(actor: ActorContext) -> AsyncIterator[AsyncConnection] .	00
02	Database	async def apply_catalogue_ schema(dsn: str) -> None; schema comparison is owned by collect_schema_receipt in prompt03.	01
03	Database	async def setup_ checkpoints(dsn: str) -> None; async def collect_schema_ receipt(dsn: str) -> SchemaReceipt; no application service invokes setup_ checkpoints.	02
04	Gateway/worker	async def create_upload(actor: ActorContext, request: UploadRequest) -> UploadTicket; async def sign_ download(request: DownloadRequest) -> DownloadTicket; async def remove_object(bucket: str, object_key: str) -> None.	03
05	Catalogue	def build_scenarios(dataset: CatalogDataset) -> list[DemoScenario]; def render_ nameplate(model: str, serial: str, destination: Path) -> None; def validate_ scenarios(scenarios: list[DemoScenario], dataset: CatalogDataset) -> list[str].	04
06	Search	async def embed_ version(version_id: UUID, batch_ size: int=16) -> EmbeddingReceipt; preserve SearchService.identify(request: IdentifyRequest) -> IdentifyResponse and SearchService.search_ parts(request: PartSearchRequest) -> PartSearchResponse.	05
07	Gateway	async def authenticate(token: str) -> ActorContext; async def require_access(actor: ActorContext, enquiry_id: UUID, action: AccessAction) -> None; async def create_demo(actor: ActorContext) -> DemoSession; async def reset_demo(actor: ActorContext, key: UUID) -> DemoSession.	06
08	Gateway	async def create_enquiry(actor:	07

Prompt	Owner	Entry contract	Build prerequisites
		ActorContext, body: EnquiryCreate, key: UUID) -> EnquiryView; async def add_message(actor: ActorContext, enquiry_id: UUID, body: MessageCreate, key: UUID) -> MessageView; async def confirm_upload(actor: ActorContext, attachment_id: UUID) -> AttachmentView.	
09	Orchestrator	def next_step(state: EnquiryState) -> WorkflowStep; def build_graph(handlers: WorkflowHandlers, checkpointer: AsyncPostgresSaver) -> CompiledStateGraph; async def consume_event(event_id: UUID) -> EventOutcome.	08
10	Orchestrator	async def extract_intake(messages: list[MessageView], previous: IntakeResult / None, actor: ActorContext) -> IntakeResult.	09
11	Orchestrator	async def run_parts(item: ItemState, context: ToolContext) -> PartsProposal; async def identify(request: IdentifyRequest) -> IdentifyResponse; async def search_parts(request: PartSearchRequest) -> PartSearchResponse; async def plan_fulfilment(request: FulfilmentRequest) -> FulfilmentResponse.	10
12	Orchestrator	def verify_facts(facts: VerifiedFacts, evidence: VerificationEvidence) -> VerificationResult; async def verify_current(enquiry_id: UUID, revision: int) -> VerificationResult.	11
13	Orchestrator	async def draft_reply(facts: VerifiedFacts, locale: Locale) -> ReplySlots; def render_reply(slots: ReplySlots, facts: VerifiedFacts, locale: Locale) -> str.	12
14	Orchestrator	async def approve(actor: ActorContext, request: ApprovalRequest, key: UUID) -> ApprovalResult; async def reject(actor: ActorContext, request: RejectionRequest, key: UUID) -> ReviewResult; async def revise(actor: ActorContext, request: RevisionRequest) -> EnquiryView.	13
15	Worker	async def claim_job(job_id: UUID, worker_id: str) -> JobLease / None; async def heartbeat(lease: JobLease) -> bool; async def finish_job(lease: JobLease, result: JobResult) -> bool; async def sweep_once(now: datetime) -> int.	14
16	Worker	async def read_nameplate(job: JobLease, image: bytes) -> NameplateResult; def validate_image(image: bytes, media_type: str) -> ImageInfo.	15
17	Worker	async def export_enquiry(job: JobLease, request: ExportRequest) -> ExportResult; def render_export_html(snapshot: ApprovedSnapshot) -> str.	16
18	Web	export function BrandLogo(props: {compact?: boolean}): React.JSX.Element; export async function api<T>(path: string, options: RequestInit): Promise<T>; export async function startAnonymousDemo(): Promise<DemoSession>.	17

Prompt	Owner	Entry contract	Build prerequisites
19	Web	export function Chat(props: {enquiryId: string}): React.JSX.Element; export function OcrConfirmation(props: {extraction: NameplateResult; onConfirm: (value: ConfirmExtraction) => Promise<void>}): React.JSX.Element.	18
20	Web	export function Workspace(props: {enquiryId: string}): React.JSX.Element; export function ApprovalControls(props: {review: ReviewView; onChanged: () => Promise<void>}): React.JSX.Element.	19
F16	worker	def assess_photo(request: F16Request) -> F16Result	16, 19
F17	orchestrator	async def resolve_items(actor: ActorContext, request: F17Request) -> F17Result	09, 10, 11, 12, 13, 14, 19, 20
F19	gateway	async def list_history(actor: ActorContext, request: F19Request) -> F19Result	07, 08, 19
F55	gateway	async def save_machine(actor: ActorContext, request: F55Request) -> F55Result	07, 19
F56	gateway	async def repeat_enquiry(actor: ActorContext, request: F56Request) -> F56Result	09, 19, F19, F55
F59	catalog	async def publish_version(actor: ActorContext, request: F59Request) -> F59Result	02, 05, 06
F57	gateway	async def browse_ catalogue(actor: ActorContext, request: F57Request) -> F57Result	02, 06, 18, F59
F58	worker	async def validate_import(actor: ActorContext, request: F58Request) -> F58Result	02, 05, 15, F59, F57
F60	gateway	async def update_missing_ part(actor: ActorContext, request: F60Request) -> F60Result	03, 20
F35	search	async def check_stock_ freshness(actor: ActorContext, request: F35Request) -> F35Result	06, 12, 14, 20
F29	gateway	async def open_evidence(actor: ActorContext, request: F29Request) -> F29Result	04, 06, 20
F31	web	export function compare_ candidates(request: F31Request): F31Result	06, 20
F38	orchestrator	async def correct_ extraction(actor: ActorContext, request: F38Request) -> F38Result	09, 14, 20
F42	gateway	async def assign_enquiry(actor: ActorContext, request: F42Request) -> F42Result	03, 07, 20
F41	gateway	async def append_note(actor: ActorContext, request: F41Request) -> F41Result	03, 07, 20
F43	gateway	async def classify_ deadline(actor: ActorContext, request: F43Request) -> F43Result	20
F44	orchestrator	async def summarize_ enquiry(actor: ActorContext, request: F44Request) -> F44Result	09, 13, 20
F36	gateway	async def list_inbox(actor: ActorContext, request: F36Request) -> F36Result	07, 20, F42, F43
F07	web	export function resolve_ locale(request: F07Request): F07Result	18, 19, 20
F08	web	export function resolve_ theme(request: F08Request): F08Result	18
F06	web	export function get_	18, 19, 20, F07, F08

Prompt	Owner	Entry contract	Build prerequisites
		layout(request: F06Request): F06Result	
F48	orchestrator	def validate_tool_call(request: F48Request) -> F48Result	10, 11, 12, 13, 16
21	Quality	def evaluate_case(case: EvalCase, actual: EvalOutput) -> EvalResult; async def run_suite(cases: list[EvalCase], live: bool=False) -> EvalRun.	F48
23	Operations	async def get_metrics(actor: ActorContext, query: MetricsQuery) -> MetricsView; def validate_allowance(amount: Decimal, expires_at: datetime, paid_budget: Decimal) -> CreditAllowance; async def readiness_report() -> ReadinessReport.	21
F63	gateway	async def metrics_dashboard(actor: ActorContext, request: F63Request) -> F63Result	21, 23
F65	gateway	async def curate_correction(actor: ActorContext, request: F65Request) -> F65Result	14, 21, 23, F63
22	Quality	def assert_disposable_environment(config: TestConfig) -> None; test suites instantiate existing service entry points without new business logic.	F65
24	Infrastructure	def verify_images(compose_file: Path) -> list[str]: reject mutable application tags,missing healthchecks,public internal ports and embedded credentials.	22
25	Infrastructure	def check_ci_configuration(root: Path) -> list[str]: inspect permissions,action pinning,prod-test protection and deployment guards.	24
26	Infrastructure	def preflight(account: AccountReceipt, sizing: SizingReceipt) -> DeploymentDecision; bootstrap.sh accepts a validated environment file path; deploy.sh accepts one hexadecimal release SHA.	25
27	Infrastructure	async def run_hosted_acceptance(base_url: str, api_url: str) -> HostedReceipt; reject non-HTTPS URLs outside localhost.	26
F68	docs	Documentation contract: README.md and docs/RECRUITER_GUIDE.md	21, 23, 22, 24, 25, 26, 27
F69	docs	Documentation contract: docs/DEMO_SCRIPT.md and docs/RECORDING_CHECKLIST.md	27, F68

Event and transaction boundaries

Intake transaction commits before dispatch. OCR completion transaction fences lease/revision/generation and inserts workflow event. Event consumer uses persisted checkpointer and ignores obsolete events. Approval transaction holds enquiry lock,rechecks permissions,current catalogue and latest stock,then inserts decision,recommendation and audit. Storage cleanup is asynchronous after demo invalidation; it is never described as a cross-system atomic delete.

Status transitions

received -> processing; processing -> needs_information or awaiting_review or failed; needs_information -> processing after validated reply; awaiting_review -> approved or rejected; relevant correction -> processing with incremented revision. A failed dependency can return to processing through an authorized idempotent retry. Approved historical messages remain historical; a changed enquiry needs new review before another recommendation.

011 docs/CONTRACTS.md

Contract rules

`spec/contracts.py` is the exact common Pydantic contract source copied by prompt 01. Use model_validate_json for wire JSON; strict Python validation otherwise. Decimal quantities are serialized as decimal strings, UUIDs as canonical strings, timestamps as UTC ISO8601. Reject extra fields.

Feature-specific request/result fields are declared in each F prompt. Define their BaseModel classes in the named feature implementation file, inheriting common Contract. Python optional fields without defaults remain required nullable fields. TypeScript equivalents use the same field names and explicit null. Each UI component uses the concrete request/result props printed in its feature prompt. No generic feature-ID token is left for the implementation agent to guess.

CandidateView: item_id UUID, revision integer, part_id integer, part_number string, name_en string, name_ar string, score number, reasons string array, source_reference string, evidence_id UUID, stock StockEvidence array. ComparisonRow: field string, values array of string or null. PartView: id integer, part_number string, name_en string, name_ar string, category string, is_active boolean, current_part_id integer or null. ImportErrorView: row_number integer, column string, code string, message string.

WorkflowHandlers is a dataclass with `intake`, `identify`, `search`, `choose`, `plan`, `draft`, `verify` asynchronous callables taking EnquiryState and returning EnquiryState. Blocking states are graph interrupts, not executable specialist functions. Production startup fails readiness until every required handler is registered. Test doubles are injected only by tests.

Public error JSON: `{"error":{"code":"STALE_REVISION","message":"Reload the enquiry and review the current revision.","request_id":"00000000-0000-4000-8000-000000000001"}}`. 400 malformed request; 401 unauthenticated; 403 forbidden role; 404 unavailable/foreign resource; 409 stale/conflicting action; 413 size limit; 415 unsupported content; 422 field validation; 429 quota; 503 dependency unavailable. Never include stack traces or credentials.

Pagination: opaque base64url JSON containing created_at and id, validated before SQL binding; limit 1 through 100 with default20. Cursor is not authorization. Idempotent writes use UUID Idempotency-Key and compare canonical payload hash. Same key/different input returns409. Request validation precedes any provider call.

Legacy SearchService request/response models remain in services/search/app/schemas.py. Add catalogue_version_id and per-item correlation without renaming legacy fields. The adapter translates common contracts into those models. Do not duplicate search domain types in gateway.

Specialized supplementary routes and files are listed in each feature prompt. Use idempotency_records for session reset,notes,saved-machine writes,assignment and catalogue publication where no domain-specific key exists. A reservation response is stored atomically with business change; abandoned request rows have response NULL and may only be resumed under row lock.

012 docs/TECH_SPEC.md

Technical specification

1. Stack and conventions

Python3.12.12,Node22.22.0,Next.js,React,TypeScript,Tailwind,shadcn-derived components,next-intl,FastAPI,LangGraph,Supabase and OpenAI. Preserve asyncpg for legacy search/catalogue; psycopg supports new transactional repositories/checkpointer. All decimals are strings over JSON. UTC timestamps; bilingual UI with RTL. Root uv workspace uses one uv.lock. Run service tests separately because legacy import packages are named app.

2. Exhaustive application file ownership

The table lists planned application files,not files claimed to be implemented in this ZIP. Original code comes from references/legacy-backend.zip. Generated assets are enumerated in brand/assets-manifest.json; prompt18 copies them byte-for-byte into apps/web/public/brand. Python package __init__.py markers required for new package directories are created by their first owning prompt and listed by the final manifest expansion. The machine-readable authority is spec/file-ownership.json.

Path	Creating prompt	Later modifying prompts
.env.example	00	none
.github/workflows/ci.yml	25	none
.github/workflows/deploy-backend.yml	25	26
.gitignore	00	none
.nvmrc	00	none
.python-version	00	none
Caddyfile	24	none
README.md	00	F68
apps/web/app/chat/[id]/page.tsx	19	none
apps/web/app/demo/page.tsx	19	none
apps/web/app/history/page.tsx	F19	none
apps/web/app/layout.tsx	18	F06, F07, F08
apps/web/app/login/page.tsx	18	none
apps/web/app/page.tsx	18	F06, F07, F08, F16, F17, F19, F36, F55, F56, F57, F58, F59, F60, F63, F65
apps/web/app/staff/analytics/page.tsx	F63	none
apps/web/app/staff/catalogue/page.tsx	F57	none
apps/web/app/staff/enquiries/[id]/page.tsx	20	F29, F31, F35, F38, F41, F42, F43, F44
apps/web/app/staff/inbox/page.tsx	20	F36
apps/web/components/BrandLogo.tsx	18	none
apps/web/components/chat/Chat.tsx	19	F16, F17
apps/web/components/chat/OcrConfirmation.tsx	19	none
apps/web/components/chat/Progress.tsx	19	none
apps/web/components/chat/UploadCard.tsx	19	none
apps/web/components/features/AssignmentControl.tsx	F42	none
apps/web/components/features/CandidateComparison.tsx	F31	none
apps/web/components/features/CatalogueBrowser.tsx	F57	none
apps/web/components/features/CatalogueVersions.tsx	F59	none
apps/web/components/features/ConversationSummary.tsx	F44	none
apps/web/components/features/CorrectionEditor.tsx	F38	none
apps/web/components/features/CorrectionFeedback.tsx	F65	none
apps/web/components/features/EvidenceViewer.tsx	F29	none
apps/web/components/features/HistoryList.tsx	F19	none
apps/web/components/features/ImportWizard.tsx	F58	none
apps/web/components/features/InboxFilters.tsx	F36	none
apps/web/components/features/InternalNotes.tsx	F41	none
apps/web/components/features/	F17	none

Path	Creating prompt	Later modifying prompts
ItemList.tsx		
apps/web/components/features/LocaleSwitcher.tsx	F07	none
apps/web/components/features/MetricsDashboard.tsx	F63	none
apps/web/components/features/MissingPartsQueue.tsx	F60	none
apps/web/components/features/OverdueBadge.tsx	F43	none
apps/web/components/features/PhotoQualityNotice.tsx	F16	none
apps/web/components/features/RepeatEnquiry.tsx	F56	none
apps/web/components/features/ResponsiveShell.tsx	F06	none
apps/web/components/features/SavedMachines.tsx	F55	none
apps/web/components/features/StockFreshness.tsx	F35	none
apps/web/components/features/ThemeToggle.tsx	F08	none
apps/web/components/review/ApprovalControls.tsx	20	none
apps/web/components/review/Timeline.tsx	20	none
apps/web/components/review/Workspace.tsx	20	none
apps/web/eslint.config.mjs	18	none
apps/web/i18n/request.ts	F07	none
apps/web/lib/api.ts	18	none
apps/web/lib/auth.ts	18	none
apps/web/lib/contracts.ts	18	none
apps/web/lib/features/f06_responsive_layouts.ts	F06	none
apps/web/lib/features/f07_arabic_and_english_interface.ts	F07	none
apps/web/lib/features/f08_light_and_dark_themes.ts	F08	none
apps/web/lib/features/f16_photo_quality_feedback.ts	F16	none
apps/web/lib/features/f17_multi_part_enquiries.ts	F17	none
apps/web/lib/features/f19_enquiry_history.ts	F19	none
apps/web/lib/features/f29_evidence_page_viewer.ts	F29	none
apps/web/lib/features/f31_candidate_comparison.ts	F31	none
apps/web/lib/features/f35_stock_freshness.ts	F35	none
apps/web/lib/features/f36_staff_inbox.ts	F36	none
apps/web/lib/features/f38_correct_extracted_details.ts	F38	none
apps/web/lib/features/f41_internal_notes.ts	F41	none
apps/web/lib/features/f42_assignments.ts	F42	none
apps/web/lib/features/f43_overdue_indicators.ts	F43	none
apps/web/lib/features/f44_conversation_summary.ts	F44	none
apps/web/lib/features/f55_saved_machines.ts	F55	none
apps/web/lib/features/f56_repeat_enquiry.ts	F56	none
apps/web/lib/features/f57_catalogue_browser.ts	F57	none
apps/web/lib/features/f58_catalogue_imports.ts	F58	none
apps/web/lib/features/f59_catalogue_versioning.ts	F59	none
apps/web/lib/features/f60_missing_parts_queue.ts	F60	none
apps/web/lib/features/f63_metrics_dashboard.ts	F63	none
apps/web/lib/features/f65_correction_feedback.ts	F65	none
apps/web/messages/ar.json	F07	none

Path	Creating prompt	Later modifying prompts
apps/web/messages/en.json	F07	none
apps/web/next.config.ts	18	F07
apps/web/package.json	18	none
apps/web/playwright.config.ts	18	none
apps/web/pnpm-lock.yaml	18	none
apps/web/postcss.config.mjs	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/Preview.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-1024.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-128.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-144.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-152.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-16.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-180.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-192.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-24.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-256.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-32.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-384.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-48.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-512.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-64.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-96.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-1024.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-128.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-144.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-152.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-16.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-180.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-192.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-24.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-256.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-	18	none

Path	Creating prompt	Later modifying prompts
white-32.png		
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-384.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-48.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-512.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-64.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-96.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-1280.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-1740.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-320.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-640.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-1280.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-1740.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-320.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-640.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-hdpi-72.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-mdpi-48.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xhdpi-96.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xxhdpi-144.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xxxhdpi-192.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/google-play-512.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/ios-app-store-1024.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/social/partsiq-og-1200x630.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/social/partsiq-profile-1080.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/source/original-logo.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/source/transparent-logo.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/android-chrome-192x192.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/android-chrome-512x512.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/apple-touch-icon.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-16x16.png	18	none
apps/web/public/brand/assets/PartsIQ-	18	none

Path	Creating prompt	Later modifying prompts
Asset-Pack/web/favicon-32x32.png		
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-48x48.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-96x96.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon.ico	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/maskable-icon-192x192.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/maskable-icon-512x512.png	18	none
apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/site.webmanifest	18	none
apps/web/public/brand/fonts/NotoNaskhArabic-OFL.txt	18	none
apps/web/public/brand/fonts/NotoNaskhArabic.ttf	18	none
apps/web/public/brand/fonts/NotoSans-OFL.txt	18	none
apps/web/public/brand/fonts/NotoSans.ttf	18	none
apps/web/public/brand/fonts/NotoSansArabic-OFL.txt	18	none
apps/web/public/brand/fonts/NotoSansArabic.ttf	18	none
apps/web/public/brand/logo-original.png	18	none
apps/web/styles/tokens.css	18	F06, F08
apps/web/tests/features/f01.test.tsx	18	none
apps/web/tests/features/f06.test.tsx	F06	none
apps/web/tests/features/f07.test.tsx	F07	none
apps/web/tests/features/f08.test.tsx	F08	none
apps/web/tests/features/f15.test.tsx	19	none
apps/web/tests/features/f18.test.tsx	20	none
apps/web/tests/features/f27.test.tsx	20	none
apps/web/tests/features/f31.test.tsx	F31	none
apps/web/tests/features/f37.test.tsx	20	none
apps/web/tests/features/f53.test.tsx	22	none
apps/web/tests/setup.ts	18	none
apps/web/tsconfig.json	18	none
apps/web/vitest.config.ts	18	none
data/.gitkeep	00	none
db/migrations/001_initial_schema.sql	00	none
db/migrations/002_search_and_fulfilment.sql	00	none
docker-compose.prod.yml	24	none
docker-compose.yml	00	none
docs/DEMO_SCRIPT.md	F69	none
docs/RECORDING_CHECKLIST.md	F69	none
docs/RECRUITER_GUIDE.md	F68	none
infra/ec2/bootstrap.sh	26	none
infra/ec2/deploy.sh	26	none
infra/ec2/preflight.py	26	none
infra/ec2/rollback.sh	26	none
infra/images.lock.json	24	none
infra/vercel/README.md	27	none
infra/verify_images.py	24	none
libs/partsiq_common/partsiq_common/__init__.py	00	none
libs/partsiq_common/partsiq_common/ai/__init__.py	01	none
libs/partsiq_common/partsiq_common/ai/budget.py	01	23
libs/partsiq_common/partsiq_common/ai/client.py	01	23
libs/partsiq_common/partsiq_common/ai/prices.json	01	none
libs/partsiq_common/partsiq_common/app_factory.py	00	none
libs/partsiq_common/partsiq_common/audit.py	01	none
libs/partsiq_common/partsiq_common/config.py	00	none
libs/partsiq_common/partsiq_common/contracts.py	01	none

Path	Creating prompt	Later modifying prompts
libs/partsiq_common/partsiq_common/db.py	00	none
libs/partsiq_common/partsiq_common/embeddings.py	00	06
libs/partsiq_common/partsiq_common/errors.py	01	none
libs/partsiq_common/partsiq_common/health.py	00	none
libs/partsiq_common/partsiq_common/logging.py	00	none
libs/partsiq_common/partsiq_common/middleware.py	00	none
libs/partsiq_common/partsiq_common/security_context.py	01	none
libs/partsiq_common/partsiq_common/telemetry.py	01	23
libs/partsiq_common/partsiq_common/text.py	00	none
libs/partsiq_common/pyproject.toml	00	none
pyproject.toml	00	none
services/gateway/Dockerfile	00	24
services/gateway/app/__init__.py	00	none
services/gateway/app/api_v1.py	08	F19, F29, F36, F41, F42, F43, F55, F56, F57, F60, F63, F65, F58, F59, F38, F44, 16
services/gateway/app/auth.py	07	none
services/gateway/app/config.py	00	none
services/gateway/app/demo.py	07	none
services/gateway/app/enquiries.py	08	none
services/gateway/app/features/__init__.py	F19	none
services/gateway/app/features/f19_enquiry_history.py	F19	none
services/gateway/app/features/f29_evidence_page_viewer.py	F29	none
services/gateway/app/features/f36_staff_inbox.py	F36	none
services/gateway/app/features/f38_correction_routes.py	F38	none
services/gateway/app/features/f41_internal_notes.py	F41	none
services/gateway/app/features/f42_assignments.py	F42	none
services/gateway/app/features/f43_overdue_indicators.py	F43	none
services/gateway/app/features/f44_summary_routes.py	F44	none
services/gateway/app/features/f55_saved_machines.py	F55	none
services/gateway/app/features/f56_repeat_enquiry.py	F56	none
services/gateway/app/features/f57_catalogue_browser.py	F57	none
services/gateway/app/features/f58_import_routes.py	F58	none
services/gateway/app/features/f59_publish_routes.py	F59	none
services/gateway/app/features/f60_missing_parts_queue.py	F60	none
services/gateway/app/features/f63_metrics_dashboard.py	F63	none
services/gateway/app/features/f65_correction_feedback.py	F65	none
services/gateway/app/main.py	00	07, 08
services/gateway/app/metrics.py	23	none
services/gateway/app/ocr_confirmation.py	16	none
services/gateway/app/permissions.py	07	none
services/gateway/app/proxy.py	00	08
services/gateway/app/rate_limit.py	00	08
services/gateway/app/routes.py	00	07
services/gateway/app/routing.py	00	none
services/gateway/app/security.py	00	07
services/gateway/app/storage.py	04	none
services/gateway/app/uploads.py	08	none
services/gateway/pytest.ini	00	none
services/gateway/requirements-dev.txt	00	none
services/gateway/requirements.txt	00	none

Path	Creating prompt	Later modifying prompts
services/gateway/tests/__init__.py	00	none
services/gateway/tests/features/test_f03.py	20	none
services/gateway/tests/features/test_f04.py	22	none
services/gateway/tests/features/test_f05.py	19	none
services/gateway/tests/features/test_f13.py	19	none
services/gateway/tests/features/test_f19.py	F19	none
services/gateway/tests/features/test_f29.py	F29	none
services/gateway/tests/features/test_f36.py	F36	none
services/gateway/tests/features/test_f41.py	F41	none
services/gateway/tests/features/test_f42.py	F42	none
services/gateway/tests/features/test_f43.py	F43	none
services/gateway/tests/features/test_f54.py	22	none
services/gateway/tests/features/test_f55.py	F55	none
services/gateway/tests/features/test_f56.py	F56	none
services/gateway/tests/features/test_f57.py	F57	none
services/gateway/tests/features/test_f60.py	F60	none
services/gateway/tests/features/test_f63.py	F63	none
services/gateway/tests/features/test_f65.py	F65	none
services/gateway/tests/test_routing.py	00	none
services/gateway/tests/test_security.py	00	none
services/orchestrator/Dockerfile	24	none
services/orchestrator/app/__init__.py	09	none
services/orchestrator/app/approval.py	14	F35
services/orchestrator/app/controller.py	09	none
services/orchestrator/app/events.py	09	none
services/orchestrator/app/features/__init__.py	F17	none
services/orchestrator/app/features/f17_multi_part_enquiries.py	F17	none
services/orchestrator/app/features/f38_correct_extracted_details.py	F38	none
services/orchestrator/app/features/f44_conversation_summary.py	F44	none
services/orchestrator/app/features/f48_input_isolation.py	F48	none
services/orchestrator/app/graph.py	09	10, 11, 12, 13, 14
services/orchestrator/app/handlers.py	09	10, 11, 12, 13, 14, F17, F38, F44, F48
services/orchestrator/app/main.py	09	none
services/orchestrator/app/prompts/intake.txt	10	none
services/orchestrator/app/prompts/reply.txt	13	none
services/orchestrator/app/rendering.py	13	none
services/orchestrator/app/revisions.py	14	none
services/orchestrator/app/routes.py	09	F17, F38, F44, F48
services/orchestrator/app/runner.py	09	none
services/orchestrator/app/search_client.py	11	none
services/orchestrator/app/specialists/__init__.py	10	none
services/orchestrator/app/specialists/intake.py	10	none
services/orchestrator/app/specialists/parts.py	11	none
services/orchestrator/app/specialists/reply.py	13	none

Path	Creating prompt	Later modifying prompts
services/orchestrator/app/state.py	09	none
services/orchestrator/app/tools.py	11	none
services/orchestrator/app/verification.py	12	none
services/orchestrator/pyproject.toml	00	none
services/orchestrator/tests/features/test_f09.py	19	none
services/orchestrator/tests/features/test_f10.py	10	none
services/orchestrator/tests/features/test_f11.py	19	none
services/orchestrator/tests/features/test_f12.py	09	none
services/orchestrator/tests/features/test_f17.py	F17	none
services/orchestrator/tests/features/test_f26.py	11	none
services/orchestrator/tests/features/test_f38.py	F38	none
services/orchestrator/tests/features/test_f39.py	20	none
services/orchestrator/tests/features/test_f40.py	14	none
services/orchestrator/tests/features/test_f44.py	F44	none
services/orchestrator/tests/features/test_f45.py	13	none
services/orchestrator/tests/features/test_f46.py	12	none
services/orchestrator/tests/features/test_f47.py	14	none
services/orchestrator/tests/features/test_f48.py	F48	none
services/orchestrator/tests/features/test_f50.py	09	none
services/search/Dockerfile	00	06, 24
services/search/app/__init__.py	00	none
services/search/app/config.py	00	06
services/search/app/embedding_ingestion.py	06	none
services/search/app/features/__init__.py	F35	none
services/search/app/features/f35_stock_freshness.py	F35	none
services/search/app/fulfilment.py	00	none
services/search/app/identification.py	00	none
services/search/app/lexical.py	00	none
services/search/app/main.py	00	06
services/search/app/repository.py	00	06
services/search/app/routes.py	00	06, F35
services/search/app/schemas.py	00	06
services/search/app/service.py	00	06
services/search/pytest.ini	00	none
services/search/requirements-dev.txt	00	none
services/search/requirements-embeddings.txt	00	none
services/search/requirements.txt	00	none
services/search/tests/__init__.py	00	none
services/search/tests/features/test_f20.py	06	none
services/search/tests/features/test_f21.py	06	none
services/search/tests/features/test_f22.py	06	none
services/search/tests/features/test_f23.py	06	none
services/search/tests/features/test_f24.py	06	none
services/search/tests/features/test_f25.py	06	none
services/search/tests/features/test_f28.py	12	none
services/search/tests/features/test_f30.py	06	none
services/search/tests/features/test_f32.py	20	none
services/search/tests/features/test_f33.py	06	none

Path	Creating prompt	Later modifying prompts
services/search/tests/features/test_f34.py	06	none
services/search/tests/features/test_f35.py	F35	none
services/search/tests/test_characterization.py	06	none
services/search/tests/test_fulfilment.py	00	none
services/search/tests/test_identification.py	00	none
services/search/tests/test_lexical.py	00	none
services/search/tests/test_text.py	00	none
services/worker/Dockerfile	24	none
services/worker/app/__init__.py	04	none
services/worker/app/dispatcher.py	15	16, 17, F16, F58, F59
services/worker/app/features/__init__.py	F16	none
services/worker/app/features/f16_photo_quality_feedback.py	F16	none
services/worker/app/features/f58_catalogue_imports.py	F58	none
services/worker/app/internal.py	15	none
services/worker/app/jobs/__init__.py	16	none
services/worker/app/jobs/catalog_publish.py	F59	none
services/worker/app/jobs/export_pdf.py	17	none
services/worker/app/jobs/nameplate.py	16	none
services/worker/app/leases.py	15	none
services/worker/app/main.py	15	none
services/worker/app/static/brand/fonts/NotoNaskhArabic-OFL.txt	17	none
services/worker/app/static/brand/fonts/NotoNaskhArabic.ttf	17	none
services/worker/app/static/brand/fonts/NotoSans-OFL.txt	17	none
services/worker/app/static/brand/fonts/NotoSans.ttf	17	none
services/worker/app/static/brand/fonts/NotoSansArabic-OFL.txt	17	none
services/worker/app/static/brand/fonts/NotoSansArabic.ttf	17	none
services/worker/app/static/brand/logo-original.png	17	none
services/worker/app/storage.py	04	none
services/worker/app/sweeper.py	15	none
services/worker/app/templates/enquiry_summary.html	17	none
services/worker/pyproject.toml	00	none
services/worker/tests/features/test_f14.py	16	none
services/worker/tests/features/test_f16.py	F16	none
services/worker/tests/features/test_f51.py	15	none
services/worker/tests/features/test_f52.py	15	none
services/worker/tests/features/test_f58.py	F58	none
services/worker/tests/features/test_f61.py	20	none
supabase/config.toml	02	none
supabase/migrations/20261011000100_legacy_catalogue.sql	02	none
supabase/migrations/20261011000200_legacy_search.sql	02	none
supabase/migrations/20261011000300_application.sql	03	none
supabase/migrations/20261011000400_rls.sql	03	none
supabase/migrations/20261011000500_storage.sql	04	none
tests/acceptance/test_f02.py	19	none
tests/acceptance/test_f49.py	14	none
tests/acceptance/test_f59.py	F59	none
tests/acceptance/test_f62.py	21	none
tests/acceptance/test_f64.py	23	none
tests/acceptance/test_f66.py	25	none

Path	Creating prompt	Later modifying prompts
tests/acceptance/test_f67.py	23	none
tests/acceptance/test_f68.py	F68	none
tests/acceptance/test_f69.py	F69	none
tests/e2e/hosted.spec.ts	27	none
tests/e2e/security.spec.ts	22	none
tests/evals/cases.json	21	none
tests/evals/test_scenarios.py	21	none
tests/foundations/test_00.py	00	none
tests/foundations/test_01.py	01	none
tests/foundations/test_02.py	02	none
tests/foundations/test_03.py	03	none
tests/foundations/test_04.py	04	none
tests/foundations/test_05.py	05	none
tests/foundations/test_06.py	06	none
tests/foundations/test_07.py	07	none
tests/foundations/test_08.py	08	none
tests/foundations/test_09.py	09	none
tests/foundations/test_10.py	10	none
tests/foundations/test_11.py	11	none
tests/foundations/test_12.py	12	none
tests/foundations/test_13.py	13	none
tests/foundations/test_14.py	14	none
tests/foundations/test_15.py	15	none
tests/foundations/test_16.py	16	none
tests/foundations/test_17.py	17	none
tests/foundations/test_18.py	18	none
tests/foundations/test_19.py	19	none
tests/foundations/test_20.py	20	none
tests/foundations/test_21.py	21	none
tests/foundations/test_22.py	22	none
tests/foundations/test_23.py	23	none
tests/foundations/test_24.py	24	none
tests/foundations/test_25.py	25	none
tests/foundations/test_26.py	26	none
tests/foundations/test_27.py	27	none
tests/integration/conftest.py	03	none
tests/integration/test_budget_races.py	22	none
tests/integration/test_rls.py	22	none
tests/integration/test_security_recovery.py	22	none
tools/apply_catalogue_schema.py	02	none
tools/catalog/Dockerfile	00	none
tools/catalog/__init__.py	00	none
tools/catalog/catalog/__init__.py	00	none
tools/catalog/catalog/__main__.py	00	F59
tools/catalog/catalog/blueprint.py	00	none
tools/catalog/catalog/dataset.py	00	none
tools/catalog/catalog/embedder.py	00	06
tools/catalog/catalog/generator.py	00	none
tools/catalog/catalog/loader.py	00	none
tools/catalog/catalog/validator.py	00	none
tools/catalog/catalog/versioning.py	F59	none
tools/catalog/pytest.ini	00	none
tools/catalog/requirements-embeddings.txt	00	none
tools/catalog/tests/__init__.py	00	none
tools/catalog/tests/test_catalog.py	00	none
tools/check_readiness.py	23	none
tools/check_release_eligibility.py	00	none
tools/checkpoints_setup.py	03	none
tools/ci_checks.py	25	none
tools/credit_allowance.py	23	none
tools/demo/build_scenarios.py	05	none
tools/demo/catalogue_evidence/manifest.json	05	none
tools/demo/nameplates/manifest.json	05	none
tools/demo/render_assets.py	05	none
tools/demo/scenarios.json	05	none
tools/demo/scenarios/incompatible_number.json	05	none
tools/demo/scenarios/local_	05	none

Path	Creating prompt	Later modifying prompts
shortage.json		
tools/demo/scenarios/missing_serial.json	05	none
tools/demo/scenarios/mixed_photo.json	05	none
tools/evals/__init__.py	21	none
tools/evals/run.py	21	none
tools/hosted_acceptance.py	27	none
tools/run_checks.py	00	none
tools/schema_receipt.py	03	none
tools/verify_env.py	00	none
uv.lock	00	none

3. Exact dependency pins

Direct release dates were retrieved from official PyPI/npm registries on2026-10-11. Cutoff2026-09-26. Registry receipts include source URL,integrity and dependency requirements. Resolved references are not a claim of full application tests or a vulnerability-free system. Recheck security advisories before internet exposure; a required emergency patch newer than the cutoff needs an explicitly recorded exception.

Package	Exact version	Release date	Installation owner	Registry evidence
fastapi	0.135.1	2026-03-01	01	https://pypi.org/pypi/fastapi/0.135.1/json
uvicorn	0.41.0	2026-02-16	01	https://pypi.org/pypi/uvicorn/0.41.0/json
pydantic	2.12.5	2025-11-26	01	https://pypi.org/pypi/pydantic/2.12.5/json
pydantic-settings	2.13.1	2026-02-19	01	https://pypi.org/pypi/pydantic-settings/2.13.1/json
httpx	0.28.1	2024-12-06	01	https://pypi.org/pypi/httpx/0.28.1/json
asyncpg	0.31.0	2025-11-24	01	https://pypi.org/pypi/asyncpg/0.31.0/json
psycopg	3.3.3	2026-02-18	01	https://pypi.org/pypi/psycopg/3.3.3/json
psycopg-pool	3.3.0	2025-12-01	01	https://pypi.org/pypi/psycopg-pool/3.3.0/json
langgraph	1.1.3	2026-03-18	09	https://pypi.org/pypi/langgraph/1.1.3/json
langgraph-checkpoint-postgres	3.0.4	2026-01-31	09	https://pypi.org/pypi/langgraph-checkpoint-postgres/3.0.4/json
redis	7.3.0	2026-03-06	01	https://pypi.org/pypi/redis/7.3.0/json
openai	3.19.2	2026-09-24	01	https://pypi.org/pypi/openai/3.19.2/json
structlog	25.5.0	2025-10-27	01	https://pypi.org/pypi/structlog/25.5.0/json
tenacity	9.1.2	2025-04-02	01	https://pypi.org/pypi/tenacity/9.1.2/json
PyJWT	2.12.1	2026-03-13	01	https://pypi.org/pypi/PyJWT/2.12.1/json
cryptography	46.0.5	2026-02-10	01	https://pypi.org/pypi/cryptography/46.0.5/json
sentence-transformers	5.2.3	2026-02-17	06	https://pypi.org/pypi/sentence-transformers/5.2.3/json
transformers	4.57.6	2026-01-16	06	https://pypi.org/pypi/transformers/4.57.6/json
torch	2.10.0+cpu (base release2.10.0)	2026-01-21	06	https://pypi.org/pypi/torch/2.10.0/json
numpy	2.4.2	2026-01-31	06	https://pypi.org/pypi/numpy/2.4.2/json
Jinja2	3.1.6	2025-03-05	17	https://pypi.org/pypi/Jinja2/3.1.6/json
Pillow	12.1.1	2026-02-11	05	https://pypi.org/pypi/Pillow/12.1.1/json
playwright	1.58.0	2026-01-30	17	https://pypi.org/pypi/playwright/1.58.0/json
openpyxl	3.1.5	2024-06-28	F58	https://pypi.org/pypi/openpyxl/3.1.5/json
PyYAML	6.0.3	2025-09-25	00	https://pypi.org/pypi/PyYAML/6.0.3/json
pytest	9.0.2	2025-12-06	00	https://pypi.org/pypi/pytest/9.0.2/json
pytest-asyncio	1.3.0	2025-11-10	00	https://pypi.org/pypi/

Package	Exact version	Release date	Installation owner	Registry evidence
				pytest-asyncio/1.3.0/json
ruff	0.15.6	2026-03-12	00	https://pypi.org/pypi/ruff/0.15.6/json
mypy	1.19.1	2025-12-15	00	https://pypi.org/pypi/mypy/1.19.1/json
uv	0.10.9	2026-03-06	00	https://pypi.org/pypi/uv/0.10.9/json
langfuse	3.14.5	2026-02-23	23	https://pypi.org/pypi/langfuse/3.14.5/json
python-docx	1.2.0	2025-06-16	F68	https://pypi.org/pypi/python-docx/1.2.0/json
pypdf	6.8.0	2026-03-09	17	https://pypi.org/pypi/pypdf/6.8.0/json
python-multipart	0.0.22	2026-01-25	01	https://pypi.org/pypi/python-multipart/0.0.22/json
next	16.2.12	2026-07-25	18	https://registry.npmjs.org/next
react	19.2.6	2026-05-06	18	https://registry.npmjs.org/react
react-dom	19.2.6	2026-05-06	18	https://registry.npmjs.org/react-dom
typescript	5.9.3	2025-09-30	18	https://registry.npmjs.org/typescript
tailwindcss	4.1.18	2025-12-11	18	https://registry.npmjs.org/tailwindcss
@tailwindcss/postcss	4.1.18	2025-12-11	18	https://registry.npmjs.org/%40tailwindcss%2Fpostcss
postcss	8.5.6	2025-06-16	18	https://registry.npmjs.org/postcss
@supabase/supabase-js	2.99.1	2026-03-11	18	https://registry.npmjs.org/%40supabase%2Fsupabase-js
@supabase/ssr	0.8.0	2025-11-26	18	https://registry.npmjs.org/%40supabase%2Fssr
next-intl	4.8.3	2026-02-16	18	https://registry.npmjs.org/next-intl
zod	4.6.5	2026-09-13	18	https://registry.npmjs.org/zod
vitest	4.1.11	2026-08-18	18	https://registry.npmjs.org/vitest
@playwright/test	1.58.2	2026-02-06	18	https://registry.npmjs.org/%40playwright%2Ftest
eslint	9.39.1	2025-11-03	18	https://registry.npmjs.org/eslint
eslint-config-next	16.2.12	2026-07-25	18	https://registry.npmjs.org/eslint-config-next
@testing-library/react	16.3.0	2025-04-02	18	https://registry.npmjs.org/%40testing-library%2Freact
@testing-library/dom	10.4.1	2025-07-27	18	https://registry.npmjs.org/%40testing-library%2Fdom
@testing-library/jest-dom	6.9.1	2025-10-01	18	https://registry.npmjs.org/%40testing-library%2Fjest-dom
@testing-library/user-event	14.6.1	2025-01-21	18	https://registry.npmjs.org/%40testing-library%2Fuser-event
jsdom	27.4.0	2025-12-26	18	https://registry.npmjs.org/jsdom
@vitejs/plugin-react	5.1.4	2026-02-10	18	https://registry.npmjs.org/%40vitejs%2Fplugin-react
@types/react	19.2.14	2026-02-11	18	https://

Package	Exact version	Release date	Installation owner	Registry evidence
				registry.npmjs.org/%40types%2Freact
@types/react-dom	19.2.3	2025-11-12	18	https://registry.npmjs.org/%40types%2Freact-dom
@types/node	22.19.15	2026-03-06	18	https://registry.npmjs.org/%40types%2Fnode
clsx	2.1.1	2024-04-23	18	https://registry.npmjs.org/clsx
tailwind-merge	3.5.0	2026-02-18	18	https://registry.npmjs.org/tailwind-merge
class-variance-authority	0.7.1	2024-11-26	18	https://registry.npmjs.org/class-variance-authority
lucide-react	0.577.0	2026-03-04	18	https://registry.npmjs.org/lucide-react
radix-ui	1.4.3	2025-08-13	18	https://registry.npmjs.org/radix-ui
shadcn	4.20.1	2026-09-02	18	https://registry.npmjs.org/shadcn
pnpm	10.30.3	2026-02-26	18	https://registry.npmjs.org/pnpm
supabase	2.75.0	2026-02-02	02	https://registry.npmjs.org/supabase

Infrastructure and GitHub action SHAs are in spec/infrastructure-receipts.json. Python base3.12.12 and Node22.22.0 must use immutable image manifests recorded by prompt24. BGE-M3 revision is immutable; model weights are downloaded only in the search image. Managed Supabase PostgreSQL/extension versions are collected from the actual project by schema receipt rather than falsely fixed by this kit.

4. Exact data model and policies

The following SQL is authoritative for all48 application-owned tables,columns,constraints,indexes,triggers,grants and RLS policies. It preserves legacy names and extends them. auth.users,storage.buckets and storage.objects are Supabase-managed. The four LangGraph checkpoint tables have exact extracted DDL below; prompt03 runs pinned library setup in autocommit mode and then applies explicit checkpoint RLS/grants. Application routes never create those library tables ad hoc.

Gateway identity is transaction-local app.actor_id,set only after JWT verification. Private policy helper functions run as their migration owner with fixed search_path and PUBLIC execution revoked; untrusted browser roles have no application-table grants. Internal worker/orchestrator roles have system access to the tables they own; application handlers still fence revision/session/lease. Runtime credentials are not database owners and have no BYPASSRLS.

JSON fields use schemas in spec/contracts.py: extraction=IntakeResult,revision snapshot=ApprovedSnapshot or preapproval EnquiryState,candidate reasons=array of strings,plan allocations=array of Allocation,stock snapshot=array of StockEvidence,draft prose=ReplySlots,verified_facts=VerifiedFacts,verification failures=array of VerificationFailure. Import payload uses legacy CatalogDataset entity records. Job payload fields are checked against kind before dispatch. Audit payload is bounded/redacted. No JSON blob is permission to invent critical fields.

001_initial_schema.sql

```
-- PartsIQ initial schema.
-- Executed automatically by the postgres container on first initialisation (empty data volume).
-- Design principle: the database is the single source of truth for part compatibility.
-- AI components may rank and describe candidates, but never decide whether a part fits a machine.

BEGIN;

CREATE EXTENSION IF NOT EXISTS vector;
CREATE EXTENSION IF NOT EXISTS pg_trgm;
CREATE EXTENSION IF NOT EXISTS btree_gist;

-- ------------------------------------------------------------------------
-- Shared trigger functions
-- ------------------------------------------------------------------------

CREATE FUNCTION set_updated_at() RETURNS trigger
LANGUAGE plpgsql AS $$
BEGIN
    NEW.updated_at := now();
```

```
    RETURN NEW;
END;
$$;

-- ------------------------------------------------------------------------------
-- Equipment catalogue
-- ------------------------------------------------------------------------------

CREATE TABLE equipment_families (
    id          SMALLSERIAL PRIMARY KEY,
    code        TEXT NOT NULL UNIQUE CHECK (code ~ '^[A-Z0-9_]{2,32}$'),
    name_en     TEXT NOT NULL,
    name_ar     TEXT NOT NULL
);

CREATE TABLE equipment_models (
    id            SERIAL PRIMARY KEY,
    family_id     SMALLINT NOT NULL REFERENCES equipment_families (id),
    brand         TEXT NOT NULL,
    model_code    TEXT NOT NULL,
    display_name  TEXT NOT NULL,
    rated_output  TEXT,
    created_at    TIMESTAMPTZ NOT NULL DEFAULT now(),
    UNIQUE (brand, model_code)
);

CREATE INDEX equipment_models_model_code_trgm_idx
    ON equipment_models USING gin (model_code gin_trgm_ops);

-- A configuration is a serial-number range of a model that shares one bill of materials.
-- Overlapping ranges for the same model are rejected so serial lookup is always unambiguous.
CREATE TABLE model_configurations (
    id            SERIAL PRIMARY KEY,
    model_id      INTEGER NOT NULL REFERENCES equipment_models (id) ON DELETE CASCADE,
    config_code   TEXT NOT NULL,
    description   TEXT NOT NULL,
    serial_start  BIGINT NOT NULL CHECK (serial_start >= 0),
    serial_end    BIGINT NOT NULL,
    CHECK (serial_start <= serial_end),
    UNIQUE (model_id, config_code),
    EXCLUDE USING gist (
        model_id WITH =,
        int8range(serial_start, serial_end, '[]') WITH &&
    )
);

-- ------------------------------------------------------------------------------
-- Parts
-- ------------------------------------------------------------------------------

CREATE TABLE parts (
    id               SERIAL PRIMARY KEY,
    part_number      TEXT NOT NULL UNIQUE,
    name_en          TEXT NOT NULL,
    name_ar          TEXT NOT NULL,
    description_en   TEXT NOT NULL DEFAULT '',
    category         TEXT NOT NULL,
    unit_of_measure  TEXT NOT NULL DEFAULT 'EA',
    is_active        BOOLEAN NOT NULL DEFAULT TRUE,
    created_at       TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at       TIMESTAMPTZ NOT NULL DEFAULT now(),
    search_document  TSVECTOR GENERATED ALWAYS AS (
        to_tsvector(
            'simple'::regconfig,
            part_number || ' ' || name_en || ' ' || name_ar || ' ' || description_en
        )
    ) STORED
);

CREATE INDEX parts_search_document_idx ON parts USING gin (search_document);
CREATE INDEX parts_part_number_trgm_idx ON parts USING gin (part_number gin_trgm_ops);
CREATE INDEX parts_category_idx ON parts (category);

CREATE TRIGGER parts_set_updated_at
    BEFORE UPDATE ON parts
    FOR EACH ROW EXECUTE FUNCTION set_updated_at();

-- Informal names used by customers and staff, e.g. "black filter" or Arabic colloquial terms.
CREATE TABLE part_aliases (
    id        SERIAL PRIMARY KEY,
    part_id   INTEGER NOT NULL REFERENCES parts (id) ON DELETE CASCADE,
    alias     TEXT NOT NULL,
    language  TEXT NOT NULL CHECK (language IN ('en', 'ar')),
    UNIQUE (part_id, alias)
);

CREATE INDEX part_aliases_alias_trgm_idx ON part_aliases USING gin (alias gin_trgm_ops);

-- Embeddings are stored separately so the embedding model can be replaced without altering parts.
-- Dimension 1024 matches multilingual models such as BGE-M3, which support Arabic and English.
CREATE TABLE part_embeddings (
```

```
    part_id      INTEGER NOT NULL REFERENCES parts (id) ON DELETE CASCADE,
    model_name   TEXT NOT NULL,
    source_text  TEXT NOT NULL,
    embedding    VECTOR(1024) NOT NULL,
    created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
    PRIMARY KEY (part_id, model_name)
);

CREATE INDEX part_embeddings_hnsw_idx
    ON part_embeddings USING hnsw (embedding vector_cosine_ops);

-- The authoritative compatibility matrix. Every row must cite the document it was derived from.
CREATE TABLE part_compatibility (
    part_id            INTEGER NOT NULL REFERENCES parts (id) ON DELETE CASCADE,
    configuration_id   INTEGER NOT NULL REFERENCES model_configurations (id) ON DELETE CASCADE,
    quantity_per_unit  NUMERIC(10, 3) NOT NULL DEFAULT 1 CHECK (quantity_per_unit > 0),
    source_reference   TEXT NOT NULL,
    notes              TEXT,
    PRIMARY KEY (part_id, configuration_id)
);

CREATE INDEX part_compatibility_configuration_idx ON part_compatibility (configuration_id);

-- Each superseded part has exactly one direct successor; chains are resolved by recursive query.
CREATE TABLE part_supersessions (
    old_part_id     INTEGER PRIMARY KEY REFERENCES parts (id) ON DELETE CASCADE,
    new_part_id     INTEGER NOT NULL REFERENCES parts (id) ON DELETE CASCADE,
    effective_date  DATE NOT NULL,
    reason          TEXT,
    CHECK (old_part_id <> new_part_id)
);

CREATE INDEX part_supersessions_new_part_idx ON part_supersessions (new_part_id);

CREATE FUNCTION prevent_supersession_cycle() RETURNS trigger
LANGUAGE plpgsql AS $$
BEGIN
    IF EXISTS (
        WITH RECURSIVE chain (part_id) AS (
            SELECT NEW.new_part_id
            UNION
            SELECT s.new_part_id
            FROM part_supersessions AS s
            JOIN chain AS c ON s.old_part_id = c.part_id
            WHERE s.old_part_id <> NEW.old_part_id
        )
        SELECT 1 FROM chain WHERE part_id = NEW.old_part_id
    ) THEN
        RAISE EXCEPTION 'Supersession % -> % would create a cycle', NEW.old_part_id, NEW.new_part_id
            USING ERRCODE = 'check_violation';
    END IF;
    RETURN NEW;
END;
$$;

CREATE TRIGGER part_supersessions_no_cycles
    BEFORE INSERT OR UPDATE ON part_supersessions
    FOR EACH ROW EXECUTE FUNCTION prevent_supersession_cycle();

-- ---------------------------------------------------------------------------
-- Inventory
-- ---------------------------------------------------------------------------

CREATE TABLE warehouses (
    id      SMALLSERIAL PRIMARY KEY,
    code    TEXT NOT NULL UNIQUE CHECK (code ~ '^[A-Z0-9_]{2,16}$'),
    name    TEXT NOT NULL,
    region  TEXT NOT NULL
);

CREATE TABLE stock_levels (
    part_id            INTEGER NOT NULL REFERENCES parts (id) ON DELETE CASCADE,
    warehouse_id       SMALLINT NOT NULL REFERENCES warehouses (id) ON DELETE CASCADE,
    quantity_on_hand   INTEGER NOT NULL DEFAULT 0 CHECK (quantity_on_hand >= 0),
    quantity_reserved  INTEGER NOT NULL DEFAULT 0 CHECK (quantity_reserved >= 0),
    updated_at         TIMESTAMPTZ NOT NULL DEFAULT now(),
    PRIMARY KEY (part_id, warehouse_id),
    CHECK (quantity_reserved <= quantity_on_hand)
);

CREATE TRIGGER stock_levels_set_updated_at
    BEFORE UPDATE ON stock_levels
    FOR EACH ROW EXECUTE FUNCTION set_updated_at();

-- ---------------------------------------------------------------------------
-- Enquiry workflow
-- ---------------------------------------------------------------------------

CREATE TYPE enquiry_status AS ENUM (
    'received',
    'processing',
```

```
    'needs_information',
    'awaiting_review',
    'approved',
    'rejected',
    'failed'
);

CREATE TABLE enquiries (
    id                  UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    channel             TEXT NOT NULL CHECK (channel IN ('chat', 'email', 'staff')),
    customer_reference  TEXT,
    raw_text            TEXT NOT NULL,
    detected_language   TEXT CHECK (detected_language IN ('en', 'ar', 'mixed')),
    extracted           JSONB NOT NULL DEFAULT '{}'::jsonb,
    configuration_id    INTEGER REFERENCES model_configurations (id),
    status              enquiry_status NOT NULL DEFAULT 'received',
    failure_reason      TEXT,
    created_at          TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at          TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX enquiries_status_created_idx ON enquiries (status, created_at DESC);

CREATE TRIGGER enquiries_set_updated_at
    BEFORE UPDATE ON enquiries
    FOR EACH ROW EXECUTE FUNCTION set_updated_at();

CREATE TABLE enquiry_attachments (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    enquiry_id   UUID NOT NULL REFERENCES enquiries (id) ON DELETE CASCADE,
    storage_key  TEXT NOT NULL UNIQUE,
    media_type   TEXT NOT NULL CHECK (media_type IN ('image/jpeg', 'image/png', 'image/webp')),
    size_bytes   INTEGER NOT NULL CHECK (size_bytes > 0 AND size_bytes <= 10485760),
    created_at   TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX enquiry_attachments_enquiry_idx ON enquiry_attachments (enquiry_id);

CREATE TABLE enquiry_candidates (
    enquiry_id     UUID NOT NULL REFERENCES enquiries (id) ON DELETE CASCADE,
    rank           SMALLINT NOT NULL CHECK (rank BETWEEN 1 AND 10),
    part_id        INTEGER NOT NULL REFERENCES parts (id),
    score          NUMERIC(6, 5) NOT NULL CHECK (score >= 0 AND score <= 1),
    match_reasons  JSONB NOT NULL DEFAULT '[]'::jsonb,
    PRIMARY KEY (enquiry_id, rank),
    UNIQUE (enquiry_id, part_id)
);

CREATE TABLE review_decisions (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    enquiry_id   UUID NOT NULL REFERENCES enquiries (id) ON DELETE CASCADE,
    reviewer     TEXT NOT NULL,
    decision     TEXT NOT NULL CHECK (decision IN ('approved', 'edited', 'rejected')),
    final_reply  TEXT,
    comment      TEXT,
    decided_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
    CHECK (decision = 'rejected' OR final_reply IS NOT NULL)
);

CREATE INDEX review_decisions_enquiry_idx ON review_decisions (enquiry_id, decided_at DESC);

CREATE TABLE review_approved_parts (
    decision_id  UUID NOT NULL REFERENCES review_decisions (id) ON DELETE CASCADE,
    part_id      INTEGER NOT NULL REFERENCES parts (id),
    quantity     NUMERIC(10, 3) NOT NULL CHECK (quantity > 0),
    PRIMARY KEY (decision_id, part_id)
);

-- ---------------------------------------------------------------------------
-- Append-only audit trail
-- ---------------------------------------------------------------------------

CREATE TABLE audit_log (
    id           BIGSERIAL PRIMARY KEY,
    occurred_at  TIMESTAMPTZ NOT NULL DEFAULT now(),
    actor_type   TEXT NOT NULL CHECK (actor_type IN ('system', 'ai', 'user')),
    actor_id     TEXT NOT NULL,
    action       TEXT NOT NULL,
    entity_type  TEXT NOT NULL,
    entity_id    TEXT NOT NULL,
    request_id   TEXT,
    payload      JSONB NOT NULL DEFAULT '{}'::jsonb
);

CREATE INDEX audit_log_entity_idx ON audit_log (entity_type, entity_id, occurred_at DESC);

CREATE FUNCTION prevent_audit_mutation() RETURNS trigger
LANGUAGE plpgsql AS $$
BEGIN
    RAISE EXCEPTION 'audit_log is append-only' USING ERRCODE = 'insufficient_privilege';
END;
```

```
$$;

CREATE TRIGGER audit_log_no_update_delete
    BEFORE UPDATE OR DELETE ON audit_log
    FOR EACH ROW EXECUTE FUNCTION prevent_audit_mutation();

CREATE TRIGGER audit_log_no_truncate
    BEFORE TRUNCATE ON audit_log
    FOR EACH STATEMENT EXECUTE FUNCTION prevent_audit_mutation();

COMMIT;
```

002_search_and_fulfilment.sql

```
-- Search and fulfilment support.

BEGIN;

-- Canonical part-number key so that "vp of 1042", "VP-OF-1042" and "VPOF1042" all match.
-- Must remain consistent with partsiq_common.text.part_number_key.
ALTER TABLE parts
    ADD COLUMN part_number_key TEXT GENERATED ALWAYS AS (
        upper(regexp_replace(part_number, '[^A-Za-z0-9]', '', 'g'))
    ) STORED;

CREATE INDEX parts_part_number_key_idx ON parts (part_number_key);

-- Planned transfer lead times between warehouses, used to recommend where to fulfil from.
CREATE TABLE warehouse_transfer_routes (
    from_warehouse_id  SMALLINT NOT NULL REFERENCES warehouses (id) ON DELETE CASCADE,
    to_warehouse_id    SMALLINT NOT NULL REFERENCES warehouses (id) ON DELETE CASCADE,
    transfer_days      SMALLINT NOT NULL CHECK (transfer_days BETWEEN 1 AND 60),
    PRIMARY KEY (from_warehouse_id, to_warehouse_id),
    CHECK (from_warehouse_id <> to_warehouse_id)
);

COMMIT;
```

003_application.sql

```
-- Additive application schema. Apply after immutable legacy migrations 001 and 002.
BEGIN;
CREATE SCHEMA IF NOT EXISTS app_private;
REVOKE ALL ON SCHEMA app_private FROM PUBLIC;
CREATE TABLE profiles (
 id uuid PRIMARY KEY REFERENCES auth.users(id),
 display_name text NOT NULL CHECK(length(display_name) BETWEEN 1 AND 120),
 preferred_locale text NOT NULL DEFAULT 'en' CHECK(preferred_locale IN ('en','ar')),
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE TABLE staff_roles (
 user_id uuid NOT NULL REFERENCES profiles(id),
 role text NOT NULL CHECK(role IN ('reviewer','catalogue_admin','admin')),
 granted_by uuid REFERENCES profiles(id),
 granted_at timestamptz NOT NULL DEFAULT now(),
 PRIMARY KEY(user_id,role)
);
CREATE TABLE demo_sessions (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 owner_id uuid NOT NULL REFERENCES profiles(id),
 generation integer NOT NULL DEFAULT 1 CHECK(generation>0),
 expires_at timestamptz NOT NULL,
 revoked_at timestamptz,
 created_at timestamptz NOT NULL DEFAULT now(),
 CHECK(expires_at>created_at)
);
CREATE INDEX demo_sessions_owner_idx ON demo_sessions(owner_id,expires_at DESC);
CREATE TABLE catalogue_versions (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 label text NOT NULL UNIQUE,
 state text NOT NULL CHECK(state IN ('staging','published','retired')),
 content_sha256 text NOT NULL CHECK(content_sha256 ~ '^[a-f0-9]{64}$'),
 created_by uuid REFERENCES profiles(id),
 created_at timestamptz NOT NULL DEFAULT now(),
 published_at timestamptz,
 CHECK((state='staging' AND published_at IS NULL) OR (state IN ('published','retired') AND published_at
IS NOT NULL))
);
CREATE UNIQUE INDEX catalogue_one_published_idx ON catalogue_versions((state)) WHERE state='published';
CREATE TABLE catalogue_records (
 version_id uuid NOT NULL REFERENCES catalogue_versions(id),
 entity_type text NOT NULL CHECK(entity_type IN ('equipment_families','equipment_models','model_configur
ations','parts','part_aliases','part_compatibility','part_supersessions','warehouses','warehouse_transfe
r_routes')),
 entity_key text NOT NULL,
 payload jsonb NOT NULL CHECK(jsonb_typeof(payload)='object'),
 PRIMARY KEY(version_id,entity_type,entity_key)
);
CREATE TABLE catalogue_evidence (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 version_id uuid NOT NULL REFERENCES catalogue_versions(id),
```

```
 source_reference text NOT NULL CHECK(length(trim(source_reference))>0),
 storage_key text NOT NULL,
 page_number integer NOT NULL CHECK(page_number>0),
 sha256 text NOT NULL CHECK(sha256 ~ '^[a-f0-9]{64}$'),
 synthetic boolean NOT NULL DEFAULT true,
 UNIQUE(version_id,source_reference)
);
ALTER TABLE part_compatibility ADD CONSTRAINT compatibility_nonempty_source
CHECK(length(trim(source_reference))>0);
ALTER TABLE part_embeddings ADD COLUMN model_revision text;
ALTER TABLE part_embeddings ADD COLUMN source_sha256 text;
ALTER TABLE enquiries ADD COLUMN owner_id uuid REFERENCES profiles(id);
ALTER TABLE enquiries ADD COLUMN session_id uuid REFERENCES demo_sessions(id);
ALTER TABLE enquiries ADD COLUMN session_generation integer;
ALTER TABLE enquiries ADD COLUMN revision integer NOT NULL DEFAULT 1 CHECK(revision>0);
ALTER TABLE enquiries ADD COLUMN catalogue_version_id uuid REFERENCES catalogue_versions(id);
ALTER TABLE enquiries ADD COLUMN destination_warehouse_id smallint REFERENCES warehouses(id);
ALTER TABLE enquiries ADD COLUMN assigned_to uuid REFERENCES profiles(id);
ALTER TABLE enquiries ADD COLUMN priority text NOT NULL DEFAULT 'normal' CHECK(priority IN
('normal','urgent'));
ALTER TABLE enquiries ADD COLUMN due_at timestamptz;
ALTER TABLE enquiries ADD COLUMN idempotency_key uuid;
ALTER TABLE enquiries ADD CONSTRAINT enquiry_demo_pair CHECK((session_id IS NULL AND session_generation
IS NULL) OR (session_id IS NOT NULL AND session_generation>0));
CREATE UNIQUE INDEX enquiries_owner_idempotency_idx ON enquiries(owner_id,idempotency_key) WHERE
idempotency_key IS NOT NULL;
CREATE INDEX enquiries_owner_created_idx ON enquiries(owner_id,created_at DESC,id);
CREATE INDEX enquiries_assigned_status_idx ON enquiries(assigned_to,status,created_at DESC,id);
CREATE INDEX enquiries_demo_idx ON enquiries(session_id,session_generation);
CREATE TABLE enquiry_items (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 ordinal smallint NOT NULL CHECK(ordinal BETWEEN 1 AND 20),
 requested_text text NOT NULL CHECK(length(trim(requested_text))>0),
 quantity numeric(10,3) NOT NULL CHECK(quantity>0 AND quantity<=100000),
 model_text text,
 serial_text text,
 configuration_id integer REFERENCES model_configurations(id),
 selected_part_id integer REFERENCES parts(id),
 status text NOT NULL DEFAULT 'unresolved' CHECK(status IN
('unresolved','needs_information','ready','unavailable')),
 revision integer NOT NULL CHECK(revision>0),
 UNIQUE(enquiry_id,ordinal),
 UNIQUE(enquiry_id,id)
);
ALTER TABLE enquiry_candidates ADD COLUMN item_id uuid;
ALTER TABLE enquiry_candidates ADD COLUMN revision integer NOT NULL DEFAULT 1;
ALTER TABLE enquiry_candidates ADD COLUMN catalogue_version_id uuid REFERENCES catalogue_versions(id);
ALTER TABLE enquiry_candidates ADD COLUMN evidence_id uuid REFERENCES catalogue_evidence(id);
ALTER TABLE enquiry_candidates DROP CONSTRAINT enquiry_candidates_pkey;
ALTER TABLE enquiry_candidates DROP CONSTRAINT enquiry_candidates_enquiry_id_part_id_key;
-- Historical rows receive a legacy item before the new per-item keys are enforced.
INSERT INTO enquiry_items(enquiry_id,ordinal,requested_text,quantity,revision)
 SELECT id,1,CASE WHEN length(trim(raw_text))>0 THEN raw_text ELSE 'Historical enquiry' END,1,revision
FROM enquiries;
UPDATE enquiry_candidates c SET item_id=i.id FROM enquiry_items i WHERE i.enquiry_id=c.enquiry_id AND
i.ordinal=1;
ALTER TABLE enquiry_candidates ALTER COLUMN item_id SET NOT NULL;
ALTER TABLE enquiry_candidates ADD CONSTRAINT candidate_item_fk FOREIGN KEY(enquiry_id,item_id)
REFERENCES enquiry_items(enquiry_id,id);
ALTER TABLE enquiry_candidates ADD PRIMARY KEY(item_id,revision,rank);
ALTER TABLE enquiry_candidates ADD UNIQUE(item_id,revision,part_id);
ALTER TABLE enquiry_attachments ADD COLUMN status text NOT NULL DEFAULT 'pending' CHECK(status IN
('pending','validated','rejected','deleted'));
ALTER TABLE enquiry_attachments ADD COLUMN sha256 text;
ALTER TABLE enquiry_attachments ADD COLUMN revision integer NOT NULL DEFAULT 1;
CREATE TABLE messages (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL CHECK(revision>0),
 kind text NOT NULL CHECK(kind IN ('customer','clarification','status','recommendation')),
 body text NOT NULL CHECK(length(body) BETWEEN 1 AND 12000),
 author_id uuid REFERENCES profiles(id),
 decision_id uuid REFERENCES review_decisions(id),
 idempotency_key uuid NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(),
 UNIQUE(enquiry_id,idempotency_key),
 CHECK((kind='recommendation' AND decision_id IS NOT NULL) OR (kind<>'recommendation' AND decision_id IS
 NULL))
);
CREATE UNIQUE INDEX messages_one_publication_idx ON messages(enquiry_id,revision) WHERE
kind='recommendation';
CREATE INDEX messages_enquiry_created_idx ON messages(enquiry_id,created_at,id);
CREATE TABLE enquiry_revisions (
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL CHECK(revision>0),
 snapshot jsonb NOT NULL CHECK(jsonb_typeof(snapshot)='object'),
 changed_by uuid REFERENCES profiles(id),
 reason text NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(),
 PRIMARY KEY(enquiry_id,revision)
```

```
);
CREATE TABLE extractions (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 attachment_id uuid REFERENCES enquiry_attachments(id),
 revision integer NOT NULL,
 model_text text,
 serial_text text,
 issues jsonb NOT NULL DEFAULT '[]' CHECK(jsonb_typeof(issues)='array'),
 confirmed_by uuid REFERENCES profiles(id),
 confirmed_at timestamptz,
 created_at timestamptz NOT NULL DEFAULT now(),
 CHECK((confirmed_by IS NULL)=(confirmed_at IS NULL))
);
CREATE INDEX extractions_enquiry_idx ON extractions(enquiry_id,revision);
CREATE TABLE fulfilment_plans (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL,
 item_id uuid NOT NULL,
 requested_quantity numeric(10,3) NOT NULL CHECK(requested_quantity>0),
 allocations jsonb NOT NULL CHECK(jsonb_typeof(allocations)='array'),
 shortage numeric(10,3) NOT NULL CHECK(shortage>=0),
 stock_snapshot jsonb NOT NULL CHECK(jsonb_typeof(stock_snapshot)='array'),
 calculated_at timestamptz NOT NULL DEFAULT now(),
 FOREIGN KEY(enquiry_id,item_id) REFERENCES enquiry_items(enquiry_id,id),
 UNIQUE(item_id,revision)
);
CREATE TABLE drafts (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL,
 locale text NOT NULL CHECK(locale IN ('en','ar')),
 prose jsonb NOT NULL CHECK(jsonb_typeof(prose)='object'),
 verified_facts jsonb NOT NULL CHECK(jsonb_typeof(verified_facts)='object'),
 rendered_text text NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(),
 UNIQUE(enquiry_id,revision)
);
CREATE TABLE verification_results (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
 enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL,
 draft_id uuid NOT NULL REFERENCES drafts(id),
 passed boolean NOT NULL,
 failures jsonb NOT NULL CHECK(jsonb_typeof(failures)='array'),
 facts_sha256 text NOT NULL CHECK(facts_sha256 ~ '^[a-f0-9]{64}$'),
 created_at timestamptz NOT NULL DEFAULT now(),
 UNIQUE(enquiry_id,revision)
);
ALTER TABLE review_decisions ADD COLUMN revision integer NOT NULL DEFAULT 1;
ALTER TABLE review_decisions ADD COLUMN reviewer_id uuid REFERENCES profiles(id);
ALTER TABLE review_decisions ADD COLUMN verification_id uuid REFERENCES verification_results(id);
ALTER TABLE review_decisions ADD COLUMN idempotency_key uuid;
CREATE UNIQUE INDEX review_decisions_idempotency_idx ON review_decisions(enquiry_id,idempotency_key)
WHERE idempotency_key IS NOT NULL;
ALTER TABLE review_approved_parts ADD COLUMN item_id uuid REFERENCES enquiry_items(id);
CREATE TABLE internal_notes (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 author_id uuid NOT NULL REFERENCES profiles(id), body text NOT NULL CHECK(length(trim(body)) BETWEEN 1
AND 4000),
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX internal_notes_enquiry_idx ON internal_notes(enquiry_id,created_at,id);
CREATE TABLE assignment_events (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 assigned_to uuid REFERENCES profiles(id), assigned_by uuid NOT NULL REFERENCES profiles(id),
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE TABLE saved_machines (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), owner_id uuid NOT NULL REFERENCES profiles(id),
 session_id uuid REFERENCES demo_sessions(id), session_generation integer,
 label text NOT NULL CHECK(length(trim(label)) BETWEEN 1 AND 120),
 model_id integer NOT NULL REFERENCES equipment_models(id), serial_text text NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(), archived_at timestamptz,
 CHECK((session_id IS NULL AND session_generation IS NULL) OR (session_id IS NOT NULL AND
session_generation>0))
);
CREATE INDEX saved_machines_owner_idx ON saved_machines(owner_id,created_at DESC);
CREATE TABLE missing_part_requests (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 item_id uuid NOT NULL, status text NOT NULL DEFAULT 'open' CHECK(status IN
('open','investigating','resolved','unavailable')),
 staff_note text NOT NULL DEFAULT '', resolved_by uuid REFERENCES profiles(id),
 created_at timestamptz NOT NULL DEFAULT now(), updated_at timestamptz NOT NULL DEFAULT now(),
 FOREIGN KEY(enquiry_id,item_id) REFERENCES enquiry_items(enquiry_id,id), UNIQUE(item_id)
);
CREATE TABLE jobs (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid REFERENCES enquiries(id),
 revision integer, session_generation integer,
 kind text NOT NULL CHECK(kind IN
```

```
('nameplate','export_pdf','catalog_import','catalog_publish','storage_cleanup')),
 status text NOT NULL DEFAULT 'pending' CHECK(status IN
('pending','running','succeeded','failed','cancelled')),
 payload jsonb NOT NULL CHECK(jsonb_typeof(payload)='object'),
 result jsonb, dedupe_key text NOT NULL UNIQUE,
 attempts integer NOT NULL DEFAULT 0 CHECK(attempts>=0), max_attempts integer NOT NULL DEFAULT 3
CHECK(max_attempts BETWEEN 1 AND 5),
 available_at timestamptz NOT NULL DEFAULT now(), lease_token uuid, lease_expires_at timestamptz,
 last_error_code text, created_at timestamptz NOT NULL DEFAULT now(), updated_at timestamptz NOT NULL
DEFAULT now(),
 CHECK((status='running' AND lease_token IS NOT NULL AND lease_expires_at IS NOT NULL) OR
status<>'running')
);
CREATE INDEX jobs_claim_idx ON jobs(status,available_at,lease_expires_at);
CREATE TABLE workflow_events (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
revision integer NOT NULL,
 kind text NOT NULL CHECK(kind IN ('start','message','ocr_completed','approved','retry')),
 payload jsonb NOT NULL DEFAULT '{}', dedupe_key text NOT NULL UNIQUE,
 processed_at timestamptz, lease_token uuid, lease_expires_at timestamptz,
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX workflow_events_pending_idx ON workflow_events(created_at) WHERE processed_at IS NULL;
CREATE TABLE exports (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL, decision_id uuid NOT NULL REFERENCES review_decisions(id),
 job_id uuid NOT NULL REFERENCES jobs(id), storage_key text UNIQUE, sha256 text,
 status text NOT NULL CHECK(status IN ('pending','ready','failed','deleted')),
 created_at timestamptz NOT NULL DEFAULT now(), UNIQUE(enquiry_id,revision)
);
CREATE TABLE import_batches (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), created_by uuid NOT NULL REFERENCES profiles(id),
 version_id uuid REFERENCES catalogue_versions(id), job_id uuid REFERENCES jobs(id),
 storage_key text NOT NULL, sha256 text NOT NULL, status text NOT NULL CHECK(status IN
('uploaded','validating','invalid','valid','published')),
 row_count integer NOT NULL DEFAULT 0 CHECK(row_count BETWEEN 0 AND 50000),
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE TABLE import_rows (
 batch_id uuid NOT NULL REFERENCES import_batches(id), row_number integer NOT NULL CHECK(row_number>0),
 entity_type text NOT NULL, payload jsonb NOT NULL, errors jsonb NOT NULL DEFAULT '[]',
 PRIMARY KEY(batch_id,row_number)
);
CREATE TABLE ai_budget_accounts (
 id text PRIMARY KEY, cap_usd numeric(14,8) NOT NULL CHECK(cap_usd>=0),
 reserved_usd numeric(14,8) NOT NULL DEFAULT 0 CHECK(reserved_usd>=0),
 consumed_usd numeric(14,8) NOT NULL DEFAULT 0 CHECK(consumed_usd>=0),
 expires_at timestamptz NOT NULL, enabled boolean NOT NULL DEFAULT false,
 CHECK(reserved_usd+consumed_usd<=cap_usd)
);
CREATE TABLE ai_usage (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid REFERENCES enquiries(id),
 request_key uuid NOT NULL UNIQUE, budget_account_id text NOT NULL REFERENCES ai_budget_accounts(id),
 model text NOT NULL, stage text NOT NULL, status text NOT NULL CHECK(status IN
('reserved','completed','uncertain','released')),
 reserved_usd numeric(14,8) NOT NULL CHECK(reserved_usd>=0), actual_usd numeric(14,8)
CHECK(actual_usd>=0),
 input_tokens integer CHECK(input_tokens>=0), output_tokens integer CHECK(output_tokens>=0),
 duration_ms integer CHECK(duration_ms>=0), provider_request_id text,
 created_at timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX ai_usage_created_idx ON ai_usage(created_at,stage);
CREATE TABLE eval_cases (
 id text PRIMARY KEY, scenario text NOT NULL, input jsonb NOT NULL, expected jsonb NOT NULL,
 catalogue_version_id uuid NOT NULL REFERENCES catalogue_versions(id),
 language text NOT NULL CHECK(language IN ('en','ar','mixed')), approved_by uuid REFERENCES profiles(id)
);
CREATE TABLE eval_runs (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), git_sha text NOT NULL, model text NOT NULL,
 dataset_sha256 text NOT NULL, started_at timestamptz NOT NULL DEFAULT now(), completed_at timestamptz,
 status text NOT NULL CHECK(status IN ('running','passed','failed'))
);
CREATE TABLE eval_results (
 run_id uuid NOT NULL REFERENCES eval_runs(id), case_id text NOT NULL REFERENCES eval_cases(id),
 actual jsonb NOT NULL, passed boolean NOT NULL, duration_ms integer NOT NULL CHECK(duration_ms>=0),
 PRIMARY KEY(run_id,case_id)
);
CREATE TABLE correction_records (
 id uuid PRIMARY KEY DEFAULT gen_random_uuid(), enquiry_id uuid NOT NULL REFERENCES enquiries(id),
 revision integer NOT NULL, corrected_by uuid NOT NULL REFERENCES profiles(id),
 before_value jsonb NOT NULL, after_value jsonb NOT NULL, reason text NOT NULL,
 eval_case_id text REFERENCES eval_cases(id), created_at timestamptz NOT NULL DEFAULT now()
);
CREATE TABLE enquiry_summaries (
 enquiry_id uuid NOT NULL REFERENCES enquiries(id), revision integer NOT NULL,
 summary text NOT NULL, source_message_ids uuid[] NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(), PRIMARY KEY(enquiry_id,revision)
);
CREATE TABLE retention_policies (
 scope text PRIMARY KEY CHECK(scope IN ('demo','client')),
 enquiry_days integer CHECK(enquiry_days BETWEEN 1 AND 3650),
```

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 upload_days integer CHECK(upload_days BETWEEN 1 AND 3650),
 audit_days integer CHECK(audit_days BETWEEN 1 AND 3650),
 configured_by uuid REFERENCES profiles(id), configured_at timestamptz NOT NULL DEFAULT now()
);
INSERT INTO retention_policies(scope,enquiry_days,upload_days,audit_days) VALUES
('demo',1,1,NULL),('client',NULL,NULL,NULL);
-- NULL client values disable live-client onboarding, not indefinite silent consent.
ALTER TABLE audit_log ADD COLUMN enquiry_id uuid REFERENCES enquiries(id);
ALTER TABLE audit_log ADD COLUMN revision integer;
CREATE INDEX audit_log_enquiry_idx ON audit_log(enquiry_id,occurred_at DESC);
CREATE FUNCTION app_private.lock_supersessions() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN PERFORM pg_advisory_xact_lock(710069); RETURN NEW; END; $$;
CREATE TRIGGER aa_supersessions_lock BEFORE INSERT OR UPDATE ON part_supersessions FOR EACH ROW EXECUTE
FUNCTION app_private.lock_supersessions();
CREATE FUNCTION app_private.immutable_record() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN RAISE EXCEPTION 'immutable historical record' USING ERRCODE='42501'; END; $$;
CREATE TRIGGER catalogue_records_immutable BEFORE UPDATE OR DELETE ON catalogue_records FOR EACH ROW
EXECUTE FUNCTION app_private.immutable_record();
CREATE TRIGGER revisions_immutable BEFORE UPDATE OR DELETE ON enquiry_revisions FOR EACH ROW EXECUTE
FUNCTION app_private.immutable_record();
CREATE TABLE idempotency_records (
 actor_id uuid NOT NULL REFERENCES profiles(id),
 operation text NOT NULL,
 key uuid NOT NULL,
 request_sha256 text NOT NULL CHECK(request_sha256 ~ '^[a-f0-9]{64}$'),
 response jsonb,
 created_at timestamptz NOT NULL DEFAULT now(),
 PRIMARY KEY(actor_id,operation,key)
);
COMMIT;
```

004_rls.sql

```
-- Runtime roles are NOLOGIN groups. The operator creates distinct LOGIN roles and grants one group per
service.
BEGIN;
CREATE ROLE partsiq_gateway NOLOGIN;
CREATE ROLE partsiq_search NOLOGIN;
CREATE ROLE partsiq_orchestrator NOLOGIN;
CREATE ROLE partsiq_worker NOLOGIN;
CREATE ROLE partsiq_catalogue NOLOGIN;
GRANT USAGE ON SCHEMA public,app_private TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker,partsiq_catalogue;
CREATE FUNCTION app_private.actor() RETURNS uuid LANGUAGE sql STABLE AS $$
 SELECT nullif(current_setting('app.actor_id',true),'')::uuid;
$$;
CREATE FUNCTION app_private.has_role(wanted text) RETURNS boolean LANGUAGE sql STABLE SECURITY DEFINER
SET search_path=public,pg_temp AS $$
 SELECT EXISTS(SELECT 1 FROM staff_roles WHERE user_id=app_private.actor() AND role=wanted);
$$;
CREATE FUNCTION app_private.demo_active(sid uuid, gen integer, uid uuid) RETURNS boolean LANGUAGE sql
STABLE SECURITY DEFINER SET search_path=public,pg_temp AS $$
 SELECT EXISTS(SELECT 1 FROM demo_sessions WHERE id=sid AND owner_id=uid AND generation=gen AND
revoked_at IS NULL AND expires_at>now());
$$;
CREATE FUNCTION app_private.can_read_enquiry(eid uuid) RETURNS boolean LANGUAGE sql STABLE SECURITY
DEFINER SET search_path=public,pg_temp AS $$
 SELECT EXISTS(SELECT 1 FROM enquiries e WHERE e.id=eid AND (
 (e.owner_id=app_private.actor() AND (e.session_id IS NULL OR
app_private.demo_active(e.session_id,e.session_generation,e.owner_id)))
 OR (e.session_id IS NULL AND (app_private.has_role('admin') OR (e.assigned_to=app_private.actor() AND
app_private.has_role('reviewer'))))));
$$;
CREATE FUNCTION app_private.can_review(eid uuid) RETURNS boolean LANGUAGE sql STABLE SECURITY DEFINER
SET search_path=public,pg_temp AS $$
 SELECT EXISTS(SELECT 1 FROM enquiries e WHERE e.id=eid AND (
 (e.session_id IS NOT NULL AND e.owner_id=app_private.actor() AND
app_private.demo_active(e.session_id,e.session_generation,e.owner_id))
 OR (e.session_id IS NULL AND (app_private.has_role('admin') OR (e.assigned_to=app_private.actor() AND
app_private.has_role('reviewer'))))));
$$;
REVOKE ALL ON ALL FUNCTIONS IN SCHEMA app_private FROM PUBLIC;
GRANT EXECUTE ON ALL FUNCTIONS IN SCHEMA app_private TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker,partsiq_catalogue;
ALTER TABLE equipment_families ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON equipment_families FROM anon,authenticated;
GRANT SELECT ON equipment_families TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY equipment_families_read ON equipment_families FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON equipment_families TO partsiq_catalogue;
CREATE POLICY equipment_families_publisher ON equipment_families TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE equipment_models ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON equipment_models FROM anon,authenticated;
GRANT SELECT ON equipment_models TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY equipment_models_read ON equipment_models FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON equipment_models TO partsiq_catalogue;
CREATE POLICY equipment_models_publisher ON equipment_models TO partsiq_catalogue USING (true) WITH
CHECK (true);
```

```
ALTER TABLE model_configurations ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON model_configurations FROM anon,authenticated;
GRANT SELECT ON model_configurations TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY model_configurations_read ON model_configurations FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON model_configurations TO partsiq_catalogue;
CREATE POLICY model_configurations_publisher ON model_configurations TO partsiq_catalogue USING (true)
WITH CHECK (true);
ALTER TABLE parts ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON parts FROM anon,authenticated;
GRANT SELECT ON parts TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY parts_read ON parts FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON parts TO partsiq_catalogue;
CREATE POLICY parts_publisher ON parts TO partsiq_catalogue USING (true) WITH CHECK (true);
ALTER TABLE part_aliases ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON part_aliases FROM anon,authenticated;
GRANT SELECT ON part_aliases TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY part_aliases_read ON part_aliases FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON part_aliases TO partsiq_catalogue;
CREATE POLICY part_aliases_publisher ON part_aliases TO partsiq_catalogue USING (true) WITH CHECK
(true);
ALTER TABLE part_embeddings ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON part_embeddings FROM anon,authenticated;
GRANT SELECT ON part_embeddings TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY part_embeddings_read ON part_embeddings FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON part_embeddings TO partsiq_catalogue;
CREATE POLICY part_embeddings_publisher ON part_embeddings TO partsiq_catalogue USING (true) WITH CHECK
(true);
GRANT INSERT,UPDATE ON part_embeddings TO partsiq_search;
CREATE POLICY part_embeddings_encoder ON part_embeddings FOR INSERT TO partsiq_search WITH CHECK(true);
CREATE POLICY part_embeddings_encoder_update ON part_embeddings FOR UPDATE TO partsiq_search USING(true)
 WITH CHECK(true);
ALTER TABLE part_compatibility ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON part_compatibility FROM anon,authenticated;
GRANT SELECT ON part_compatibility TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY part_compatibility_read ON part_compatibility FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON part_compatibility TO partsiq_catalogue;
CREATE POLICY part_compatibility_publisher ON part_compatibility TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE part_supersessions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON part_supersessions FROM anon,authenticated;
GRANT SELECT ON part_supersessions TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY part_supersessions_read ON part_supersessions FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON part_supersessions TO partsiq_catalogue;
CREATE POLICY part_supersessions_publisher ON part_supersessions TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE warehouses ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON warehouses FROM anon,authenticated;
GRANT SELECT ON warehouses TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY warehouses_read ON warehouses FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON warehouses TO partsiq_catalogue;
CREATE POLICY warehouses_publisher ON warehouses TO partsiq_catalogue USING (true) WITH CHECK (true);
ALTER TABLE stock_levels ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON stock_levels FROM anon,authenticated;
GRANT SELECT ON stock_levels TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY stock_levels_read ON stock_levels FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON stock_levels TO partsiq_catalogue;
CREATE POLICY stock_levels_publisher ON stock_levels TO partsiq_catalogue USING (true) WITH CHECK
(true);
ALTER TABLE enquiries ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON enquiries FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON enquiries TO partsiq_gateway,partsiq_orchestrator,partsiq_worker;
CREATE POLICY enquiries_gateway ON enquiries TO partsiq_gateway USING(app_private.can_read_enquiry(id))
WITH CHECK(owner_id=app_private.actor() OR app_private.can_review(id));
CREATE POLICY enquiries_internal ON enquiries TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
ALTER TABLE enquiry_attachments ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON enquiry_attachments FROM anon,authenticated;
GRANT SELECT ON enquiry_attachments TO partsiq_gateway;
CREATE POLICY enquiry_attachments_gateway_read ON enquiry_attachments FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT INSERT ON enquiry_attachments TO partsiq_gateway;
CREATE POLICY enquiry_attachments_gateway_insert ON enquiry_attachments FOR INSERT TO partsiq_gateway
WITH CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT UPDATE ON enquiry_attachments TO partsiq_gateway;
CREATE POLICY enquiry_attachments_gateway_update ON enquiry_attachments FOR UPDATE TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id)) WITH CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON enquiry_attachments TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY enquiry_attachments_internal ON enquiry_attachments TO partsiq_orchestrator,partsiq_worker
 USING(true) WITH CHECK(true);
ALTER TABLE enquiry_candidates ENABLE ROW LEVEL SECURITY;
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REVOKE ALL ON enquiry_candidates FROM anon,authenticated;
GRANT SELECT ON enquiry_candidates TO partsiq_gateway;
CREATE POLICY enquiry_candidates_gateway_read ON enquiry_candidates FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON enquiry_candidates TO partsiq_orchestrator;
GRANT SELECT ON enquiry_candidates TO partsiq_worker;
CREATE POLICY enquiry_candidates_internal ON enquiry_candidates TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE review_decisions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON review_decisions FROM anon,authenticated;
GRANT SELECT ON review_decisions TO partsiq_gateway;
CREATE POLICY review_decisions_gateway_read ON review_decisions FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON review_decisions TO partsiq_orchestrator;
GRANT SELECT ON review_decisions TO partsiq_worker;
CREATE POLICY review_decisions_internal ON review_decisions TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE review_approved_parts ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON review_approved_parts FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON review_approved_parts TO partsiq_orchestrator;
GRANT SELECT ON review_approved_parts TO partsiq_worker;
CREATE POLICY review_approved_parts_internal ON review_approved_parts TO
partsiq_orchestrator,partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE audit_log ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON audit_log FROM anon,authenticated;
GRANT INSERT ON audit_log TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker,partsiq_catalogue;
CREATE POLICY audit_log_append ON audit_log FOR INSERT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker,partsiq_catalogue WITH CHECK(true);
GRANT SELECT ON audit_log TO partsiq_gateway;
CREATE POLICY audit_log_read ON audit_log FOR SELECT TO partsiq_gateway
USING(app_private.has_role('admin') OR app_private.can_review(enquiry_id));
ALTER TABLE warehouse_transfer_routes ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON warehouse_transfer_routes FROM anon,authenticated;
GRANT SELECT ON warehouse_transfer_routes TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY warehouse_transfer_routes_read ON warehouse_transfer_routes FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON warehouse_transfer_routes TO partsiq_catalogue;
CREATE POLICY warehouse_transfer_routes_publisher ON warehouse_transfer_routes TO partsiq_catalogue
USING (true) WITH CHECK (true);
ALTER TABLE profiles ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON profiles FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON profiles TO partsiq_gateway;
CREATE POLICY profiles_owner ON profiles TO partsiq_gateway USING(id=app_private.actor()) WITH
CHECK(id=app_private.actor());
GRANT SELECT,UPDATE ON profiles TO partsiq_worker;
CREATE POLICY profiles_cleanup ON profiles TO partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE staff_roles ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON staff_roles FROM anon,authenticated;
GRANT SELECT,INSERT,DELETE ON staff_roles TO partsiq_gateway;
CREATE POLICY staff_roles_read ON staff_roles FOR SELECT TO partsiq_gateway
USING(user_id=app_private.actor() OR app_private.has_role('admin'));
CREATE POLICY staff_roles_insert ON staff_roles FOR INSERT TO partsiq_gateway WITH
CHECK(app_private.has_role('admin'));
CREATE POLICY staff_roles_delete ON staff_roles FOR DELETE TO partsiq_gateway
USING(app_private.has_role('admin'));
ALTER TABLE demo_sessions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON demo_sessions FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON demo_sessions TO partsiq_gateway;
CREATE POLICY demo_sessions_owner ON demo_sessions TO partsiq_gateway
USING(owner_id=app_private.actor()) WITH CHECK(owner_id=app_private.actor());
GRANT SELECT,UPDATE ON demo_sessions TO partsiq_worker;
CREATE POLICY demo_sessions_cleanup ON demo_sessions TO partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE catalogue_versions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON catalogue_versions FROM anon,authenticated;
GRANT SELECT ON catalogue_versions TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY catalogue_versions_read ON catalogue_versions FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON catalogue_versions TO partsiq_catalogue;
CREATE POLICY catalogue_versions_publisher ON catalogue_versions TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE catalogue_records ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON catalogue_records FROM anon,authenticated;
GRANT SELECT ON catalogue_records TO partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY catalogue_records_read ON catalogue_records FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON catalogue_records TO partsiq_catalogue;
CREATE POLICY catalogue_records_publisher ON catalogue_records TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE catalogue_evidence ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON catalogue_evidence FROM anon,authenticated;
GRANT SELECT ON catalogue_evidence TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker;
CREATE POLICY catalogue_evidence_read ON catalogue_evidence FOR SELECT TO
partsiq_gateway,partsiq_search,partsiq_orchestrator,partsiq_worker USING (true);
GRANT SELECT,INSERT,UPDATE ON catalogue_evidence TO partsiq_catalogue;
CREATE POLICY catalogue_evidence_publisher ON catalogue_evidence TO partsiq_catalogue USING (true) WITH
CHECK (true);
ALTER TABLE enquiry_items ENABLE ROW LEVEL SECURITY;
```

```
REVOKE ALL ON enquiry_items FROM anon,authenticated;
GRANT SELECT ON enquiry_items TO partsiq_gateway;
CREATE POLICY enquiry_items_gateway_read ON enquiry_items FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT INSERT ON enquiry_items TO partsiq_gateway;
CREATE POLICY enquiry_items_gateway_insert ON enquiry_items FOR INSERT TO partsiq_gateway WITH
CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT UPDATE ON enquiry_items TO partsiq_gateway;
CREATE POLICY enquiry_items_gateway_update ON enquiry_items FOR UPDATE TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id)) WITH CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON enquiry_items TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY enquiry_items_internal ON enquiry_items TO partsiq_orchestrator,partsiq_worker USING(true)
 WITH CHECK(true);
ALTER TABLE messages ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON messages FROM anon,authenticated;
GRANT SELECT ON messages TO partsiq_gateway;
CREATE POLICY messages_gateway_read ON messages FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT INSERT ON messages TO partsiq_gateway;
CREATE POLICY messages_gateway_insert ON messages FOR INSERT TO partsiq_gateway WITH
CHECK(app_private.can_read_enquiry(enquiry_id) AND kind='customer');
GRANT SELECT,INSERT,UPDATE ON messages TO partsiq_orchestrator;
GRANT SELECT ON messages TO partsiq_worker;
CREATE POLICY messages_internal ON messages TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
ALTER TABLE enquiry_revisions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON enquiry_revisions FROM anon,authenticated;
GRANT SELECT ON enquiry_revisions TO partsiq_gateway;
CREATE POLICY enquiry_revisions_gateway_read ON enquiry_revisions FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON enquiry_revisions TO partsiq_orchestrator;
GRANT SELECT ON enquiry_revisions TO partsiq_worker;
CREATE POLICY enquiry_revisions_internal ON enquiry_revisions TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE extractions ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON extractions FROM anon,authenticated;
GRANT SELECT ON extractions TO partsiq_gateway;
CREATE POLICY extractions_gateway_read ON extractions FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON extractions TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY extractions_internal ON extractions TO partsiq_orchestrator,partsiq_worker USING(true)
WITH CHECK(true);
ALTER TABLE fulfilment_plans ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON fulfilment_plans FROM anon,authenticated;
GRANT SELECT ON fulfilment_plans TO partsiq_gateway;
CREATE POLICY fulfilment_plans_gateway_read ON fulfilment_plans FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON fulfilment_plans TO partsiq_orchestrator;
GRANT SELECT ON fulfilment_plans TO partsiq_worker;
CREATE POLICY fulfilment_plans_internal ON fulfilment_plans TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE drafts ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON drafts FROM anon,authenticated;
GRANT SELECT ON drafts TO partsiq_gateway;
CREATE POLICY drafts_gateway_read ON drafts FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON drafts TO partsiq_orchestrator;
GRANT SELECT ON drafts TO partsiq_worker;
CREATE POLICY drafts_internal ON drafts TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
ALTER TABLE verification_results ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON verification_results FROM anon,authenticated;
GRANT SELECT ON verification_results TO partsiq_gateway;
CREATE POLICY verification_results_gateway_read ON verification_results FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON verification_results TO partsiq_orchestrator;
GRANT SELECT ON verification_results TO partsiq_worker;
CREATE POLICY verification_results_internal ON verification_results TO
partsiq_orchestrator,partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE internal_notes ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON internal_notes FROM anon,authenticated;
GRANT SELECT ON internal_notes TO partsiq_gateway;
CREATE POLICY internal_notes_gateway_read ON internal_notes FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT INSERT ON internal_notes TO partsiq_gateway;
CREATE POLICY internal_notes_gateway_insert ON internal_notes FOR INSERT TO partsiq_gateway WITH
CHECK(app_private.can_review(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON internal_notes TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY internal_notes_internal ON internal_notes TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE assignment_events ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON assignment_events FROM anon,authenticated;
GRANT SELECT ON assignment_events TO partsiq_gateway;
CREATE POLICY assignment_events_gateway_read ON assignment_events FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT INSERT ON assignment_events TO partsiq_gateway;
CREATE POLICY assignment_events_gateway_insert ON assignment_events FOR INSERT TO partsiq_gateway WITH
CHECK(app_private.can_review(enquiry_id) AND app_private.has_role('admin'));
GRANT SELECT,INSERT,UPDATE ON assignment_events TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY assignment_events_internal ON assignment_events TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
```

```
ALTER TABLE saved_machines ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON saved_machines FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON saved_machines TO partsiq_gateway;
CREATE POLICY saved_machines_owner ON saved_machines TO partsiq_gateway
USING(owner_id=app_private.actor() AND (session_id IS NULL OR
app_private.demo_active(session_id,session_generation,owner_id))) WITH
CHECK(owner_id=app_private.actor() AND (session_id IS NULL OR
app_private.demo_active(session_id,session_generation,owner_id)));
GRANT SELECT,UPDATE ON saved_machines TO partsiq_worker;
CREATE POLICY saved_machines_cleanup ON saved_machines TO partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE missing_part_requests ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON missing_part_requests FROM anon,authenticated;
GRANT SELECT ON missing_part_requests TO partsiq_gateway;
CREATE POLICY missing_part_requests_gateway_read ON missing_part_requests FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT INSERT ON missing_part_requests TO partsiq_gateway;
CREATE POLICY missing_part_requests_gateway_insert ON missing_part_requests FOR INSERT TO
partsiq_gateway WITH CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT UPDATE ON missing_part_requests TO partsiq_gateway;
CREATE POLICY missing_part_requests_gateway_update ON missing_part_requests FOR UPDATE TO
partsiq_gateway USING(app_private.can_read_enquiry(enquiry_id)) WITH
CHECK(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON missing_part_requests TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY missing_part_requests_internal ON missing_part_requests TO
partsiq_orchestrator,partsiq_worker USING(true) WITH CHECK(true);
ALTER TABLE jobs ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON jobs FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON jobs TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY jobs_internal ON jobs TO partsiq_orchestrator,partsiq_worker USING(true) WITH CHECK(true);
GRANT INSERT,SELECT ON jobs TO partsiq_gateway;
CREATE POLICY jobs_gateway ON jobs TO partsiq_gateway USING(app_private.can_read_enquiry(enquiry_id))
WITH CHECK(app_private.can_read_enquiry(enquiry_id));
ALTER TABLE workflow_events ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON workflow_events FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON workflow_events TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY workflow_events_internal ON workflow_events TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
GRANT INSERT,SELECT ON workflow_events TO partsiq_gateway;
CREATE POLICY workflow_events_gateway ON workflow_events TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id)) WITH CHECK(app_private.can_read_enquiry(enquiry_id));
ALTER TABLE exports ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON exports FROM anon,authenticated;
GRANT SELECT ON exports TO partsiq_gateway;
CREATE POLICY exports_gateway_read ON exports FOR SELECT TO partsiq_gateway
USING(app_private.can_read_enquiry(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON exports TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY exports_internal ON exports TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
ALTER TABLE import_batches ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON import_batches FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON import_batches TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY import_batches_internal ON import_batches TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
GRANT SELECT,INSERT,UPDATE ON import_batches TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY import_batches_admin ON import_batches TO partsiq_gateway
USING(app_private.has_role('admin') OR app_private.has_role('catalogue_admin')) WITH
CHECK(app_private.has_role('admin') OR app_private.has_role('catalogue_admin'));
CREATE POLICY import_batches_catalogue ON import_batches TO partsiq_catalogue USING(true) WITH
CHECK(true);
ALTER TABLE import_rows ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON import_rows FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON import_rows TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY import_rows_internal ON import_rows TO partsiq_orchestrator,partsiq_worker USING(true)
WITH CHECK(true);
GRANT SELECT,INSERT,UPDATE ON import_rows TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY import_rows_admin ON import_rows TO partsiq_gateway USING(app_private.has_role('admin') OR
 app_private.has_role('catalogue_admin')) WITH CHECK(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin'));
CREATE POLICY import_rows_catalogue ON import_rows TO partsiq_catalogue USING(true) WITH CHECK(true);
ALTER TABLE ai_budget_accounts ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON ai_budget_accounts FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON ai_budget_accounts TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY ai_budget_accounts_internal ON ai_budget_accounts TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE ai_usage ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON ai_usage FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON ai_usage TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY ai_usage_internal ON ai_usage TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
ALTER TABLE eval_cases ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON eval_cases FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON eval_cases TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY eval_cases_internal ON eval_cases TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
GRANT SELECT,INSERT,UPDATE ON eval_cases TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY eval_cases_admin ON eval_cases TO partsiq_gateway USING(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin')) WITH CHECK(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin'));
CREATE POLICY eval_cases_catalogue ON eval_cases TO partsiq_catalogue USING(true) WITH CHECK(true);
ALTER TABLE eval_runs ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON eval_runs FROM anon,authenticated;
```

```
GRANT SELECT,INSERT,UPDATE ON eval_runs TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY eval_runs_internal ON eval_runs TO partsiq_orchestrator,partsiq_worker USING(true) WITH
CHECK(true);
GRANT SELECT,INSERT,UPDATE ON eval_runs TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY eval_runs_admin ON eval_runs TO partsiq_gateway USING(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin')) WITH CHECK(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin'));
CREATE POLICY eval_runs_catalogue ON eval_runs TO partsiq_catalogue USING(true) WITH CHECK(true);
ALTER TABLE eval_results ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON eval_results FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON eval_results TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY eval_results_internal ON eval_results TO partsiq_orchestrator,partsiq_worker USING(true)
WITH CHECK(true);
GRANT SELECT,INSERT,UPDATE ON eval_results TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY eval_results_admin ON eval_results TO partsiq_gateway USING(app_private.has_role('admin')
OR app_private.has_role('catalogue_admin')) WITH CHECK(app_private.has_role('admin') OR
app_private.has_role('catalogue_admin'));
CREATE POLICY eval_results_catalogue ON eval_results TO partsiq_catalogue USING(true) WITH CHECK(true);
ALTER TABLE correction_records ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON correction_records FROM anon,authenticated;
GRANT SELECT ON correction_records TO partsiq_gateway;
CREATE POLICY correction_records_gateway_read ON correction_records FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT INSERT ON correction_records TO partsiq_gateway;
CREATE POLICY correction_records_gateway_insert ON correction_records FOR INSERT TO partsiq_gateway WITH
 CHECK(app_private.can_review(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON correction_records TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY correction_records_internal ON correction_records TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE enquiry_summaries ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON enquiry_summaries FROM anon,authenticated;
GRANT SELECT ON enquiry_summaries TO partsiq_gateway;
CREATE POLICY enquiry_summaries_gateway_read ON enquiry_summaries FOR SELECT TO partsiq_gateway
USING(app_private.can_review(enquiry_id));
GRANT SELECT,INSERT,UPDATE ON enquiry_summaries TO partsiq_orchestrator;
GRANT SELECT ON enquiry_summaries TO partsiq_worker;
CREATE POLICY enquiry_summaries_internal ON enquiry_summaries TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
ALTER TABLE retention_policies ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON retention_policies FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON retention_policies TO partsiq_orchestrator,partsiq_worker;
CREATE POLICY retention_policies_internal ON retention_policies TO partsiq_orchestrator,partsiq_worker
USING(true) WITH CHECK(true);
GRANT SELECT,INSERT,UPDATE ON retention_policies TO partsiq_gateway,partsiq_catalogue;
CREATE POLICY retention_policies_admin ON retention_policies TO partsiq_gateway
USING(app_private.has_role('admin') OR app_private.has_role('catalogue_admin')) WITH
CHECK(app_private.has_role('admin') OR app_private.has_role('catalogue_admin'));
CREATE POLICY retention_policies_catalogue ON retention_policies TO partsiq_catalogue USING(true) WITH
CHECK(true);
GRANT USAGE ON ALL SEQUENCES IN SCHEMA public TO
partsiq_gateway,partsiq_orchestrator,partsiq_worker,partsiq_catalogue;
CREATE FUNCTION app_private.validate_publication() RETURNS trigger LANGUAGE plpgsql SECURITY DEFINER SET
 search_path=public,pg_temp AS $$
BEGIN
 IF NEW.kind='recommendation' AND NOT EXISTS(
 SELECT 1 FROM review_decisions d JOIN enquiries e ON e.id=d.enquiry_id
 JOIN verification_results v ON v.id=d.verification_id
 WHERE d.id=NEW.decision_id AND d.enquiry_id=NEW.enquiry_id AND d.revision=NEW.revision
 AND e.revision=NEW.revision AND d.decision='approved' AND v.passed AND v.revision=NEW.revision
 ) THEN RAISE EXCEPTION 'approval required for current revision' USING ERRCODE='23514'; END IF;
 RETURN NEW;
END; $$;
REVOKE ALL ON FUNCTION app_private.validate_publication() FROM PUBLIC;
CREATE TRIGGER messages_publication_guard BEFORE INSERT ON messages FOR EACH ROW EXECUTE FUNCTION
app_private.validate_publication();
GRANT USAGE ON SEQUENCE audit_log_id_seq TO partsiq_search;
ALTER TABLE idempotency_records ENABLE ROW LEVEL SECURITY;
REVOKE ALL ON idempotency_records FROM anon,authenticated;
GRANT SELECT,INSERT,UPDATE ON idempotency_records TO
partsiq_gateway,partsiq_orchestrator,partsiq_worker,partsiq_catalogue;
CREATE POLICY idempotency_actor ON idempotency_records TO partsiq_gateway
USING(actor_id=app_private.actor()) WITH CHECK(actor_id=app_private.actor());
CREATE POLICY idempotency_internal ON idempotency_records TO
partsiq_orchestrator,partsiq_worker,partsiq_catalogue USING(true) WITH CHECK(true);
GRANT SELECT ON ai_usage TO partsiq_gateway;
CREATE POLICY ai_usage_metrics ON ai_usage FOR SELECT TO partsiq_gateway
USING(app_private.has_role('admin') OR app_private.can_review(enquiry_id));
COMMIT;
```

005_storage.sql

```
BEGIN;
INSERT INTO storage.buckets(id,name,public,file_size_limit,allowed_mime_types) VALUES
 ('nameplates','nameplates',false,10485760,ARRAY['image/jpeg','image/png','image/webp']),
 ('catalogue-evidence','catalogue-evidence',false,20971520,ARRAY['application/pdf']),
 ('imports','imports',false,10485760,ARRAY['text/csv','application/vnd.openxmlformats-officedocument.spr
eadsheetml.sheet']),
 ('exports','exports',false,20971520,ARRAY['application/pdf']);
CREATE FUNCTION public.partsiq_storage_allowed(object_name text, target_bucket text, write_access
boolean) RETURNS boolean
LANGUAGE plpgsql STABLE SECURITY DEFINER SET search_path=public,pg_temp AS $$
```

```
DECLARE actor uuid:=auth.uid(); allowed boolean:=false;
BEGIN
 IF actor IS NULL THEN RETURN false; END IF;
 IF target_bucket='nameplates' THEN
 SELECT EXISTS(SELECT 1 FROM enquiry_attachments a JOIN enquiries e ON e.id=a.enquiry_id
 WHERE a.storage_key=object_name AND e.owner_id=actor
 AND (e.session_id IS NULL OR app_private.demo_active(e.session_id,e.session_generation,actor))
 AND ((write_access AND a.status='pending') OR (NOT write_access AND a.status='validated'))) INTO
allowed;
 END IF;
 RETURN allowed;
END; $$;
REVOKE ALL ON FUNCTION public.partsiq_storage_allowed(text,text,boolean) FROM PUBLIC;
GRANT EXECUTE ON FUNCTION public.partsiq_storage_allowed(text,text,boolean) TO authenticated;
CREATE POLICY nameplate_insert ON storage.objects FOR INSERT TO authenticated
 WITH CHECK(bucket_id='nameplates' AND public.partsiq_storage_allowed(name,bucket_id,true));
CREATE POLICY nameplate_read ON storage.objects FOR SELECT TO authenticated
 USING(bucket_id='nameplates' AND public.partsiq_storage_allowed(name,bucket_id,false));
-- Other buckets have no end-user policy. Worker signs authorized downloads only after gateway scope
validation.
COMMIT;
```

006_checkpoint_library.sql

```
-- Exact library migration statements from langgraph-checkpoint-postgres3.0.4.
-- Applied by pinned saver setup in dedicated checkpoints schema.
CREATE SCHEMA IF NOT EXISTS checkpoints;
SET search_path=checkpoints,public,extensions;
CREATE TABLE IF NOT EXISTS checkpoint_migrations (
    v INTEGER PRIMARY KEY
);
CREATE TABLE IF NOT EXISTS checkpoints (
    thread_id TEXT NOT NULL,
    checkpoint_ns TEXT NOT NULL DEFAULT '',
    checkpoint_id TEXT NOT NULL,
    parent_checkpoint_id TEXT,
    type TEXT,
    checkpoint JSONB NOT NULL,
    metadata JSONB NOT NULL DEFAULT '{}',
    PRIMARY KEY (thread_id, checkpoint_ns, checkpoint_id)
);
CREATE TABLE IF NOT EXISTS checkpoint_blobs (
    thread_id TEXT NOT NULL,
    checkpoint_ns TEXT NOT NULL DEFAULT '',
    channel TEXT NOT NULL,
    version TEXT NOT NULL,
    type TEXT NOT NULL,
    blob BYTEA,
    PRIMARY KEY (thread_id, checkpoint_ns, channel, version)
);
CREATE TABLE IF NOT EXISTS checkpoint_writes (
    thread_id TEXT NOT NULL,
    checkpoint_ns TEXT NOT NULL DEFAULT '',
    checkpoint_id TEXT NOT NULL,
    task_id TEXT NOT NULL,
    idx INTEGER NOT NULL,
    channel TEXT NOT NULL,
    type TEXT,
    blob BYTEA NOT NULL,
    PRIMARY KEY (thread_id, checkpoint_ns, checkpoint_id, task_id, idx)
);
ALTER TABLE checkpoint_blobs ALTER COLUMN blob DROP not null;
SELECT 1;

    CREATE INDEX CONCURRENTLY IF NOT EXISTS checkpoints_thread_id_idx ON checkpoints(thread_id);


    CREATE INDEX CONCURRENTLY IF NOT EXISTS checkpoint_blobs_thread_id_idx ON
checkpoint_blobs(thread_id);


    CREATE INDEX CONCURRENTLY IF NOT EXISTS checkpoint_writes_thread_id_idx ON
checkpoint_writes(thread_id);

ALTER TABLE checkpoint_writes ADD COLUMN IF NOT EXISTS task_path TEXT NOT NULL DEFAULT '';
```

007_checkpoint_permissions.sql

```
-- Run after library saver setup,never within its concurrent-index transaction.
BEGIN;
REVOKE ALL ON SCHEMA checkpoints FROM PUBLIC,anon,authenticated;
GRANT USAGE ON SCHEMA checkpoints TO partsiq_orchestrator;
GRANT SELECT,INSERT,UPDATE,DELETE ON ALL TABLES IN SCHEMA checkpoints TO partsiq_orchestrator;
ALTER TABLE checkpoints.checkpoint_migrations ENABLE ROW LEVEL SECURITY;
ALTER TABLE checkpoints.checkpoints ENABLE ROW LEVEL SECURITY;
ALTER TABLE checkpoints.checkpoint_blobs ENABLE ROW LEVEL SECURITY;
ALTER TABLE checkpoints.checkpoint_writes ENABLE ROW LEVEL SECURITY;
CREATE POLICY checkpoints_migration_read ON checkpoints.checkpoint_migrations FOR SELECT TO
partsiq_orchestrator USING(true);
CREATE POLICY checkpoints_state_runtime ON checkpoints.checkpoints TO partsiq_orchestrator USING(true)
WITH CHECK(true);
```

```
CREATE POLICY checkpoints_blobs_runtime ON checkpoints.checkpoint_blobs TO partsiq_orchestrator
USING(true) WITH CHECK(true);
CREATE POLICY checkpoints_writes_runtime ON checkpoints.checkpoint_writes TO partsiq_orchestrator
USING(true) WITH CHECK(true);
REVOKE INSERT,UPDATE,DELETE ON checkpoints.checkpoint_migrations FROM partsiq_orchestrator;
COMMIT;
```

5. Environment variables

Values marked operator input/secret are deployment inputs,not invented credentials. Disabled optional capabilities do not require their credentials. Startup validation must explain missing names without printing secret values.

Name	Default/input	Used by	Purpose
APP_ENV	local	all	local/test/production; no prod in tests
NEXT_PUBLIC_SUPABASE_URL	operator input	web	dev/prod Auth endpoint
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	operator input	web	publishable key only
NEXT_PUBLIC_API_BASE_URL	operator input	web	HTTPS gateway
SUPABASE_URL	operator input	gateway,worker	project URL
SUPABASE_JWT_ISSUER	operator input	gateway	exact issuer
SUPABASE_JWKS_URL	operator input	gateway	trusted fixed JWKS URL
SUPABASE_STORAGE_KEY	operator secret	worker only	private object signing/cleanup; not database REST
DATABASE_URL	operator secret	each service	separate restricted login
MIGRATION_DATABASE_URL	operator secret	migration only	DDL credential
DATABASE_POOL_MAX	4	each service	bounded connections
ALLOWED_ORIGINS	operator input	gateway	exact Vercel origins
REDIS_URL	redis://redis:6379/0	backend	private dispatch and rate limits
INTERNAL_SERVICE_TOKEN	operator secret	internal services	random32-byte minimum
SEARCH_URL	http://search:8000	gateway,orchestrator	internal search
ORCHESTRATOR_URL	http://orchestrator:8000	gateway	internal workflow
WORKER_URL	http://worker:8000	gateway	authorized signing/import control
OPENAI_API_KEY	operator secret	orchestrator,worker	owner key; never in browser
OPENAI_MODEL_TEXT	gpt-4.1-mini-2025-04-14	orchestrator	pinned structured-output model
OPENAI_MODEL_VISION	gpt-4.1-mini-2025-04-14	worker	pinned vision model
AI_ENABLED	false	AI services	enable only after credit preflight
AI_FREE_CREDIT_USD	0	AI services	owner-confirmed dedicated allowance
AI_CREDIT_EXPIRES_AT	operator input	AI services	required UTC expiry when enabled
AI_PAID_BUDGET_USD	0	AI services	must stay0
AI_DAILY_ALLOWANCE_USD	0	AI services	required allowance within confirmed total
AI_SESSION_ALLOWANCE_USD	0	AI services	required per-session limit
BUDGET_TIMEZONE	Asia/Kolkata	AI services	daily reset timezone
WORKFLOW_MAX_STEPS	12	orchestrator	bounded transition count
TOOL_MAX_CALLS	4	orchestrator	per-stage tool bound
UPLOAD_MAX_BYTES	10485760	gateway,worker	10MiB
UPLOAD_MAX_PIXELS	24000000	worker	decode cap
RATE_LIMIT_PER_MINUTE	20	gateway	per actor plus IP limiter
DEMO_TTL_HOURS	24	gateway,sweeper	session expiry
STOCK_MAX_AGE_SECONDS	3600	search,approval	demo freshness default
NORMAL_REVIEW_HOURS	24	gateway	noncontractual deadline
URGENT_REVIEW_HOURS	2	gateway	noncontractual deadline
EMBEDDING_MODEL_REVISION	5617a9f61b028005a4858fdac845db406aefb181	search only	BAAI/bge-m3 revision
LANGFUSE_ENABLED	false	common telemetry	optional free tracing
LANGFUSE_PUBLIC_KEY	operator input	backend	not browser
LANGFUSE_SECRET_KEY	operator secret	backend	redacted traces only
LANGFUSE_BASE_URL	operator input	backend	selected free cloud region
API_DOMAIN	operator input	Caddy	owned or no-cost DNS name
RELEASE_SHA	operator input	deployment	immutable Git commit
AWS_REGION	ap-south-1	deployment	verify free-plan service availability
AWS_ACCOUNT_ID	operator input	deployment	never invent account ID
AWS_ROLE_ARN	operator input	CI	OIDC deploy role
EC2_INSTANCE_ID	operator input	deployment	SSM target
PRODUCTION_PROJECT_REF	operator input	tests,deploy	test safety denylist

6. Routes and signatures

Common typed shapes are in spec/contracts.py and docs/CONTRACTS.md. Every F implementation prompt defines its own request/result,exact route and component. Foundation08 owns POST /v1/enquiries,POST /v1/enquiries/{id}/messages,GET /v1/enquiries/{id},GET /v1/enquiries/{id}/messages,POST /v1/uploads and POST /v1/uploads/{id}/confirm. Foundation07 owns POST /v1/demo/sessions and POST /v1/demo/sessions/reset. Foundation14 owns POST /internal/approvals and gateway routes POST /v1/staff/enquiries/{id}/approve,/edit,/reject. Foundation17 owns POST /v1/enquiries/{id}/export and authorized download retrieval. Foundation20 consumes GET /v1/staff/enquiries/{id}/review. Search preserves identify,parts,part-detail,warehouses and fulfilment APIs behind /internal. Health GET /health reports process alive; /ready reports required dependency status without secrets.

7. Limits

Text12000chars,20items,upload10MiB,24million decoded pixels,import50000rows,tool4calls/stage,workflow12transitions,job3attempts,lease45seconds,heartbeat10seconds,signing60seconds. These are implementation defaults,not measured throughput claims. Performance tests report median/p95 with concurrency1 and3,warm/cold search separated; no latency guarantee is fabricated for free hardware.

013 docs/DEPLOYMENT.md

Deployment from local build to recruiter demo

1. Local development first

Use Python3.12.12,Node22.22.0,uv0.10.9,pnpm10.30.3 and the pinned Supabase CLI. Execute00 through20 before adding features. Start local Supabase with `pnpm exec supabase start`; record its local ports and keys privately. `pnpm exec supabase db reset` is allowed only against the disposable local instance. Start Compose with `docker compose up --build`. Keep gateway bound to127.0.0.1. Run feature checks and the four scenarios without live provider calls using test doubles. Test doubles must never appear as live AI in recruiter UI.

2. Supabase dev and prod

Create two free projects if account quota permits. Use different database passwords and restricted service credentials. Apply numbered migrations to dev first,then compare schema receipt. Use migration-only elevated access to initialize checkpointer schema. Configure private buckets and Auth anonymous sign-in; real-customer self-registration remains disabled in the product. Invite staff administratively; seed the first admin using a one-time audited migration/admin command. Never let users choose their staff role.

Set dedicated runtime LOGIN roles with one service-group grant each. Store connection secrets outside git. Use direct/session-compatible connections for checkpoint and transactional operations; do not assume transaction pooling supports every driver/prepared-statement pattern. Verify SSL and bounded pool sizes against actual Supabase connection allowance. Gateway injects app.actor_id only inside a transaction.

Supabase Free may pause after inactivity. Before recruiter outreach,open the project,restore if needed,and run hosted readiness. Do not manufacture traffic merely to claim permanent availability. Free tier is not a backup service-level guarantee: keep reproducible synthetic seed data and a tested export/restore procedure.

3. OpenAI free-credit preflight

Keep the owner's OpenAI provider. A valid API key does not expose a guaranteed credit balance. The operator confirms account/project allowance,expiry and model access in the provider console. Prefer a dedicated project/key,disable paid auto-recharge,and ensure unrelated applications cannot consume the same allocation unnoticed.

AI_ENABLED defaults false. Set AI_FREE_CREDIT_USD to an amount no larger than the verified remaining credit after a safety margin,AI_CREDIT_EXPIRES_AT to its actual expiry,and session/day allowances below the total. AI_PAID_BUDGET_USD remains0. The local budget service reserves upper-bound request cost before each attempt. Unknown pricing,expired allowance,provider insufficient_quota or uncertain billing stops new calls. On timeout keep reservation until reconciled; do not retry indefinitely.

The app cannot independently guarantee provider billing status or prevent spending by other keys. No claim of automatically detecting the exact provider balance is permitted. If the account cannot be configured to prevent paid spending,keep live AI disabled; deterministic search/manual entry/review remain available. Do not replace OpenAI or silently switch providers.

Pinned language and vision snapshot: gpt-4.1-mini-2025-04-14. Validate access with a tiny explicitly budgeted probe. Pricing reference checked2026-10-11: input0.40USD/million tokens,output1.60USD/million; cached discounts are not assumed. Image input token accounting must use the documented patch calculation and an upper-bound estimate. No web-search or paid ancillary tools are enabled.

4. AWS Free Plan eligibility and sizing

Before provisioning,record account plan,remaining credits,expiry,region and available instance types. New-account Free Plan availability is conditional and time-limited; it is not permanent free EC2. Do not upgrade to a paid account or provision an excluded service automatically. If existing-account eligibility differs,the operator must verify its actual allowance.

Default region proposal is ap-south-1,subject to eligibility. No fixed free instance is claimed adequate. BGE-M3 plus Chromium and all services can exceed micro-instance RAM. Measure search encoder peak memory,PDF peak memory and total idle usage locally. Select an eligible instance with headroom; keep search process count1 and worker concurrency bounded. If no eligible instance can run the measured stack at zero out-of-pocket cost,use local Compose. Do not hide an architectural reduction or claim full semantic functionality on hardware that cannot run it.

Use encrypted disk,SSM administration,no public SSH,security group ingress80/443 only. Avoid NAT Gateway,load balancer or paid domain purchase. A DNS name under an existing domain or a no-cost DNS

provider is an operator input. Configure HTTPS and exact CORS origin; raw HTTP public demo is not acceptable. Public IPv4,storage,egress and snapshots can consume credits; include them in the account check.

5. Images and secrets

Prompt24 builds images from pinned manifests in spec/images.lock.json and records image digests in release metadata. Build BGE weights into search image using immutable model revision. Store Compose secret environment file root-readable only on the host; never pass secrets as Docker build args. Preserve Caddy data for certificates. Redis is private and recoverable from DB jobs; use bounded persistence/logs. Health probes distinguish lexical readiness from semantic warmup.

6. CI and deployment

Prompt25 creates checks and a disabled production deployment workflow. Prompt26 provisions prerequisites and enables it only after successful preflight. GitHub OIDC role trust uses exact repo and production environment subject; only SSM commands to the named instance and image pull permissions are allowed. GHCR images are addressed by digest or immutable release SHA. Production environment approval is the deployment gate,not approval of AI-generated customer recommendations.

Deployment steps: acquire deploy lock;record current release;verify environment and credit expiry;take permitted database backup/export;apply additive migrations using migration identity;start new images;run health/readiness and four synthetic scenarios;switch release marker only when checks pass. On failure,return to previous compatible images and retain diagnostic logs. Never automatically reverse data migrations or delete audit history.

7. Vercel

Import the built repository with root directory apps/web. Install using frozen lockfile; build with `pnpm build`. Configure production publishable Supabase values and gateway HTTPS URL. Bind previews to dev project/API or disable backend access until configured. Configure no OpenAI,database or internal-service secrets in NEXT_PUBLIC variables. Confirm anonymous Auth,staff login,CORS,upload,Arabic rendering,approval and PDF on hosted deployment.

Vercel Hobby is used for the personal recruiter portfolio subject to its current terms/limits. A commercial client deployment must reassess plan eligibility rather than inherit an unsupported free-plan promise.

8. Readiness before sending recruiters

Restore paused Supabase if needed; verify AWS credits/expiry; confirm AI allowance; run four scenarios; inspect Arabic/mobile screens;verify source/README links;record date and actual latency. Provide a short recorded walkthrough as fallback. Do not present prerecorded results as live model processing.

9. Render alternative only

If the owner later selects Render,replace EC2/Caddy deployment instructions with a documented web/background-worker/Redis topology and new cost analysis. Keep Supabase and API contracts. Do not claim Render's free services can host the full persistent worker plus BGE workload without measuring current limits. This alternative is not activated by the kit.

10. Sources

https://aws.amazon.com/free/free-tier-faqs/

https://supabase.com/docs/guides/platform/free-project-pausing

https://developers.openai.com/api/docs/models/gpt-4.1-mini

https://help.openai.com/en/articles/8264644-setting-up-and-managing-prepaid-api-billing

https://langfuse.com/pricing

https://vercel.com/docs/plans/hobby

014 docs/TEST_CHECKLIST.md

Test checklist

Build-kit checks

`python tools/kit/verify_kit.py`: 69 features,97 prompt files,template headings,dependency order,file ownership,release receipts and schema RLS coverage. `python -m compileall spec/contracts.py tools/kit`: syntax check for shipped specification/tooling. These checks do not certify runtime features.

Per-prompt application checks

Run `python tools/run_checks.py --prompt 00` for bootstrap,then the current ID. The runner dispatches exact files from spec/features.json and spec/prompts.json. Python services execute in separate processes with their own app path; no mixed legacy app import cache. Frontend runs `pnpm --dir apps/web exec vitest run` on named tests,`pnpm --dir apps/web typecheck`,and `pnpm --dir apps/web lint`. Use `uv run --project services/search pytest` for legacy characterization in its isolated working directory. Expected exit0; log actual counts.

Database and permission checks

Use disposable local Supabase,not bare Postgres alone. Apply all migrations; collect schema receipt; compare tables,columns,indexes,FKs,checks and RLS. Insert overlapping serial ranges and supersession cycles and expect rejection. Exercise concurrent cycle insertion under lock. Check unknown identity,customerA/customerB,assigned reviewer/unassigned reviewer/admin and own/foreign demo. Reuse one DB connection across actors to prove transaction-local context isolation. Test private Storage upload/read with real Auth tokens.

Workflow checks

Start interrupted enquiry;restart process;resume. Inject duplicate event,expired lease,old revision and reset generation. Confirm stale results cannot commit. Kill Redis and confirm pending jobs survive; gateway rate limiter fails closed. Provider timeout leaves conservative budget reservation. Two simultaneous approvals create one recommendation; changed stock/catalogue returns409.

Quality checks

Run deterministic labelled suite with no provider calls. Report candidate recall,top-choice correctness,clarification correctness and incompatible-offer count separately. Include denominator,dataset hash and catalogue version. Optional live model eval runs only with dedicated confirmed free credit. Never claim a measured accuracy from written expected outcomes.

UI/PDF checks

360/768/1280 widths,200percent zoom,keyboard focus,screen-reader names,Arabic RTL,mixed serials,long content,empty/error/loading states. Export long bilingual approved enquiry; inspect every page for clipping,overlap and font loss. No internal notes in customer or PDF response.

Hosted checks

After27,run hosted scenario suite over HTTPS against demo only. No destructive production fixtures. Record real environment,commit,latency and result. Free-plan expiry,paused project or unavailable credentials are blocked checks. A build-kit validation pass does not substitute for them.

015 docs/ROLLBACK.md

Rollback

Code

One commit per prompt. Tag pre-ID before work. Inspect the commit hash and diff before reverting. `git revert --no-edit HEAD` is valid only when HEAD is the intended task; otherwise use the verified task hash. Never hard-reset unrelated changes. Rerun affected feature tests and update handover.

Database

Never edit applied migrations. Take an authorized export/backup before production schema changes. Prefer forward repair/additive compatibility migration. Candidate key migration preserves historical rows by creating item records. Its reversal requires proving no multi-item records would be lost; otherwise do not reverse. Restore only into an isolated recovery database first,compare receipts and obtain operator approval before replacing live state. Audit history is not deleted as routine rollback.

Catalogue

Publication is a transaction. Invalid batch never becomes active. To restore a prior catalogue,create a new validated publication from its immutable records,retain history and invalidate pending recommendations. Do not rewrite old version rows or pretend prior approvals used the new version.

Deployment

Retain previous release's image digests and environment schema. On failed healthcheck,switch to previous schema-compatible images. If current migration is not backward compatible,halt and follow the migration-specific recovery plan. Preserve logs and current DB state; do not reset volumes. Free-credit expiry disables live AI and triggers local/recorded-demo fallback,not paid upgrade.

Demo reset

Reset is deliberate expiry of a generation; it is not an audit rollback. Old jobs are cancelled/fenced. Storage cleanup is asynchronous and retryable. The user starts a new scenario rather than resurrecting obsolete processing results.

016 docs/FIELD_GUIDE_MAP.md

Fifteen field-guide habits

Habit	Concrete kit implementation
1 Handover	HANDOVER.md read first and updated after every prompt
2 Decisions	DECISIONS.md records reason,consequence and source approval
3 Explicit comments	Prompt logic asks for callers,assumptions and intent
4 Flow	FLOW.md binds entries to prompt-owned functions
5 Feature/bug traces	69 feature files and completed-example bug template
6 Architecture	Eight questions answered in ARCHITECTURE.md
7 Constraints	Numbered immutable boundaries and approved exceptions
8 Test checklist	Named paths,commands and expected outcomes
9 Rollback	Pre-task tags,compensating migration and release rollback
10 Read diff	Required before marking an implementation complete
11 Ask why	PLAN CHECK describes reasoning before edits
12 Small requests	One implementation prompt and commit at a time
13 Handoff	Five-line session summary
14 Context version	Actual model/version if exposed; otherwise not exposed
15 Mental model	Agent explains change in plain English; recruiter guide rehearses architecture

The supplied guide is preserved in references/AI-Collaboration-Field-Guide.pdf. Its approval guidance does not override the owner's already-granted authorization for this sequence.

017 docs/HANDOVER.md

Current handover

State: build kit generated; application implementation has not executed. Original search/gateway/catalogue code is bundled in references/legacy-backend.zip.

Completed: approved scope,corrected architecture,69 feature traces,foundation/feature prompts,exact schema specification,brand assets and dependency release verification.

Next: execute prompt00 in a new repository/worktree; follow prompts/README.md.

Watch: AWS Free Plan eligibility and actual BGE/Chromium sizing; owner-confirmed OpenAI credit; assigned-only permissions; no fake production-readiness claims.

Evidence: docs/SELF_AUDIT.md and spec validation receipts describe kit checks only. Runtime and hosted acceptance remain implementation work.

018 docs/SESSION_LOG.md

Session log

2026-10-11 - Build-kit generation

1. Work: incorporated owner decisions and generated the complete implementation kit.
2. Decisions: retain OpenAI; paid budget0;assigned-only review;69 grouped features;public synthetic demo.
3. Verification: see SELF_AUDIT.md for actual artifact checks and their limits.
4. Next: coding agent executes00;no application deployment performed during kit creation.
5. Model/version: not exposed by this authoring environment; no version invented.

019 docs/SELF_AUDIT.md

Build-kit self-audit

Audit date:2026-10-11. This report separates document/specification checks from application execution. The ZIP is a build kit; no full PartsIQ app was built or deployed during this task.

Check	Result	Fix applied or verification boundary
Every planned application file has one creator	PASS: ownership manifest structurally checked	Added package markers,asset copies and missing adapters; later modifications explicitly owned
Every direct package has exact pin and named installation prompt	PASS:66 official registry release receipts	Corrected incompatible Supabase SSR/client pairing
Transitive dependency versions are fixed	Resolved lock references included	Python and JS lock resolution is not a target-runtime build or security certification
Every feature maps to prompt and test	PASS:69 feature IDs,69 traces,69 F prompt files	Foundation-covered features use verification prompts,not duplicated domain code
Foundation prompts present	PASS:28,numbered00 through27	Corrected legacy boundaries,workflow and deployment ordering
Prompt template sections	PASS:all97 prompt files	Exact14 template headings retained
FLOW entries name prompt contracts	PASS:generated from prompt contract registry	Preserved legacy signatures;new entries appear in owning prompt
Paths/table/environment names agree	Structural mapping checked;runtime receipt required later	TECH_SPEC includes exact SQL,contracts,environment and ownership inventory
No unfinished prose markers	PASS:automated Markdown scan	Removed deferred-schema/package outline and unfinished instructions
No forward build dependencies	PASS:manifest DAG checked	Customer/catalogue/staff work interleaved before evaluations and deployment
Constraint contradictions	Reviewed and corrected	Clarification exception,search-only encoder,worker events,private Storage,assigned review and credit gates explicit
SQL syntax	PASS:all seven SQL specification files parsed by PostgreSQL parser	This is syntax validation,not live Supabase policy execution
Typed common contracts	PASS:64 Pydantic model schemas generated	Actual service integration remains implementation work
Brand assets	PASS:manifest hashes and image dimensions recorded	Preserved supplied artwork;bundled licensed fonts
Application tests and RLS behaviour	NOT RUN in kit-authoring task	Named tests must execute during coding prompts;do not mark features done yet
AWS/OpenAI/Vercel account preflight	NOT RUN:no account activation requested/performed	Operator verifies free credits,model access,host capacity and actual deployment settings
Full vulnerability audit	NOT CLAIMED	Target runtime security gate required before public exposure
Word guide	Generated from every current document,prompt and feature trace	Render/structural QA recorded in final artifact receipt
ZIP integrity	Verified at packaging	CRC and SHA256 manifest checked;no secrets included

Free-key clarification: application code can cap its own confirmed allowance,but cannot infer a guaranteed provider balance from a key or stop unrelated account usage. Live calls remain disabled until the owner supplies a dedicated confirmed allowance/expiry. No paid fallback is authorized.

020 brand/BRAND.md

PartsIQ brand system

Use supplied falcon/gear/magnifier bilingual mark. Preserve aspect ratio;do not redraw the Arabic text. Primary navy#2B2D7F and gold#C59A45 are standardized tokens;the raster artwork also contains gradients and antialiasing. Gold is decorative/accent,not small text on white. Use navy for primary action,neutral surfaces and semantic state colours from tokens.json. Dark-mode muted/border overrides preserve readable contrast.

Typography: Noto Sans for English,Noto Sans Arabic for Arabic UI,Noto Naskh Arabic for Arabic PDF. Licensed font files and OFL notices are bundled. Use font-display:swap;self-host fonts;never depend on external PDF font fetch. Use bdi for part numbers and serials.

Use full lockup at minimum160CSSpx width in header/landing;icon-only at24/32px when space is constrained. Leave clear space equal to one-quarter icon width. Never stretch,crop critical mark elements,place on a busy photo or imply endorsement by a dealer. Small favicon uses supplied icon crop,not illegible full bilingual lockup.

Exports: favicon ICO16/32/48;PNG16/32/48/96;Apple180;PWA192/512;maskable192/512;social1200x630;profile1080;full logo320/640/1280. Existing supplied pack is preserved under assets. Asset manifest records actual dimensions and SHA256;use it rather than guessing file names. No authentic vector source was supplied,so no SVG is falsely presented as original artwork.

Prompt18 copies every manifest-listed asset and fonts into apps/web/public/brand. PDF worker copies the same font bytes and logo into its image. Brand exports are already bundled;new raster editing is unnecessary.

021 prompts/README.md

Execution order

Execute implementation prompts in this table,not alphabetical F order. After a foundation,run its covered F verification prompts. They introduce no additional runtime dependencies. Every feature has a trace and test mapping in spec/features.json.

Order	Prompt	Task	Dependencies
1	[00](00-bootstrap-and-preserve-legacy.md)	Bootstrap and preserve legacy	none
2	[01](01-shared-contracts-and-clients.md)	Shared contracts and clients	00
3	[02](02-catalogue-migration.md)	Catalogue migration	01
4	[03](03-workflow-schema-and-database-permissions.md)	Workflow schema and database permissions	02
5	[04](04-private-storage.md)	Private storage	03
6	[05](05-synthetic-catalogue-and-demo-fixtures.md)	Synthetic catalogue and demo fixtures	04
7	[06](06-extend-legacy-search.md)	Extend legacy search	05
8	[07](07-authentication-and-demo-sessions.md)	Authentication and demo sessions	06
9	[08](08-enquiry-intake-and-limits.md)	Enquiry intake and limits	07
10	[09](09-deterministic-durable-workflow.md)	Deterministic durable workflow	08
11	[10](10-intake-specialist.md)	Intake specialist	09
12	[11](11-parts-specialist-and-tools.md)	Parts specialist and tools	10
13	[12](12-deterministic-verification.md)	Deterministic verification	11
14	[13](13-reply-specialist-and-renderer.md)	Reply specialist and renderer	12
15	[14](14-approval-and-revision-transactions.md)	Approval and revision transactions	13
16	[15](15-worker-leases-and-sweeper.md)	Worker leases and sweeper	14
17	[16](16-nameplate-vision-job.md)	Nameplate vision job	15
18	[17](17-approved-enquiry-pdf.md)	Approved enquiry PDF	16
19	[18](18-frontend-shell-and-authentication.md)	Frontend shell and authentication	17
20	[19](19-customer-chat-and-demo.md)	Customer chat and demo	18
21	[20](20-staff-workspace-and-timeline.md)	Staff workspace and timeline	19
22	[F16](F16-photo-quality-feedback.md)	Photo quality feedback	16, 19
23	[F17](F17-multi-part-enquiries.md)	Multi-part enquiries	09, 10, 11, 12, 13, 14, 19, 20
24	[F19](F19-enquiry-history.md)	Enquiry history	07, 08, 19
25	[F55](F55-saved-machines.md)	Saved machines	07, 19
26	[F56](F56-repeat-enquiry.md)	Repeat enquiry	09, 19, F19, F55
27	[F59](F59-catalogue-versioning.md)	Catalogue versioning	02, 05, 06
28	[F57](F57-catalogue-browser.md)	Catalogue browser	02, 06, 18, F59
29	[F58](F58-catalogue-imports.md)	Catalogue imports	02, 05, 15, F59, F57
30	[F60](F60-missing-parts-queue.md)	Missing-parts queue	03, 20
31	[F35](F35-stock-freshness.md)	Stock freshness	06, 12, 14, 20
32	[F29](F29-evidence-page-viewer.md)	Evidence page viewer	04, 06, 20
33	[F31](F31-candidate-comparison.md)	Candidate comparison	06, 20
34	[F38](F38-correct-extracted-details.md)	Correct extracted details	09, 14, 20
35	[F42](F42-assignments.md)	Assignments	03, 07, 20
36	[F41](F41-internal-notes.md)	Internal notes	03, 07, 20
37	[F43](F43-overdue-indicators.md)	Overdue indicators	20
38	[F44](F44-conversation-summary.md)	Conversation summary	09, 13, 20

Order	Prompt	Task	Dependencies
39	[F36](F36-staff-inbox.md)	Staff inbox	07, 20, F42, F43
40	[F07](F07-arabic-and-english-interface.md)	Arabic and English interface	18, 19, 20
41	[F08](F08-light-and-dark-themes.md)	Light and dark themes	18
42	[F06](F06-responsive-layouts.md)	Responsive layouts	18, 19, 20, F07, F08
43	[F48](F48-input-isolation.md)	Input isolation	10, 11, 12, 13, 16
44	[21](21-labelled-evaluations.md)	Labelled evaluations	F48
45	[23](23-observability-and-credit-limits.md)	Observability and credit limits	21
46	[F63](F63-metrics-dashboard.md)	Metrics dashboard	21, 23
47	[F65](F65-correction-feedback.md)	Correction feedback	14, 21, 23, F63
48	[22](22-security-and-failure-recovery-gate.md)	Security and failure recovery gate	F65
49	[24](24-production-containers-and-caddy.md)	Production containers and Caddy	22
50	[25](25-ci-and-controlled-deployment.md)	CI and controlled deployment	24
51	[26](26-aws-free-plan-deployment.md)	AWS free-plan deployment	25
52	[27](27-vercel-and-hosted-acceptance.md)	Vercel and hosted acceptance	26
53	[F68](F68-recruiter-documentation.md)	Recruiter documentation	21, 23, 22, 24, 25, 26, 27
54	[F69](F69-demo-walkthrough.md)	Demo walkthrough	27, F68

Dependency diagram

```
flowchart TD
 A[Foundations00 to06] --> B[Foundations07 to20]
 B --> C[Customer features]
 C --> D[Catalogue and staff features]
 D --> E[Language polish and input isolation]
 E --> F[Evaluation21 and telemetry23]
 F --> G[Analytics features and security22]
 G --> H[Containers24 and CI25]
 H --> I[AWS26 and Vercel27]
 I --> J[Recruiter guide F68 and recording F69]
```

Feature coverage

Feature	Implementation owner	Feature prompt	Test
F01 Branded landing	18	[F01](F01-branded-landing.md)	`apps/web/tests/features/f01.test.tsx`
F02 Demo scenarios and assets	19	[F02](F02-demo-scenarios-and-assets.md)	`tests/acceptance/test_f02.py`
F03 Demo role switching	20	[F03](F03-demo-role-switching.md)	`services/gateway/tests/features/test_f03.py`
F04 Demo isolation	22	[F04](F04-demo-isolation.md)	`services/gateway/tests/features/test_f04.py`
F05 Demo reset	19	[F05](F05-demo-reset.md)	`services/gateway/tests/features/test_f05.py`
F06 Responsive layouts	F06	[F06](F06-responsive-layouts.md)	`apps/web/tests/features/f06.test.tsx`
F07 Arabic and English interface	F07	[F07](F07-arabic-and-english-interface.md)	`apps/web/tests/features/f07.test.tsx`
F08 Light and dark themes	F08	[F08](F08-light-and-dark-themes.md)	`apps/web/tests/features/f08.test.tsx`
F09 Bilingual chat	19	[F09](F09-bilingual-chat.md)	`services/orchestrator/tests/features/test_f09.py`
F10 Structured extraction	10	[F10](F10-structured-extraction.md)	`services/orchestrator/tests/features/test_f10.py`
F11 Clarifying questions	19	[F11](F11-clarifying-questions.md)	`services/orchestrator/tests/features/test_f11.py`
F12 Conversation memory	09	[F12](F12-conversation-memory.md)	`services/orchestrator/tests/features/test_f12.py`
F13 Private uploads	19	[F13](F13-private-uploads.md)	`services/gateway/tests/features/test_f13.py`
F14 Nameplate OCR	16	[F14](F14-nameplate-ocr.md)	`services/worker/tests/features/test_f14.py`
F15 OCR confirmation	19	[F15](F15-ocr-confirmation.md)	`apps/web/tests/features/f15.test.tsx`
F16 Photo quality feedback	F16	[F16](F16-photo-quality-feedback.md)	`services/worker/tests/features/test_f16.py`

Feature	Implementation owner	Feature prompt	Test
F17 Multi-part enquiries	F17	[F17](F17-multi-part-enquiries.md)	services/orchestrator/tests/features/test_f17.py
F18 Progress timeline	20	[F18](F18-progress-timeline.md)	apps/web/tests/features/f18.test.tsx
F19 Enquiry history	F19	[F19](F19-enquiry-history.md)	services/gateway/tests/features/test_f19.py
F20 Machine identification	06	[F20](F20-machine-identification.md)	services/search/tests/features/test_f20.py
F21 Machine mismatch detection	06	[F21](F21-machine-mismatch-detection.md)	services/search/tests/features/test_f21.py
F22 Compatibility gate	06	[F22](F22-compatibility-gate.md)	services/search/tests/features/test_f22.py
F23 Aliases and typo matching	06	[F23](F23-aliases-and-typo-matching.md)	services/search/tests/features/test_f23.py
F24 Semantic retrieval	06	[F24](F24-semantic-retrieval.md)	services/search/tests/features/test_f24.py
F25 Ranked candidates	06	[F25](F25-ranked-candidates.md)	services/search/tests/features/test_f25.py
F26 Ambiguity handling	11	[F26](F26-ambiguity-handling.md)	services/orchestrator/tests/features/test_f26.py
F27 Match explanations	20	[F27](F27-match-explanations.md)	apps/web/tests/features/f27.test.tsx
F28 Compatibility references	12	[F28](F28-compatibility-references.md)	services/search/tests/features/test_f28.py
F29 Evidence page viewer	F29	[F29](F29-evidence-page-viewer.md)	services/gateway/tests/features/test_f29.py
F30 Old part numbers	06	[F30](F30-old-part-numbers.md)	services/search/tests/features/test_f30.py
F31 Candidate comparison	F31	[F31](F31-candidate-comparison.md)	apps/web/tests/features/f31.test.tsx
F32 Warehouse stock	20	[F32](F32-warehouse-stock.md)	services/search/tests/features/test_f32.py
F33 Fulfilment planning	06	[F33](F33-fulfilment-planning.md)	services/search/tests/features/test_f33.py
F34 Transfer estimates	06	[F34](F34-transfer-estimates.md)	services/search/tests/features/test_f34.py
F35 Stock freshness	F35	[F35](F35-stock-freshness.md)	services/search/tests/features/test_f35.py
F36 Staff inbox	F36	[F36](F36-staff-inbox.md)	services/gateway/tests/features/test_f36.py
F37 Review workspace	20	[F37](F37-review-workspace.md)	apps/web/tests/features/f37.test.tsx
F38 Correct extracted details	F38	[F38](F38-correct-extracted-details.md)	services/orchestrator/tests/features/test_f38.py
F39 Review actions	20	[F39](F39-review-actions.md)	services/orchestrator/tests/features/test_f39.py
F40 Approval invalidation	14	[F40](F40-approval-invalidation.md)	services/orchestrator/tests/features/test_f40.py
F41 Internal notes	F41	[F41](F41-internal-notes.md)	services/gateway/tests/features/test_f41.py
F42 Assignments	F42	[F42](F42-assignments.md)	services/gateway/tests/features/test_f42.py
F43 Overdue indicators	F43	[F43](F43-overdue-indicators.md)	services/gateway/tests/features/test_f43.py
F44 Conversation summary	F44	[F44](F44-conversation-summary.md)	services/orchestrator/tests/features/test_f44.py
F45 Reply drafting	13	[F45](F45-reply-drafting.md)	services/orchestrator/tests/features/test_f45.py
F46 Verification gate	12	[F46](F46-verification-gate.md)	services/orchestrator/tests/features/test_f46.py
F47 Human publication gate	14	[F47](F47-human-publication-gate.md)	services/orchestrator/tests/features/test_f47.py
F48 Input isolation	F48	[F48](F48-input-isolation.md)	services/orchestrator/tests/features/test_f48.py
F49 Audit trail	14	[F49](F49-audit-trail.md)	tests/acceptance/test_f49.py
F50 Durable workflows	09	[F50](F50-durable-workflows.md)	services/orchestrator/tests/features/test_f50.py
F51 Background jobs	15	[F51](F51-background-jobs.md)	services/worker/tests/features/test_f51.py
F52 Retry and deduplication	15	[F52](F52-retry-and-deduplication.md)	services/worker/tests/features/test_f52.py
F53 Graceful failures	22	[F53](F53-graceful-failures.md)	apps/web/tests/features/f53.test.tsx
F54 Access and permissions	22	[F54](F54-access-and-permissions.md)	services/gateway/tests/features/test_f54.py
F55 Saved machines	F55	[F55](F55-saved-machines.md)	services/gateway/tests/features/test_f55.py

Feature	Implementation owner	Feature prompt	Test
F56 Repeat enquiry	F56	[F56](F56-repeat-enquiry.md)	services/gateway/tests/features/test_f56.py
F57 Catalogue browser	F57	[F57](F57-catalogue-browser.md)	services/gateway/tests/features/test_f57.py
F58 Catalogue imports	F58	[F58](F58-catalogue-imports.md)	services/worker/tests/features/test_f58.py
F59 Catalogue versioning	F59	[F59](F59-catalogue-versioning.md)	tests/acceptance/test_f59.py
F60 Missing-parts queue	F60	[F60](F60-missing-parts-queue.md)	services/gateway/tests/features/test_f60.py
F61 Enquiry PDF	20	[F61](F61-enquiry-pdf.md)	services/worker/tests/features/test_f61.py
F62 Labelled evaluations	21	[F62](F62-labelled-evaluations.md)	tests/acceptance/test_f62.py
F63 Metrics dashboard	F63	[F63](F63-metrics-dashboard.md)	services/gateway/tests/features/test_f63.py
F64 Latency and AI cost	23	[F64](F64-latency-and-ai-cost.md)	tests/acceptance/test_f64.py
F65 Correction feedback	F65	[F65](F65-correction-feedback.md)	services/gateway/tests/features/test_f65.py
F66 Tests and CI	25	[F66](F66-tests-and-ci.md)	tests/acceptance/test_f66.py
F67 Traces	23	[F67](F67-traces.md)	tests/acceptance/test_f67.py
F68 Recruiter documentation	F68	[F68](F68-recruiter-documentation.md)	tests/acceptance/test_f68.py
F69 Demo walkthrough	F69	[F69](F69-demo-walkthrough.md)	tests/acceptance/test_f69.py

022 docs/templates/BUG_TEMPLATE.md

BUG-001 - Duplicate approval example

Reproduction
Submit the same approval twice with one idempotency key against a disposable demo enquiry.

Expected
One review decision and one recommendation message.

Actual
Record observed rows and request IDs during reproduction; this template is an example,not a reported live defect.

Hypothesis
Missing transaction lock or unique publication constraint.

Investigation
Inspect approval transaction,idempotency lookup and messages_one_publication_idx.

Fix
Use current-revision transaction and uniqueness; do not suppress all database errors.

Verification
Run concurrent approval test;record actual output and commit when this example is used for a real bug.

023 docs/templates/FEATURE_TEMPLATE.md

F19 - Enquiry history

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F19; foundations 07, 08, 19
Affected flows	FLOW.md row F19

Scope

Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.
- [ ] Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f19.py`; command `python tools/run_checks.py --feature F19`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

024 docs/templates/PROMPT_TEMPLATE.md

F19 - Enquiry history

PHASE: gateway DEPENDS ON: 07, 08, 19 RISK: high

GOAL

Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering. Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F19-enquiry-history.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f19.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f19_enquiry_history.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/HistoryList.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f19_enquiry_history.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def list_history(actor: ActorContext, request: F19Request) -> F19Result

`F19Request`: cursor: str | None = None; limit: int = 20; status: str | None = None.

`F19Result`: enquiries: list[EnquiryView]; next_cursor: str | None.

Endpoint: `GET /v1/enquiries`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function HistoryList(props: {value: F19Result | null; loading: boolean; error: ApiError | null; onAction: (request: F19Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.

2. Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

Required feature behaviours:

• F19 Enquiry history: Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering. Failure cases: Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F19` identifies the entry contract and preceding owners. Required predecessors: 07, 08, 19. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f19.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F19`: Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records. Assert return cursor-paginated owned history with status and created time.

`python tools/run_checks.py --prompt F19` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for enquiry history using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F19`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f19` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F19 enquiry history
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

025 docs/templates/SESSION_HANDOFF.md

Five-line handoff example

1. Completed: bootstrap imported legacy files with checksums.
2. Remaining: common contracts and database migrations.
3. Watch: no production credentials in local test configuration.
4. Evidence: record actual bootstrap test output and commit in the real session log.
5. Model/version: record actual exposed identifier;otherwise not exposed.

026 prompts/00-bootstrap-and-preserve-legacy.md

00 - Bootstrap and preserve legacy

PHASE: Repository DEPENDS ON: none RISK: high

GOAL

Preserve all files in references/legacy-backend.zip at their original relative paths; retain checksum manifest before modifications. Create one root uv workspace, with named common,gateway,search,orchestrator,worker and catalogue projects. Use a single uv.lock. Run service tests in separate processes because legacy services share the app import name. Local Compose publishes gateway on 127.0.0.1 only and uses local Supabase CLI endpoints. Do not copy real secrets or rewrite domain modules.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F66-tests-and-ci.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install PyYAML==6.0.3 pytest==9.0.2 pytest-asyncio==1.3.0 ruff==0.15.6 mypy==1.19.1` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.
- `python -m pip install uv==0.10.9`

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `README.md`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `pyproject.toml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `.gitignore`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `.env.example`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `.python-version`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `.nvmrc`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `docker-compose.yml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/pyproject.toml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/worker/pyproject.toml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tools/verify_env.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tools/run_checks.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tools/check_release_eligibility.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `uv.lock`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tests/foundations/test_00.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def verify_environment(env: Mapping[str,str]) -> list[str]: return missing or unsafe environment names without values; reject prod database URLs when APP_ENV is local.

Implementation sequence:

1. Preserve all files in references/legacy-backend.zip at their original relative paths; retain checksum manifest before modifications.

2. Create one root uv workspace, with named common,gateway,search,orchestrator,worker and catalogue projects.

3. Use a single uv.lock.

4. Run service tests in separate processes because legacy services share the app import name.

5. Local Compose publishes gateway on 127.0.0.1 only and uses local Supabase CLI endpoints.

6. Do not copy real secrets or rewrite domain modules.

Required feature behaviours:

• F66 Tests and CI: Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs. Failure cases: No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment.

Exact package markers and bundled assets created by this step:

• `tools/catalog/__init__.py`

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Repository; ARCHITECTURE.md section2. FLOW.md row `00` identifies the entry contract and preceding owners. Required predecessors: none. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_00.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F66`: No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment. Assert run service-isolated python tests,frontend checks,rls integration and deterministic evals.

`python tools/run_checks.py --prompt 00` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for bootstrap and preserve legacy using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `00`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-00` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(repository): 00 bootstrap and preserve legacy
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

027 prompts/01-shared-contracts-and-clients.md

01 - Shared contracts and clients

PHASE: Common DEPENDS ON: 00 RISK: high

GOAL

Implement the exact contract models from spec/contracts.py. Preserve text.py and DB pool behaviour. Wrap every OpenAI invocation with explicit timeout, output token limit and budget reservation; never retry inside both SDK and wrapper. Set SDK max_retries=0 and wrapper max attempts=2. Log safe error code and request ID. Insert audit within caller transaction for business writes; fail approval if audit fails. Set app.actor_id using set_config with is_local=true inside the transaction and clear no session-global state.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F49-audit-trail.md
- features/F64-latency-and-ai-cost.md
- features/F67-traces.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install fastapi==0.135.1 uvicorn==0.41.0 pydantic==2.12.5 pydantic-settings==2.13.1 httpx==0.28.1 asyncpg==0.31.0 psycopg[binary]==3.3.3 psycopg-pool==3.3.0 redis==7.3.0 openai==3.19.2 structlog==25.5.0 tenacity==9.1.2 PyJWT==2.12.1 cryptography==46.0.5 python-multipart==0.0.22` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `libs/partsiq_common/partsiq_common/contracts.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `libs/partsiq_common/partsiq_common/audit.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `libs/partsiq_common/partsiq_common/errors.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `libs/partsiq_common/partsiq_common/ai/client.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `libs/partsiq_common/partsiq_common/ai/budget.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `libs/partsiq_common/partsiq_common/telemetry.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `libs/partsiq_common/partsiq_common/security_context.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `libs/partsiq_common/partsiq_common/ai/prices.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tests/foundations/test_01.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def call_structured(request: AIRequest, budget: BudgetReservation) -> AIResult; async def append_audit(conn: AsyncConnection, event: AuditEvent) -> int; async def actor_transaction(actor: ActorContext) -> AsyncIterator[AsyncConnection].

Implementation sequence:

1. Implement the exact contract models from spec/contracts.py.

2. Preserve text.py and DB pool behaviour.

3. Wrap every OpenAI invocation with explicit timeout, output token limit and budget reservation; never retry inside both SDK and wrapper.

4. Set SDK max_retries=0 and wrapper max attempts=2.

5. Log safe error code and request ID.

6. Insert audit within caller transaction for business writes; fail approval if audit fails.

7. Set app.actor_id using set_config with is_local=true inside the transaction and clear no session-global state.

Required feature behaviours:

• F49 Audit trail: Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads. Failure cases: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.

• F64 Latency and AI cost: Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance. Failure cases: Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit.

• F67 Traces: Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable. Failure cases: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.

Exact package markers and bundled assets created by this step:

• `libs/partsiq_common/partsiq_common/ai/__init__.py`

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Common; ARCHITECTURE.md section2. FLOW.md row `01` identifies the entry contract and preceding owners. Required predecessors: 00. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_01.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• F49: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly. Assert append actor,request,entity,revision and minimal before/after metadata for ai suggestions and human changes.

• F64: Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit. Assert measure stages with monotonic clock.

• F67: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative. Assert propagate request/enquiry/job ids.

`python tools/run_checks.py --prompt 01` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for shared contracts and clients using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `01`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-01` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(common): 01 shared contracts and clients
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

028 prompts/02-catalogue-migration.md

02 - Catalogue migration

PHASE: Database DEPENDS ON: 01 RISK: high

GOAL

Copy spec/sql/001_initial_schema.sql and 002_search_and_fulfilment.sql verbatim into the named migrations. This preserves numeric catalogue keys, text normalization, exclusion constraints, cycle protection and warehouse transfer days. A fresh Supabase dev project is the baseline. For an existing database, compare migration checksums and schema receipt; baseline already-applied legacy migrations rather than executing CREATE TABLE twice. No production reset command is allowed.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F20-machine-identification.md
- features/F22-compatibility-gate.md
- features/F28-compatibility-references.md
- features/F30-old-part-numbers.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `npm exec --yes --package=supabase@2.75.0 -- supabase --version`

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `supabase/config.toml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `supabase/migrations/20261011000100_legacy_catalogue.sql`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `supabase/migrations/20261011000200_legacy_search.sql`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_02.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def apply_catalogue_schema(dsn: str) -> None; schema comparison is owned by collect_schema_receipt in prompt03.

Implementation sequence:

1. Copy spec/sql/001_initial_schema.sql and 002_search_and_fulfilment.sql verbatim into the named migrations.

2. This preserves numeric catalogue keys, text normalization, exclusion constraints, cycle protection and warehouse transfer days.

3. A fresh Supabase dev project is the baseline.

4. For an existing database, compare migration checksums and schema receipt; baseline already-applied legacy migrations rather than executing CREATE TABLE twice.

5. No production reset command is allowed.

Required feature behaviours:

• F20 Machine identification: Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input. Failure cases: Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override.

• F22 Compatibility gate: Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate. Failure cases: A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published.

• F28 Compatibility references: Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot. Failure cases: Missing source file blocks evidence viewer and flags review; mismatched version fails verification.

• F30 Old part numbers: Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration. Failure cases: Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected.

• `tools/apply_catalogue_schema.py`: `async def apply_catalogue_schema(dsn: str) -> None: apply the two immutable migration files through the Supabase migration runner; refuse a production reset.`

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Database; ARCHITECTURE.md section2. FLOW.md row `02` identifies the entry contract and preceding owners. Required predecessors: 01. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_02.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F20`: Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override. Assert preserve identify_machine logic for model, serial and rating.

• `F22`: A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published. Assert fetch authoritative fitment set first.

• `F28`: Missing source file blocks evidence viewer and flags review; mismatched version fails verification. Assert resolve source_reference to version-specific document and page.

• `F30`: Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected. Assert follow supersession chain with cycle protection.

`python tools/run_checks.py --prompt 02` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for catalogue migration using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `02`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-02` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(database): 02 catalogue migration
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

029 prompts/03-workflow-schema-and-database-permissions.md

03 - Workflow schema and database permissions

PHASE: Database DEPENDS ON: 02 RISK: high

GOAL

Copy application and RLS SQL from spec/sql. Migrate candidate keys to per-item/per-revision without losing historical rows. Do not backfill invented customer identities; legacy ownerless rows remain inaccessible through customer routes. Create restricted LOGIN credentials administratively and grant each one its service group. Run LangGraph saver setup with MIGRATION_DATABASE_URL into schema checkpoints, snapshot library-created schema, then remove runtime DDL privileges. Test two actors on a reused connection pool.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F04-demo-isolation.md
- features/F12-conversation-memory.md
- features/F49-audit-trail.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `supabase/migrations/20261011000300_application.sql`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `supabase/migrations/20261011000400_rls.sql`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/checkpoints_setup.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/schema_receipt.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/integration/conftest.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_03.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def setup_checkpoints(dsn: str) -> None; async def collect_schema_receipt(dsn: str) -> SchemaReceipt; no application service invokes setup_checkpoints.

Implementation sequence:

1. Copy application and RLS SQL from spec/sql.

2. Migrate candidate keys to per-item/per-revision without losing historical rows.

3. Do not backfill invented customer identities; legacy ownerless rows remain inaccessible through customer routes.

4. Create restricted LOGIN credentials administratively and grant each one its service group.

5. Run LangGraph saver setup with MIGRATION_DATABASE_URL into schema checkpoints, snapshot library-created schema, then remove runtime DDL privileges.

6. Test two actors on a reused connection pool.

Required feature behaviours:

• F04 Demo isolation: Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default. Failure cases: Expired, reset and foreign sessions fail; missing identity context returns no rows.

• F12 Conversation memory: Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions. Failure cases: Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied.

• F49 Audit trail: Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads. Failure cases: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Database; ARCHITECTURE.md section2. FLOW.md row `03` identifies the entry contract and preceding owners. Required predecessors: 02. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_03.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F04`: Expired, reset and foreign sessions fail; missing identity context returns no rows. Assert bind each session to verified anonymous supabase identity.

• `F12`: Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied. Assert persist all messages and typed current state.

• `F49`: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly. Assert append actor,request,entity,revision and minimal before/after metadata for ai suggestions and human changes.

`python tools/run_checks.py --prompt 03` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for workflow schema and database permissions using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For

UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `03`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-03` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(database): 03 workflow schema and database permissions
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

030 prompts/04-private-storage.md

04 - Private storage

PHASE: Gateway/worker DEPENDS ON: 03 RISK: high

GOAL

Create four private buckets from exact SQL. Gateway signs owner uploads using verified user token and pending attachment policy. Worker alone holds storage administration credential for export,evidence,import and cleanup operations; never use it as database REST bypass. Gateway first authorizes each read and asks internal worker signing handler with typed object ID, never arbitrary key. Signed links expire after 60 seconds; strip them from logs. Confirm actual bytes before marking validated.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F13-private-uploads.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `supabase/migrations/20261011000500_storage.sql`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/app/storage.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/storage.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_04.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def create_upload(actor: ActorContext, request: UploadRequest) -> UploadTicket; async def sign_download(request: DownloadRequest) -> DownloadTicket; async def remove_object(bucket: str, object_key: str) -> None.

Implementation sequence:

1. Create four private buckets from exact SQL.
2. Gateway signs owner uploads using verified user token and pending attachment policy.
3. Worker alone holds storage administration credential for export,evidence,import and cleanup operations; never use it as database REST bypass.
4. Gateway first authorizes each read and asks internal worker signing handler with typed object ID, never arbitrary key.

5. Signed links expire after 60 seconds; strip them from logs.

6. Confirm actual bytes before marking validated.

Required feature behaviours:

• F13 Private uploads: Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR. Failure cases: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.

Exact package markers and bundled assets created by this step:

• `services/worker/app/__init__.py`

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Gateway/worker; ARCHITECTURE.md section2. FLOW.md row `04` identifies the entry contract and preceding owners. Required predecessors: 03. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_04.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F13`: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly. Assert create pending attachment and unique object path.

`python tools/run_checks.py --prompt 04` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for private storage using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `04`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-04` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway-worker): 04 private storage
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

031 prompts/05-synthetic-catalogue-and-demo-fixtures.md

05 - Synthetic catalogue and demo fixtures

PHASE: Catalogue DEPENDS ON: 04 RISK: high

GOAL

Use existing tools/catalog/catalog generator and validator with seed 42. Preserve loader identity mapping and source labels. Generate four scenario fixtures by selecting real generated records with required properties, not hardcoded illustrative numbers. Generate synthetic nameplate PNGs and cited PDF pages with manifests of hashes. Fail generation when an expected outcome cannot be demonstrated. Base loading creates the initial published catalogue_versions row,immutable catalogue_records snapshot and catalogue_evidence rows from generated manifests in the same controlled seed transaction. Base catalogue loading does not instantiate BGE-M3; embedding is deferred to 06.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F02-demo-scenarios-and-assets.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install Pillow==12.1.1` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `tools/demo/build_scenarios.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/scenarios.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/nameplates/manifest.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/catalogue_evidence/manifest.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/render_assets.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/scenarios/mixed_photo.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/demo/scenarios/missing_serial.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tools/demo/scenarios/incompatible_number.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tools/demo/scenarios/local_shortage.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tests/foundations/test_05.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def build_scenarios(dataset: CatalogDataset) -> list[DemoScenario]; def render_nameplate(model: str, serial: str, destination: Path) -> None; def validate_scenarios(scenarios: list[DemoScenario], dataset: CatalogDataset) -> list[str].

Implementation sequence:

1. Use existing tools/catalog/catalog generator and validator with seed 42.

2. Preserve loader identity mapping and source labels.

3. Generate four scenario fixtures by selecting real generated records with required properties, not hardcoded illustrative numbers.

4. Generate synthetic nameplate PNGs and cited PDF pages with manifests of hashes.

5. Fail generation when an expected outcome cannot be demonstrated.

6. Base loading creates the initial published catalogue_versions row,immutable catalogue_records snapshot and catalogue_evidence rows from generated manifests in the same controlled seed transaction. Base catalogue loading does not instantiate BGE-M3; embedding is deferred to 06.

Required feature behaviours:

• F02 Demo scenarios and assets: Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records. Failure cases: Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Catalogue; ARCHITECTURE.md section2. FLOW.md row `05` identifies the entry contract and preceding owners. Required predecessors: 04. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_05.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F02`: Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities. Assert seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage.

`python tools/run_checks.py --prompt 05` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for synthetic catalogue and demo fixtures using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `05`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-05` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(catalogue): 05 synthetic catalogue and demo fixtures
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

032 prompts/06-extend-legacy-search.md

06 - Extend legacy search

PHASE: Search DEPENDS ON: 05 RISK: high

GOAL

Characterize current identify_machine,LexicalScorer,SearchService and plan_fulfilment first. Keep identification.py,lexical.py,fulfilment.py,repository.py,service.py,routes.py and schemas.py. Add internal token authentication to routes. Preserve weighted semantic fusion and lexical fallback. Require terminal supersession part to have explicit fitment; fix margin calculation before result truncation. Search-owned encoder reads a validated published version and upserts vectors with source hash. Use one search process and one encoder; readiness reports lexical readiness and semantic capability separately.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F20-machine-identification.md
- features/F21-machine-mismatch-detection.md
- features/F22-compatibility-gate.md
- features/F23-aliases-and-typo-matching.md
- features/F24-semantic-retrieval.md
- features/F25-ranked-candidates.md
- features/F26-ambiguity-handling.md
- features/F27-match-explanations.md
- features/F28-compatibility-references.md
- features/F30-old-part-numbers.md
- features/F32-warehouse-stock.md
- features/F33-fulfilment-planning.md
- features/F34-transfer-estimates.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install sentence-transformers==5.2.3 transformers==4.57.6 torch==2.10.0+cpu numpy==2.4.2` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

CPU wheel command: `uv pip install torch==2.10.0+cpu --index-url https://download.pytorch.org/whl/cpu`. Configure the root uv workspace with an explicit pytorch-cpu index for torch only. Never install CUDA wheels

on the EC2 CPU profile. The base release date is2026-01-21; wheel/index receipt is captured by dependency resolution.

FILES TO CREATE

• `services/search/app/embedding_ingestion.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/search/tests/test_characterization.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/search/tests/features/test_f20.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f21.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f22.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f23.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f24.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f25.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f30.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f33.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/search/tests/features/test_f34.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `tests/foundations/test_06.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def embed_version(version_id: UUID, batch_size: int=16) -> EmbeddingReceipt; preserve SearchService.identify(request: IdentifyRequest) -> IdentifyResponse and SearchService.search_parts(request: PartSearchRequest) -> PartSearchResponse.

Implementation sequence:

1. Characterize current identify_machine,LexicalScorer,SearchService and plan_fulfilment first.

2. Keep identification.py,lexical.py,fulfilment.py,repository.py,service.py,routes.py and schemas.py.

3. Add internal token authentication to routes.

4. Preserve weighted semantic fusion and lexical fallback.

5. Require terminal supersession part to have explicit fitment; fix margin calculation before result truncation.

6. Search-owned encoder reads a validated published version and upserts vectors with source hash.

7. Use one search process and one encoder; readiness reports lexical readiness and semantic capability separately.

Required feature behaviours:

• F20 Machine identification: Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input. Failure cases: Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override.

• F21 Machine mismatch detection: Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved. Failure cases: Valid serial of a different model does not match requested model; nonnumeric serial handled without crashing.

• F22 Compatibility gate: Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate. Failure cases: A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published.

• F23 Aliases and typo matching: Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context. Failure cases: Arabic part words are not removed as machine words; exact part number outranks unrelated fuzzy alias; normalization parity test matches SQL.

• F24 Semantic retrieval: Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path. Failure cases: Wrong dimension rejected; no model loaded in worker; no semantic candidate outside compatible IDs.

• F25 Ranked candidates: Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit. Failure cases: Limit one still reports ambiguity; empty set returns no candidate; NaN scores rejected.

• F26 Ambiguity handling: Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates. Failure cases: LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases.

• F27 Match explanations: Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose. Failure cases: Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability.

• F28 Compatibility references: Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot. Failure cases: Missing source file blocks evidence viewer and flags review; mismatched version fails verification.

• F30 Old part numbers: Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration. Failure cases: Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected.

• F32 Warehouse stock: Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot. Failure cases: Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected.

• F33 Fulfilment planning: Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders. Failure cases: Total allocations never exceed request or stock; fractional quantities respect unit policy; unavailable transfer route cannot be invented.

• F34 Transfer estimates: Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days. Failure cases: No route gives unknown/unavailable plan rather than invented one-day promise; estimate labelled non-guaranteed.

FILES TO MODIFY

• `libs/partsiq_common/partsiq_common/embeddings.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/Dockerfile`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/config.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/main.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/repository.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/schemas.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/service.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `tools/catalog/catalog/embedder.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Search; ARCHITECTURE.md section2. FLOW.md row `06` identifies the entry contract and preceding owners. Required predecessors: 05. HTTP routes call application functions,which invoke domain rules and

infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/search/tests/features/test_f20.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f21.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f22.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f23.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f24.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f25.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f30.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f33.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f34.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_06.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F20`: Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override. Assert preserve identify_machine logic for model, serial and rating.

• `F21`: Valid serial of a different model does not match requested model; nonnumeric serial handled without crashing. Assert compare model and serial together.

• `F22`: A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published. Assert fetch authoritative fitment set first.

• `F23`: Arabic part words are not removed as machine words; exact part number outranks unrelated fuzzy alias; normalization parity test matches SQL. Assert preserve arabic normalization, part-number keys, phrase scoring and fuzzy token matching.

• `F24`: Wrong dimension rejected; no model loaded in worker; no semantic candidate outside compatible IDs. Assert load one bge-m3 encoder per search process.

• `F25`: Limit one still reports ambiguity; empty set returns no candidate; NaN scores rejected. Assert return top candidates with lexical/semantic evidence and deterministic tie order.

• `F26`: LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases. Assert use tested score and margin thresholds in code.

• `F27`: Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability. Assert show actual alias, machine configuration, supersession chain and fitment reference.

• `F28`: Missing source file blocks evidence viewer and flags review; mismatched version fails verification. Assert resolve source_reference to version-specific document and page.

• `F30`: Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected. Assert follow supersession chain with cycle protection.

• `F32`: Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected. Assert show available quantity as on_hand minus reserved.

• F33: Total allocations never exceed request or stock; fractional quantities respect unit policy; unavailable transfer route cannot be invented. Assert allocate local stock first then reachable warehouses deterministically.

• F34: No route gives unknown/unavailable plan rather than invented one-day promise; estimate labelled non-guaranteed. Assert carry integer transfer_days from approved route.

`python tools/run_checks.py --prompt 06` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for extend legacy search using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `06`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-06` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(search): 06 extend legacy search
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

033 prompts/F20-machine-identification.md

F20 - Machine identification (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F20 coverage without implementing it twice. Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F20-machine-identification.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 02, 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f20.py` is created by 06.

FILES TO MODIFY

`features/F20-machine-identification.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 02, 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override.

ACCEPTANCE TESTS

`services/search/tests/features/test_f20.py` must assert: Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input. Negative cases: Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override.

Run `python tools/run_checks.py --feature F20`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise machine identification using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F20-machine-identification.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F20 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F20-machine-identification.md

F20 - Machine identification

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 02, 06
Affected flows	FLOW.md row 06

Scope

Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Preserve identify_machine logic for model, serial and rating; return resolved configuration or explicit missing/ambiguous status; normalize numeric serial only after preserving input.
- [ ] Boundary serials match exactly one configuration; overlapping ranges fail database insertion; conflicting hints never silently override.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

equipment_models, model_configurations. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f20.py`; command `python tools/run_checks.py --feature F20`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

034 prompts/F21-machine-mismatch-detection.md

F21 - Machine mismatch detection (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F21 coverage without implementing it twice. Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F21-machine-mismatch-detection.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f21.py` is created by 06.

FILES TO MODIFY

`features/F21-machine-mismatch-detection.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Valid serial of a different model does not match requested model; nonnumeric serial handled without crashing.

ACCEPTANCE TESTS

`services/search/tests/features/test_f21.py` must assert: Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved. Negative cases: Valid serial of a different model does not match requested model; nonnumeric serial handled without crashing.

Run `python tools/run_checks.py --feature F21`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise machine mismatch detection using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F21-machine-mismatch-detection.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F21 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F21-machine-mismatch-detection.md

F21 - Machine mismatch detection

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Compare model and serial together; return mismatch reason with safe next question; prevent retrieval while identity unresolved.
- [ ] Valid serial of a different model does not match requested model; nonnumeric serial handled without crashing.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

model_configurations. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f21.py`; command `python tools/run_checks.py --feature F21`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

035 prompts/F22-compatibility-gate.md

F22 - Compatibility gate (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F22 coverage without implementing it twice. Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F22-compatibility-gate.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 02, 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f22.py` is created by 06.

FILES TO MODIFY

`features/F22-compatibility-gate.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 02, 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published.

ACCEPTANCE TESTS

`services/search/tests/features/test_f22.py` must assert: Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate. Negative cases: A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published.

Run `python tools/run_checks.py --feature F22`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise compatibility gate using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F22-compatibility-gate.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F22 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F22-compatibility-gate.md

F22 - Compatibility gate

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 02, 06
Affected flows	FLOW.md row 06

Scope

Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Fetch authoritative fitment set first; rank only members; carry source_reference and catalogue version through every candidate.
- [ ] A perfect lexical match outside fitment is absent; empty fitment returns unavailable; blank provenance cannot be published.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

part_compatibility. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f22.py`; command `python tools/run_checks.py --feature F22`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

036 prompts/F23-aliases-and-typo-matching.md

F23 - Aliases and typo matching (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F23 coverage without implementing it twice. Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F23-aliases-and-typo-matching.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f23.py` is created by 06.

FILES TO MODIFY

`features/F23-aliases-and-typo-matching.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Arabic part words are not removed as machine words; exact part number outranks unrelated fuzzy alias; normalization parity test matches SQL.

ACCEPTANCE TESTS

`services/search/tests/features/test_f23.py` must assert: Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context. Negative cases: Arabic part words are not removed as machine words; exact part number outranks unrelated fuzzy alias; normalization parity test matches SQL.

Run `python tools/run_checks.py --feature F23`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise aliases and typo matching using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F23-aliases-and-typo-matching.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F23 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F23-aliases-and-typo-matching.md

F23 - Aliases and typo matching

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Preserve Arabic normalization, part-number keys, phrase scoring and fuzzy token matching; remove machine terms using resolved model context.
- [ ] Arabic part words are not removed as machine words; exact part number outranks unrelated fuzzy alias; normalization parity test matches SQL.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

parts, part_aliases. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f23.py`; command `python tools/run_checks.py --feature F23`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

037 prompts/F24-semantic-retrieval.md

F24 - Semantic retrieval (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F24 coverage without implementing it twice. Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F24-semantic-retrieval.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f24.py` is created by 06.

FILES TO MODIFY

`features/F24-semantic-retrieval.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Wrong dimension rejected; no model loaded in worker; no semantic candidate outside compatible IDs.

ACCEPTANCE TESTS

`services/search/tests/features/test_f24.py` must assert: Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path. Negative cases: Wrong dimension rejected; no model loaded in worker; no semantic candidate outside compatible IDs.

Run `python tools/run_checks.py --feature F24`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise semantic retrieval using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F24-semantic-retrieval.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F24 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F24-semantic-retrieval.md

F24 - Semantic retrieval

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Load one BGE-M3 encoder per search process; use normalized 1024-d vectors; retain legacy weighted fusion; fall back to lexical on unavailable embedding path.
- [ ] Wrong dimension rejected; no model loaded in worker; no semantic candidate outside compatible IDs.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

part_embeddings. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f24.py`; command `python tools/run_checks.py --feature F24`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

038 prompts/F25-ranked-candidates.md

F25 - Ranked candidates (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F25 coverage without implementing it twice. Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F25-ranked-candidates.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f25.py` is created by 06.

FILES TO MODIFY

`features/F25-ranked-candidates.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Limit one still reports ambiguity; empty set returns no candidate; NaN scores rejected.

ACCEPTANCE TESTS

`services/search/tests/features/test_f25.py` must assert: Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit. Negative cases: Limit one still reports ambiguity; empty set returns no candidate; NaN scores rejected.

Run `python tools/run_checks.py --feature F25`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise ranked candidates using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F25-ranked-candidates.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F25 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F25-ranked-candidates.md

F25 - Ranked candidates

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Return top candidates with lexical/semantic evidence and deterministic tie order; compute runner-up margin before applying response limit.
- [ ] Limit one still reports ambiguity; empty set returns no candidate; NaN scores rejected.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_candidates. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f25.py`; command `python tools/run_checks.py --feature F25`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

039 prompts/F30-old-part-numbers.md

F30 - Old part numbers (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F30 coverage without implementing it twice. Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F30-old-part-numbers.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 02, 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f30.py` is created by 06.

FILES TO MODIFY

`features/F30-old-part-numbers.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 02, 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected.

ACCEPTANCE TESTS

`services/search/tests/features/test_f30.py` must assert: Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration. Negative cases: Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected.

Run `python tools/run_checks.py --feature F30`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise old part numbers using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F30-old-part-numbers.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F30 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F30-old-part-numbers.md

F30 - Old part numbers

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 02, 06
Affected flows	FLOW.md row 06

Scope

Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Follow supersession chain with cycle protection; display old and current numbers; verify current successor has explicit fitment for resolved configuration.
- [ ] Cycles rejected; inactive terminal part returns unavailable; inherited compatibility without source is rejected.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

part_supersessions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f30.py`; command `python tools/run_checks.py --feature F30`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

040 prompts/F33-fulfilment-planning.md

F33 - Fulfilment planning (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F33 coverage without implementing it twice. Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F33-fulfilment-planning.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f33.py` is created by 06.

FILES TO MODIFY

`features/F33-fulfilment-planning.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Total allocations never exceed request or stock; fractional quantities respect unit policy; unavailable transfer route cannot be invented.

ACCEPTANCE TESTS

`services/search/tests/features/test_f33.py` must assert: Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders. Negative cases: Total allocations never exceed request or stock; fractional quantities respect unit policy; unavailable transfer route cannot be invented.

Run `python tools/run_checks.py --feature F33`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise fulfilment planning using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F33-fulfilment-planning.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F33 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F33-fulfilment-planning.md

F33 - Fulfilment planning

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Allocate local stock first then reachable warehouses deterministically; compute shortage; use quantity units; do not reserve stock or create orders.
- [ ] Total allocations never exceed request or stock; fractional quantities respect unit policy; unavailable transfer route cannot be invented.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

stock_levels, warehouse_transfer_routes. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f33.py`; command `python tools/run_checks.py --feature F33`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

041 prompts/F34-transfer-estimates.md

F34 - Transfer estimates (covered by 06)

PHASE: Coverage verification DEPENDS ON: 06 RISK: low

GOAL

Verify F34 coverage without implementing it twice. Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days.

READ FIRST

- prompts/06-extend-legacy-search.md
- features/F34-transfer-estimates.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 06 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f34.py` is created by 06.

FILES TO MODIFY

`features/F34-transfer-estimates.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06. Covered by 06. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No route gives unknown/unavailable plan rather than invented one-day promise; estimate labelled non-guaranteed.

ACCEPTANCE TESTS

`services/search/tests/features/test_f34.py` must assert: Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days. Negative cases: No route gives unknown/unavailable plan rather than invented one-day promise; estimate labelled non-guaranteed.

Run `python tools/run_checks.py --feature F34`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise transfer estimates using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F34-transfer-estimates.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 06 with a separate trace.

COMMIT

```
docs(features): verify F34 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F34-transfer-estimates.md

F34 - Transfer estimates

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	06; foundations 06
Affected flows	FLOW.md row 06

Scope

Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Carry integer transfer_days from approved route; show estimate and source warehouse; local availability means zero transfer days.
- [ ] No route gives unknown/unavailable plan rather than invented one-day promise; estimate labelled non-guaranteed.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

warehouse_transfer_routes. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f34.py`; command `python tools/run_checks.py --feature F34`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

042 prompts/07-authentication-and-demo-sessions.md

07 - Authentication and demo sessions

PHASE: Gateway DEPENDS ON: 06 RISK: high

GOAL

Verify Supabase JWT issuer,audience,expiry and allowed algorithm through cached JWKS; refresh on unknown kid once. Fetch staff membership from DB, never JWT user_metadata. Use Supabase anonymous sign-in for public demos. Session creation requires that authenticated anonymous identity; rate limit before database work. Reset under row lock with idempotency key and generation increment. Reviewer access is assigned-only for real enquiries; admin may manage assignments. Demo review capability applies only to own active session.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F03-demo-role-switching.md
- features/F04-demo-isolation.md
- features/F05-demo-reset.md
- features/F54-access-and-permissions.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/app/auth.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/app/permissions.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/app/demo.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_07.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def authenticate(token: str) -> ActorContext; async def require_access(actor: ActorContext, enquiry_id: UUID, action: AccessAction) -> None; async def create_demo(actor: ActorContext) -> DemoSession; async def reset_demo(actor: ActorContext, key: UUID) -> DemoSession.

Implementation sequence:

1. Verify Supabase JWT issuer,audience,expiry and allowed algorithm through cached JWKS; refresh on unknown kid once.

2. Fetch staff membership from DB, never JWT user_metadata.

3. Use Supabase anonymous sign-in for public demos.

4. Session creation requires that authenticated anonymous identity; rate limit before database work.

5. Reset under row lock with idempotency key and generation increment.

6. Reviewer access is assigned-only for real enquiries; admin may manage assignments.

7. Demo review capability applies only to own active session.

Required feature behaviours:

• F03 Demo role switching: Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session. Failure cases: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.

• F04 Demo isolation: Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default. Failure cases: Expired, reset and foreign sessions fail; missing identity context returns no rows.

• F05 Demo reset: Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection. Failure cases: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.

• F54 Access and permissions: Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup. Failure cases: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.

FILES TO MODIFY

• `services/gateway/app/main.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/security.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Gateway; ARCHITECTURE.md section2. FLOW.md row `07` identifies the entry contract and preceding owners. Required predecessors: 06. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_07.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F03`: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404. Assert switch the presentation between customer and reviewer using the same anonymous identity and session.

• `F04`: Expired, reset and foreign sessions fail; missing identity context returns no rows. Assert bind each session to verified anonymous supabase identity.

• `F05`: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation. Assert lock session.

• `F54`: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue. Assert verify supabase issuer,audience,signature,expiry.

`python tools/run_checks.py --prompt 07` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner

exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for authentication and demo sessions using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `07`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-07` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): 07 authentication and demo sessions
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

043 prompts/08-enquiry-intake-and-limits.md

08 - Enquiry intake and limits

PHASE: Gateway DEPENDS ON: 07 RISK: high

GOAL

Create enquiry,first message,revision snapshot and workflow event atomically. Accept Idempotency-Key UUID; repeated identical input returns existing resource, different payload returns 409. Generate object names server-side. Preserve legacy middleware for size limits and traversal blocking. Redis rate limiter rejects when Redis unavailable. Gateway writes intake/session/staff CRUD; orchestrator owns derived facts and publication. Return 202 after durable commit, not after OCR or language calls.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F13-private-uploads.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/app/enquiries.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/app/uploads.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/app/api_v1.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_08.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def create_enquiry(actor: ActorContext, body: EnquiryCreate, key: UUID) -> EnquiryView; async def add_message(actor: ActorContext, enquiry_id: UUID, body: MessageCreate, key: UUID) -> MessageView; async def confirm_upload(actor: ActorContext, attachment_id: UUID) -> AttachmentView.

Implementation sequence:

1. Create enquiry,first message,revision snapshot and workflow event atomically.
2. Accept Idempotency-Key UUID; repeated identical input returns existing resource, different payload returns 409.
3. Generate object names server-side.
4. Preserve legacy middleware for size limits and traversal blocking.
5. Redis rate limiter rejects when Redis unavailable.

6. Gateway writes intake/session/staff CRUD; orchestrator owns derived facts and publication.

7. Return 202 after durable commit, not after OCR or language calls.

Required feature behaviours:

• F13 Private uploads: Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR. Failure cases: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.

FILES TO MODIFY

• `services/gateway/app/main.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/proxy.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/rate_limit.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Gateway; ARCHITECTURE.md section2. FLOW.md row `08` identifies the entry contract and preceding owners. Required predecessors: 07. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_08.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F13`: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly. Assert create pending attachment and unique object path.

`python tools/run_checks.py --prompt 08` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for enquiry intake and limits using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `08`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-08` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): 08 enquiry intake and limits
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

044 prompts/09-deterministic-durable-workflow.md

09 - Deterministic durable workflow

PHASE: Orchestrator DEPENDS ON: 08 RISK: high

GOAL

Define EnquiryState with items keyed by item UUID,revision,session generation,catalogue version,pending question,draft and verification IDs. Ordered controller rules select intake,await_ocr,confirm,identify,clarify,search,choose,plan,draft,verify,review,complete or needs_staff. Persist interrupts. Register handler interfaces and inject test doubles in tests; do not ship runnable fake-success nodes. Start/resume endpoints enqueue idempotent DB events. Consumer claims events under lease and locks enquiry; after each stage verify revision/session still current. Checkpointer migrations run only in 03.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F10-structured-extraction.md
- features/F11-clarifying-questions.md
- features/F12-conversation-memory.md
- features/F18-progress-timeline.md
- features/F50-durable-workflows.md
- features/F67-traces.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install langgraph==1.1.3 langgraph-checkpoint-postgres==3.0.4` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `services/orchestrator/app/main.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/state.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/controller.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/graph.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/app/events.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/app/routes.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/app/handlers.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/app/runner.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `services/orchestrator/tests/features/test_f12.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/orchestrator/tests/features/test_f50.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `tests/foundations/test_09.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def next_step(state: EnquiryState) -> WorkflowStep; def build_graph(handlers: WorkflowHandlers, checkpointer: AsyncPostgresSaver) -> CompiledStateGraph; async def consume_event(event_id: UUID) -> EventOutcome.

Implementation sequence:

1. Define EnquiryState with items keyed by item UUID,revision,session generation,catalogue version,pending question,draft and verification IDs.

2. Ordered controller rules select intake,await_ocr,confirm,identify,clarify,search,choose,plan,draft,verify,review,complete or needs_staff.

3. Persist interrupts.

4. Register handler interfaces and inject test doubles in tests; do not ship runnable fake-success nodes.

5. Start/resume endpoints enqueue idempotent DB events.

6. Consumer claims events under lease and locks enquiry; after each stage verify revision/session still current.

7. Checkpointer migrations run only in 03.

Required feature behaviours:

• F10 Structured extraction: Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity. Failure cases: Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search.

• F11 Clarifying questions: Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only. Failure cases: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.

• F12 Conversation memory: Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions. Failure cases: Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied.

• F18 Progress timeline: Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff. Failure cases: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.

• F50 Durable workflows: Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart. Failure cases: Duplicate events do not repeat publication; stale revision event is consumed as obsolete; checkpoint DDL never runs under runtime role.

• F67 Traces: Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable. Failure cases: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.

Exact package markers and bundled assets created by this step:

• `services/orchestrator/app/__init__.py`

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row `09` identifies the entry contract and preceding owners. Required predecessors: 08. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f12.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f50.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_09.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F10`: Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search. Assert extract item descriptions, quantities, urgency and machine hints with strict structured outputs.

• `F11`: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question. Assert use controller reason codes to choose code-owned bilingual questions.

• `F12`: Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied. Assert persist all messages and typed current state.

• `F18`: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling. Assert show persisted stages, wait reason and last update.

• `F50`: Duplicate events do not repeat publication; stale revision event is consumed as obsolete; checkpoint DDL never runs under runtime role. Assert use postgres checkpointer and db event inbox.

• `F67`: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative. Assert propagate request/enquiry/job ids.

`python tools/run_checks.py --prompt 09` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for deterministic durable workflow using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `09`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-09` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 09 deterministic durable workflow
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

045 prompts/F12-conversation-memory.md

F12 - Conversation memory (covered by 09)

PHASE: Coverage verification DEPENDS ON: 09 RISK: low

GOAL

Verify F12 coverage without implementing it twice. Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions.

READ FIRST

- prompts/09-deterministic-durable-workflow.md
- features/F12-conversation-memory.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 09 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 03, 09 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f12.py` is created by 09.

FILES TO MODIFY

`features/F12-conversation-memory.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 03, 09. Covered by 09. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f12.py` must assert: Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions. Negative cases: Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied.

Run `python tools/run_checks.py --feature F12`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise conversation memory using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F12-conversation-memory.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 09 with a separate trace.

COMMIT

```
docs(features): verify F12 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F12-conversation-memory.md

F12 - Conversation memory

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	09; foundations 03, 09
Affected flows	FLOW.md row 09

Scope

Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Persist all messages and typed current state; compact context using retained structured facts; resume by enquiry/revision; never mix customer sessions.
- [ ] Restart retains unresolved questions; old revision events are ignored and audited; foreign thread ID denied.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

messages, enquiry_revisions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f12.py`; command `python tools/run_checks.py --feature F12`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

046 prompts/F50-durable-workflows.md

F50 - Durable workflows (covered by 09)

PHASE: Coverage verification DEPENDS ON: 09 RISK: low

GOAL

Verify F50 coverage without implementing it twice. Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart.

READ FIRST

- prompts/09-deterministic-durable-workflow.md
- features/F50-durable-workflows.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 09 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 09 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f50.py` is created by 09.

FILES TO MODIFY

`features/F50-durable-workflows.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 09. Covered by 09. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Duplicate events do not repeat publication; stale revision event is consumed as obsolete; checkpoint DDL never runs under runtime role.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f50.py` must assert: Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart. Negative cases: Duplicate events do not repeat publication; stale revision event is consumed as obsolete; checkpoint DDL never runs under runtime role.

Run `python tools/run_checks.py --feature F50`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise durable workflows using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F50-durable-workflows.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 09 with a separate trace.

COMMIT

```
docs(features): verify F50 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F50-durable-workflows.md

F50 - Durable workflows

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	09; foundations 09
Affected flows	FLOW.md row 09

Scope

Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Use Postgres checkpointer and DB event inbox; thread key includes enquiry ID and active revision; serialize processing for one enquiry; recover after restart.
- [ ] Duplicate events do not repeat publication; stale revision event is consumed as obsolete; checkpoint DDL never runs under runtime role.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

workflow_events. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f50.py`; command `python tools/run_checks.py --feature F50`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

047 prompts/10-intake-specialist.md

10 - Intake specialist

PHASE: Orchestrator DEPENDS ON: 09 RISK: high

GOAL

Use shared OpenAI structured-output wrapper with IntakeResult schema. Send customer input in data field, preserve source spans, extract up to20 items. The model proposes quantities explicitly mentioned by customer; code validates positive units and asks when absent. Disallow status,approval,stock and compatibility fields. Detect change intent and compare against confirmed state. Store extracted proposals only after schema validation; confirmed machine fields are not silently overwritten.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F09-bilingual-chat.md
- features/F10-structured-extraction.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/app/specialists/intake.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/prompts/intake.txt`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/tests/features/test_f10.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_10.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def extract_intake(messages: list[MessageView], previous: IntakeResult | None, actor: ActorContext) -> IntakeResult.

Implementation sequence:

1. Use shared OpenAI structured-output wrapper with IntakeResult schema.
2. Send customer input in data field, preserve source spans, extract up to20 items.
3. The model proposes quantities explicitly mentioned by customer; code validates positive units and asks when absent.
4. Disallow status,approval,stock and compatibility fields.
5. Detect change intent and compare against confirmed state.

6. Store extracted proposals only after schema validation; confirmed machine fields are not silently overwritten.

Required feature behaviours:

• F09 Bilingual chat: Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence. Failure cases: Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message.

• F10 Structured extraction: Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity. Failure cases: Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search.

Exact package markers and bundled assets created by this step:

• `services/orchestrator/app/specialists/__init__.py`

FILES TO MODIFY

• `services/orchestrator/app/graph.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row `10` identifies the entry contract and preceding owners. Required predecessors: 09. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f10.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_10.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F09`: Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message. Assert accept arabic, english and mixed text.

• `F10`: Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search. Assert extract item descriptions, quantities, urgency and machine hints with strict structured outputs.

`python tools/run_checks.py --prompt 10` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for intake specialist using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `10`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing

reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-10` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 10 intake specialist
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

048 prompts/F10-structured-extraction.md

F10 - Structured extraction (covered by 10)

PHASE: Coverage verification DEPENDS ON: 10 RISK: low

GOAL

Verify F10 coverage without implementing it twice. Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity.

READ FIRST

- prompts/10-intake-specialist.md
- features/F10-structured-extraction.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 10 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 09, 10 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f10.py` is created by 10.

FILES TO MODIFY

`features/F10-structured-extraction.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 09, 10. Covered by 10. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f10.py` must assert: Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity. Negative cases: Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search.

Run `python tools/run_checks.py --feature F10`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise structured extraction using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F10-structured-extraction.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 10 with a separate trace.

COMMIT

```
docs(features): verify F10 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F10-structured-extraction.md

F10 - Structured extraction

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	10; foundations 09, 10
Affected flows	FLOW.md row 10

Scope

Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Extract item descriptions, quantities, urgency and machine hints with strict structured outputs; preserve source spans; missing quantity becomes a clarification rather than assumed order quantity.
- [ ] Unknown keys and invalid quantities reject the response; unsupported model output cannot progress to search.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_items, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f10.py`; command `python tools/run_checks.py --feature F10`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

049 prompts/11-parts-specialist-and-tools.md

11 - Parts specialist and tools

PHASE: Orchestrator DEPENDS ON: 10 RISK: high

GOAL

Register only identify,search_parts,part_detail and plan_fulfilment tools with typed arguments and fixed internal URL. Deterministic code identifies configuration and restricts candidate IDs. LLM may explain or propose clarification using those results; it cannot select incompatible parts or change quantities. Limit to four tool calls per stage and twelve graph transitions before needs_staff. Validate response catalogue version,request correlation and item ID; reject stale version.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F11-clarifying-questions.md
- features/F26-ambiguity-handling.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/app/specialists/parts.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/tools.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/search_client.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/tests/features/test_f26.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_11.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def run_parts(item: ItemState, context: ToolContext) -> PartsProposal; async def identify(request: IdentifyRequest) -> IdentifyResponse; async def search_parts(request: PartSearchRequest) -> PartSearchResponse; async def plan_fulfilment(request: FulfilmentRequest) -> FulfilmentResponse.

Implementation sequence:

1. Register only identify,search_parts,part_detail and plan_fulfilment tools with typed arguments and fixed internal URL.

2. Deterministic code identifies configuration and restricts candidate IDs.

3. LLM may explain or propose clarification using those results; it cannot select incompatible parts or change quantities.

4. Limit to four tool calls per stage and twelve graph transitions before needs_staff.

5. Validate response catalogue version,request correlation and item ID; reject stale version.

Required feature behaviours:

• F11 Clarifying questions: Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only. Failure cases: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.

• F26 Ambiguity handling: Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates. Failure cases: LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases.

FILES TO MODIFY

• `services/orchestrator/app/graph.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row `11` identifies the entry contract and preceding owners. Required predecessors: 10. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f26.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_11.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F11`: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question. Assert use controller reason codes to choose code-owned bilingual questions.

• `F26`: LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases. Assert use tested score and margin thresholds in code.

`python tools/run_checks.py --prompt 11` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for parts specialist and tools using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `11`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing

reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-11` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 11 parts specialist and tools
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

050 prompts/F26-ambiguity-handling.md

F26 - Ambiguity handling (covered by 11)

PHASE: Coverage verification DEPENDS ON: 11 RISK: low

GOAL

Verify F26 coverage without implementing it twice. Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates.

READ FIRST

- prompts/11-parts-specialist-and-tools.md
- features/F26-ambiguity-handling.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 11 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06, 11 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f26.py` is created by 11.

FILES TO MODIFY

`features/F26-ambiguity-handling.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 06, 11. Covered by 11. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f26.py` must assert: Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates. Negative cases: LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases.

Run `python tools/run_checks.py --feature F26`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise ambiguity handling using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F26-ambiguity-handling.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 11 with a separate trace.

COMMIT

```
docs(features): verify F26 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F26-ambiguity-handling.md

F26 - Ambiguity handling

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	11; foundations 06, 11
Affected flows	FLOW.md row 11

Scope

Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Use tested score and margin thresholds in code; ask customer to distinguish fuel filter versus separator; allow staff to select only verified candidates.
- [ ] LLM cannot override low confidence; unavailable candidate choice returns 409; thresholds evaluated on labelled cases.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_items. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f26.py`; command `python tools/run_checks.py --feature F26`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

051 prompts/12-deterministic-verification.md

12 - Deterministic verification

PHASE: Orchestrator DEPENDS ON: 11 RISK: high

GOAL

Implement named pure checks for current revision,configuration membership,candidate membership,source evidence,catalogue version,quantity and allocation. Recompute aggregate allocation across all items sharing a part and warehouse. Require nonnegative allocations,positive requested quantity and explicit shortage. Canonicalize facts using sorted JSON and decimal strings before SHA256. Persist one result for the current draft; failures contain stable codes and item IDs. Do not ask an LLM whether verification passed.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F28-compatibility-references.md
- features/F46-verification-gate.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/app/verification.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/search/tests/features/test_f28.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/tests/features/test_f46.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_12.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def verify_facts(facts: VerifiedFacts, evidence: VerificationEvidence) -> VerificationResult; async def verify_current(enquiry_id: UUID, revision: int) -> VerificationResult.

Implementation sequence:

1. Implement named pure checks for current revision,configuration membership,candidate membership,source evidence,catalogue version,quantity and allocation.
2. Recompute aggregate allocation across all items sharing a part and warehouse.
3. Require nonnegative allocations,positive requested quantity and explicit shortage.
4. Canonicalize facts using sorted JSON and decimal strings before SHA256.
5. Persist one result for the current draft; failures contain stable codes and item IDs.

6. Do not ask an LLM whether verification passed.

Required feature behaviours:

• F28 Compatibility references: Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot. Failure cases: Missing source file blocks evidence viewer and flags review; mismatched version fails verification.

• F46 Verification gate: Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results. Failure cases: Unknown part, incompatible part, stale catalogue, oversold plan and hidden extra item each fail a specific rule.

FILES TO MODIFY

• `services/orchestrator/app/graph.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row 12 identifies the entry contract and preceding owners. Required predecessors: 11. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/search/tests/features/test_f28.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f46.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_12.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F28`: Missing source file blocks evidence viewer and flags review; mismatched version fails verification. Assert resolve source_reference to version-specific document and page.

• `F46`: Unknown part, incompatible part, stale catalogue, oversold plan and hidden extra item each fail a specific rule. Assert check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds.

`python tools/run_checks.py --prompt 12` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for deterministic verification using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 12, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing

reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-12` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 12 deterministic verification
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

052 prompts/F28-compatibility-references.md

F28 - Compatibility references (covered by 12)

PHASE: Coverage verification DEPENDS ON: 12 RISK: low

GOAL

Verify F28 coverage without implementing it twice. Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot.

READ FIRST

- prompts/12-deterministic-verification.md
- features/F28-compatibility-references.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 12 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 02, 06, 12 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f28.py` is created by 12.

FILES TO MODIFY

`features/F28-compatibility-references.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 02, 06, 12. Covered by 12. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Missing source file blocks evidence viewer and flags review; mismatched version fails verification.

ACCEPTANCE TESTS

`services/search/tests/features/test_f28.py` must assert: Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot. Negative cases: Missing source file blocks evidence viewer and flags review; mismatched version fails verification.

Run `python tools/run_checks.py --feature F28`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise compatibility references using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F28-compatibility-references.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 12 with a separate trace.

COMMIT

```
docs(features): verify F28 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F28-compatibility-references.md

F28 - Compatibility references

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	12; foundations 02, 06, 12
Affected flows	FLOW.md row 12

Scope

Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Resolve source_reference to version-specific document and page; retain citation with every candidate and approval snapshot.
- [ ] Missing source file blocks evidence viewer and flags review; mismatched version fails verification.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

catalogue_evidence, part_compatibility. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f28.py`; command `python tools/run_checks.py --feature F28`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

053 prompts/F46-verification-gate.md

F46 - Verification gate (covered by 12)

PHASE: Coverage verification DEPENDS ON: 12 RISK: low

GOAL

Verify F46 coverage without implementing it twice. Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results.

READ FIRST

- prompts/12-deterministic-verification.md
- features/F46-verification-gate.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 12 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 12 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f46.py` is created by 12.

FILES TO MODIFY

`features/F46-verification-gate.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 12. Covered by 12. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unknown part, incompatible part, stale catalogue, oversold plan and hidden extra item each fail a specific rule.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f46.py` must assert: Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results. Negative cases: Unknown part, incompatible part, stale catalogue, oversold plan and hidden extra item each fail a specific rule.

Run `python tools/run_checks.py --feature F46`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise verification gate using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F46-verification-gate.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 12 with a separate trace.

COMMIT

```
docs(features): verify F46 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F46-verification-gate.md

F46 - Verification gate

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	12; foundations 12
Affected flows	FLOW.md row 12

Scope

Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Check item/configuration/candidate membership, provenance, catalogue version, selected quantities and allocation bounds; compute canonical fact hash; store named rule results.
- [ ] Unknown part, incompatible part, stale catalogue, oversold plan and hidden extra item each fail a specific rule.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

verification_results. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f46.py`; command `python tools/run_checks.py --feature F46`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

054 prompts/13-reply-specialist-and-renderer.md

13 - Reply specialist and renderer

PHASE: Orchestrator DEPENDS ON: 12 RISK: high

GOAL

Generate only allowed prose slots with strict schema. Build part/quantity/warehouse/estimate lines in code from verified structured data. Reject identifiers or numeric claims in free prose instead of treating known catalogue membership as sufficient. Escape HTML and render with code-owned layout. Never evaluate model-created Jinja. Staff edits use the same slot schema and rerun verification. Missing AI credit allows staff to author prose while retaining mandatory fact rendering.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F45-reply-drafting.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/app/specialists/reply.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/rendering.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/prompts/reply.txt`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/tests/features/test_f45.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_13.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def draft_reply(facts: VerifiedFacts, locale: Locale) -> ReplySlots; def render_reply(slots: ReplySlots, facts: VerifiedFacts, locale: Locale) -> str.

Implementation sequence:

1. Generate only allowed prose slots with strict schema.
2. Build part/quantity/warehouse/estimate lines in code from verified structured data.
3. Reject identifiers or numeric claims in free prose instead of treating known catalogue membership as sufficient.
4. Escape HTML and render with code-owned layout.

5. Never evaluate model-created Jinja.

6. Staff edits use the same slot schema and rerun verification.

7. Missing AI credit allows staff to author prose while retaining mandatory fact rendering.

Required feature behaviours:

• F45 Reply drafting: Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference. Failure cases: Unapproved critical facts never rendered; unsupported slots rejected; provider failure permits staff-authored controlled draft.

FILES TO MODIFY

• `services/orchestrator/app/graph.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row `13` identifies the entry contract and preceding owners. Required predecessors: 12. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f45.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_13.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F45`: Unapproved critical facts never rendered; unsupported slots rejected; provider failure permits staff-authored controlled draft. Assert generate greeting,explanation and closing as structured slots.

`python tools/run_checks.py --prompt 13` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for reply specialist and renderer using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `13`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-13` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 13 reply specialist and renderer
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

055 prompts/F45-reply-drafting.md

F45 - Reply drafting (covered by 13)

PHASE: Coverage verification DEPENDS ON: 13 RISK: low

GOAL

Verify F45 coverage without implementing it twice. Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference.

READ FIRST

- prompts/13-reply-specialist-and-renderer.md
- features/F45-reply-drafting.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 13 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 13 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f45.py` is created by 13.

FILES TO MODIFY

`features/F45-reply-drafting.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 13. Covered by 13. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unapproved critical facts never rendered; unsupported slots rejected; provider failure permits staff-authored controlled draft.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f45.py` must assert: Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference. Negative cases: Unapproved critical facts never rendered; unsupported slots rejected; provider failure permits staff-authored controlled draft.

Run `python tools/run_checks.py --feature F45`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise reply drafting using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F45-reply-drafting.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 13 with a separate trace.

COMMIT

```
docs(features): verify F45 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F45-reply-drafting.md

F45 - Reply drafting

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	13; foundations 13
Affected flows	FLOW.md row 13

Scope

Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Generate greeting,explanation and closing as structured slots; code inserts all numbers and allocations; choose Arabic or English from confirmed preference.
- [ ] Unapproved critical facts never rendered; unsupported slots rejected; provider failure permits staff-authored controlled draft.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

drafts. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f45.py`; command `python tools/run_checks.py --feature F45`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

056 prompts/14-approval-and-revision-transactions.md

14 - Approval and revision transactions

PHASE: Orchestrator DEPENDS ON: 13 RISK: high

GOAL

Lock enquiry; re-read actor permission via assigned reviewer or own demo capability; compare expected revision and current catalogue; refresh stock through search; reject changed plan. Reverify edited prose and facts. Insert approved decision,approved parts,one recommendation message and audit in one DB transaction. Rejection records reason but does not publish recommendation. Corrections increment revision and append immutable snapshot; no update to historical approval. Idempotency key binds request hash.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F39-review-actions.md
- features/F40-approval-invalidation.md
- features/F47-human-publication-gate.md
- features/F49-audit-trail.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/app/approval.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/app/revisions.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/tests/features/test_f40.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/tests/features/test_f47.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/acceptance/test_f49.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_14.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def approve(actor: ActorContext, request: ApprovalRequest, key: UUID) -> ApprovalResult; async def reject(actor: ActorContext, request: RejectionRequest, key: UUID) -> ReviewResult; async def revise(actor: ActorContext, request: RevisionRequest) -> EnquiryView.

Implementation sequence:

1. Lock enquiry; re-read actor permission via assigned reviewer or own demo capability; compare expected revision and current catalogue; refresh stock through search; reject changed plan.

2. Reverify edited prose and facts.

3. Insert approved decision,approved parts,one recommendation message and audit in one DB transaction.

4. Rejection records reason but does not publish recommendation.

5. Corrections increment revision and append immutable snapshot; no update to historical approval.

6. Idempotency key binds request hash.

Required feature behaviours:

• F39 Review actions: Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key. Failure cases: Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication.

• F40 Approval invalidation: Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval. Failure cases: Old approval cannot authorize new revision; late workflow update cannot overwrite approved current state.

• F47 Human publication gate: Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates. Failure cases: No worker/specialist write tool can publish; database rejects recommendation lacking matching approved decision.

• F49 Audit trail: Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads. Failure cases: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.

FILES TO MODIFY

• `services/orchestrator/app/graph.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Orchestrator; ARCHITECTURE.md section2. FLOW.md row `14` identifies the entry contract and preceding owners. Required predecessors: 13. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f40.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f47.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/acceptance/test_f49.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_14.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F39`: Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication. Assert approve only verified current draft.

• `F40`: Old approval cannot authorize new revision; late workflow update cannot overwrite approved current state. Assert bind approval to revision and verified fact hash.

• F47: No worker/specialist write tool can publish; database rejects recommendation lacking matching approved decision. Assert only approval service inserts recommendation messages.

• F49: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly. Assert append actor,request,entity,revision and minimal before/after metadata for ai suggestions and human changes.

`python tools/run_checks.py --prompt 14` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for approval and revision transactions using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `14`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-14` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): 14 approval and revision transactions
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

057 prompts/F40-approval-invalidation.md

F40 - Approval invalidation (covered by 14)

PHASE: Coverage verification DEPENDS ON: 14 RISK: low

GOAL

Verify F40 coverage without implementing it twice. Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval.

READ FIRST

- prompts/14-approval-and-revision-transactions.md
- features/F40-approval-invalidation.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 14 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 14 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f40.py` is created by 14.

FILES TO MODIFY

`features/F40-approval-invalidation.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 14. Covered by 14. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Old approval cannot authorize new revision; late workflow update cannot overwrite approved current state.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f40.py` must assert: Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval. Negative cases: Old approval cannot authorize new revision; late workflow update cannot overwrite approved current state.

Run `python tools/run_checks.py --feature F40`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise approval invalidation using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F40-approval-invalidation.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 14 with a separate trace.

COMMIT

```
docs(features): verify F40 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F40-approval-invalidation.md

F40 - Approval invalidation

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	14; foundations 14
Affected flows	FLOW.md row 14

Scope

Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Bind approval to revision and verified fact hash; any identity,quantity,selection or catalogue-sensitive change requires new draft and approval.
- [ ] Old approval cannot authorize new revision; late workflow update cannot overwrite approved current state.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_revisions, review_decisions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f40.py`; command `python tools/run_checks.py --feature F40`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

058 prompts/F47-human-publication-gate.md

F47 - Human publication gate (covered by 14)

PHASE: Coverage verification DEPENDS ON: 14 RISK: low

GOAL

Verify F47 coverage without implementing it twice. Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates.

READ FIRST

- prompts/14-approval-and-revision-transactions.md
- features/F47-human-publication-gate.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 14 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 14 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f47.py` is created by 14.

FILES TO MODIFY

`features/F47-human-publication-gate.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 14. Covered by 14. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No worker/specialist write tool can publish; database rejects recommendation lacking matching approved decision.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f47.py` must assert: Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates. Negative cases: No worker/specialist write tool can publish; database rejects recommendation lacking matching approved decision.

Run `python tools/run_checks.py --feature F47`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise human publication gate using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F47-human-publication-gate.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 14 with a separate trace.

COMMIT

```
docs(features): verify F47 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F47-human-publication-gate.md

F47 - Human publication gate

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	14; foundations 14
Affected flows	FLOW.md row 14

Scope

Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Only approval service inserts recommendation messages; lock enquiry and recheck role,revision and verification; unique publication index prevents duplicates.
- [ ] No worker/specialist write tool can publish; database rejects recommendation lacking matching approved decision.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

messages, review_decisions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f47.py`; command `python tools/run_checks.py --feature F47`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

059 prompts/F49-audit-trail.md

F49 - Audit trail (covered by 14)

PHASE: Coverage verification DEPENDS ON: 14 RISK: low

GOAL

Verify F49 coverage without implementing it twice. Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads.

READ FIRST

- prompts/14-approval-and-revision-transactions.md
- features/F49-audit-trail.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 14 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 01, 03, 14 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f49.py` is created by 14.

FILES TO MODIFY

`features/F49-audit-trail.md`: record actual verification results only.

HOW IT CONNECTS

Owner libs/partsiq_common; FLOW rows 01, 03, 14. Covered by 14. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.

ACCEPTANCE TESTS

`tests/acceptance/test_f49.py` must assert: Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads. Negative cases: Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.

Run `python tools/run_checks.py --feature F49`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise audit trail using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F49-audit-trail.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 14 with a separate trace.

COMMIT

```
docs(features): verify F49 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F49-audit-trail.md

F49 - Audit trail

Field	Value
Status	planned
Owner module(s)	libs/partsiq_common
Implementing prompts	14; foundations 01, 03, 14
Affected flows	FLOW.md row 14

Scope

Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Append actor,request,entity,revision and minimal before/after metadata for AI suggestions and human changes; redact secrets and full sensitive payloads.
- [ ] Application UPDATE,DELETE,TRUNCATE fail; audit write failure rolls back approval; database-owner powers documented honestly.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

audit_log. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f49.py`; command `python tools/run_checks.py --feature F49`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

060 prompts/15-worker-leases-and-sweeper.md

15 - Worker leases and sweeper

PHASE: Worker DEPENDS ON: 14 RISK: high

GOAL

Persist jobs before Redis push; sweeper scans available pending jobs and running jobs with expired leases. Claim FOR UPDATE SKIP LOCKED with fresh token and increment attempts. Heartbeat at10seconds with45second lease; use bounded concurrency one OCR and one PDF initially. Complete transaction requires matching token,current enquiry revision and session generation. Worker writes workflow_events for OCR completion,never calls orchestrator HTTP. Expose authenticated internal signing route only on Docker network. Graceful shutdown stops claiming and allows current lease completion.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F05-demo-reset.md
- features/F14-nameplate-ocr.md
- features/F51-background-jobs.md
- features/F52-retry-and-deduplication.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/worker/app/main.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/dispatcher.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/leases.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/sweeper.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/internal.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/tests/features/test_f51.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/worker/tests/features/test_f52.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `tests/foundations/test_15.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def claim_job(job_id: UUID, worker_id: str) -> JobLease | None; async def heartbeat(lease: JobLease) -> bool; async def finish_job(lease: JobLease, result: JobResult) -> bool; async def sweep_once(now: datetime) -> int.

Implementation sequence:

1. Persist jobs before Redis push; sweeper scans available pending jobs and running jobs with expired leases.

2. Claim FOR UPDATE SKIP LOCKED with fresh token and increment attempts.

3. Heartbeat at10seconds with45second lease; use bounded concurrency one OCR and one PDF initially.

4. Complete transaction requires matching token,current enquiry revision and session generation.

5. Worker writes workflow_events for OCR completion,never calls orchestrator HTTP.

6. Expose authenticated internal signing route only on Docker network.

7. Graceful shutdown stops claiming and allows current lease completion.

Required feature behaviours:

• F05 Demo reset: Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection. Failure cases: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.

• F14 Nameplate OCR: Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision. Failure cases: Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status.

• F51 Background jobs: Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup. Failure cases: Redis loss does not lose DB job; unavailable Redis rate limiter fails closed; heavy work never blocks HTTP intake.

• F52 Retry and deduplication: Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token. Failure cases: Expired worker cannot commit; permanent validation failure not retried; provider timeout marked uncertain and budget held conservatively.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Worker; ARCHITECTURE.md section2. FLOW.md row 15 identifies the entry contract and preceding owners. Required predecessors: 14. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/worker/tests/features/test_f51.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/worker/tests/features/test_f52.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_15.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• F05: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation. Assert lock session.

• F14: Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status. Assert read validated nameplate via openai vision.

• F51: Redis loss does not lose DB job; unavailable Redis rate limiter fails closed; heavy work never blocks HTTP intake. Assert persist job before redis dispatch.

• F52: Expired worker cannot commit; permanent validation failure not retried; provider timeout marked uncertain and budget held conservatively. Assert claim with for update skip locked.

`python tools/run_checks.py --prompt 15` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for worker leases and sweeper using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 15, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-15` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(worker): 15 worker leases and sweeper
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

• [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

061 prompts/F51-background-jobs.md

F51 - Background jobs (covered by 15)

PHASE: Coverage verification DEPENDS ON: 15 RISK: low

GOAL

Verify F51 coverage without implementing it twice. Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup.

READ FIRST

- prompts/15-worker-leases-and-sweeper.md
- features/F51-background-jobs.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 15 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 15 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/worker/tests/features/test_f51.py` is created by 15.

FILES TO MODIFY

`features/F51-background-jobs.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/worker; FLOW rows 15. Covered by 15. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Redis loss does not lose DB job; unavailable Redis rate limiter fails closed; heavy work never blocks HTTP intake.

ACCEPTANCE TESTS

`services/worker/tests/features/test_f51.py` must assert: Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup. Negative cases: Redis loss does not lose DB job; unavailable Redis rate limiter fails closed; heavy work never blocks HTTP intake.

Run `python tools/run_checks.py --feature F51`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise background jobs using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F51-background-jobs.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 15 with a separate trace.

COMMIT

```
docs(features): verify F51 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F51-background-jobs.md

F51 - Background jobs

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	15; foundations 15
Affected flows	FLOW.md row 15

Scope

Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Persist job before Redis dispatch; dispatcher claims by ID; sweeper redispatches pending/expired jobs; worker handles OCR,PDF,imports and cleanup.
- [ ] Redis loss does not lose DB job; unavailable Redis rate limiter fails closed; heavy work never blocks HTTP intake.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

jobs. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f51.py`; command `python tools/run_checks.py --feature F51`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

062 prompts/F52-retry-and-deduplication.md

F52 - Retry and deduplication (covered by 15)

PHASE: Coverage verification DEPENDS ON: 15 RISK: low

GOAL

Verify F52 coverage without implementing it twice. Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token.

READ FIRST

- prompts/15-worker-leases-and-sweeper.md
- features/F52-retry-and-deduplication.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 15 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 15 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/worker/tests/features/test_f52.py` is created by 15.

FILES TO MODIFY

`features/F52-retry-and-deduplication.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/worker; FLOW rows 15. Covered by 15. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Expired worker cannot commit; permanent validation failure not retried; provider timeout marked uncertain and budget held conservatively.

ACCEPTANCE TESTS

`services/worker/tests/features/test_f52.py` must assert: Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token. Negative cases: Expired worker cannot commit; permanent validation failure not retried; provider timeout marked uncertain and budget held conservatively.

Run `python tools/run_checks.py --feature F52`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise retry and deduplication using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F52-retry-and-deduplication.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 15 with a separate trace.

COMMIT

```
docs(features): verify F52 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F52-retry-and-deduplication.md

F52 - Retry and deduplication

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	15; foundations 15
Affected flows	FLOW.md row 15

Scope

Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Claim with FOR UPDATE SKIP LOCKED; assign lease token; heartbeat; retry transient errors with backoff up to max attempts; fence final writes by lease token.
- [ ] Expired worker cannot commit; permanent validation failure not retried; provider timeout marked uncertain and budget held conservatively.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

jobs. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f52.py`; command `python tools/run_checks.py --feature F52`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

063 prompts/16-nameplate-vision-job.md

16 - Nameplate vision job

PHASE: Worker DEPENDS ON: 15 RISK: high

GOAL

Decode validated image with pixel cap,strip EXIF,normalize orientation and prepare bounded image. Use OpenAI vision via shared client. Extract model_text,serial_text and issues only. Do not use model confidence as accuracy. Persist extraction,mark confirmation required and append workflow event using lease fencing. Partial OCR asks for manual entry. Handle budget exhaustion as recoverable staff/manual-input route.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F14-nameplate-ocr.md
- features/F15-ocr-confirmation.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/worker/app/jobs/nameplate.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/tests/features/test_f14.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_16.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def read_nameplate(job: JobLease, image: bytes) -> NameplateResult; def validate_image(image: bytes, media_type: str) -> ImageInfo.

Implementation sequence:

1. Decode validated image with pixel cap,strip EXIF,normalize orientation and prepare bounded image.
2. Use OpenAI vision via shared client.
3. Extract model_text,serial_text and issues only.
4. Do not use model confidence as accuracy.
5. Persist extraction,mark confirmation required and append workflow event using lease fencing.
6. Partial OCR asks for manual entry.
7. Handle budget exhaustion as recoverable staff/manual-input route.

Required feature behaviours:

• F14 Nameplate OCR: Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision. Failure cases: Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status.

• F15 OCR confirmation: Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text. Failure cases: Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409.

Exact package markers and bundled assets created by this step:

• `services/worker/app/jobs/__init__.py`

• `services/gateway/app/ocr_confirmation.py`: `async def confirm_extraction(actor: ActorContext, enquiry_id: UUID, request: ConfirmExtraction) -> EnquiryView: authorize owner or assigned reviewer; call orchestrator revise; atomically record confirmation after revision comparison. POST /v1/enquiries/{id}/extractions/{extraction_id}/confirm.`

FILES TO MODIFY

• `services/worker/app/dispatcher.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: register this prompt's route or handler.

HOW IT CONNECTS

Owner: Worker; ARCHITECTURE.md section2. FLOW.md row 16 identifies the entry contract and preceding owners. Required predecessors: 15. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/worker/tests/features/test_f14.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_16.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F14`: Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status. Assert read validated nameplate via openai vision.

• `F15`: Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409. Assert show cropped or original nameplate beside extracted identifiers.

`python tools/run_checks.py --prompt 16` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for nameplate vision job using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 16, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing

reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-16` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(worker): 16 nameplate vision job
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

064 prompts/F14-nameplate-ocr.md

F14 - Nameplate OCR (covered by 16)

PHASE: Coverage verification DEPENDS ON: 16 RISK: low

GOAL

Verify F14 coverage without implementing it twice. Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision.

READ FIRST

- prompts/16-nameplate-vision-job.md
- features/F14-nameplate-ocr.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 16 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 15, 16 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/worker/tests/features/test_f14.py` is created by 16.

FILES TO MODIFY

`features/F14-nameplate-ocr.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/worker; FLOW rows 15, 16. Covered by 16. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status.

ACCEPTANCE TESTS

`services/worker/tests/features/test_f14.py` must assert: Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision. Negative cases: Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status.

Run `python tools/run_checks.py --feature F14`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise nameplate ocr using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F14-nameplate-ocr.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 16 with a separate trace.

COMMIT

```
docs(features): verify F14 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F14-nameplate-ocr.md

F14 - Nameplate OCR

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	16; foundations 15, 16
Affected flows	FLOW.md row 16

Scope

Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Read validated nameplate via OpenAI vision; return model_text,serial_text,issues and source attachment; persist result only with current lease and revision.
- [ ] Unreadable fields remain null; vision instructions in photo are untrusted; timeout is bounded and creates recoverable status.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

extractions, jobs. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f14.py`; command `python tools/run_checks.py --feature F14`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

065 prompts/17-approved-enquiry-pdf.md

17 - Approved enquiry PDF

PHASE: Worker DEPENDS ON: 16 RISK: high

GOAL

Load approved immutable revision and decision; deny live draft export. Use trusted HTML template with local brand/fonts and Chromium. Disable external browser requests,JavaScript and arbitrary file access. Render Arabic with RTL and identifiers isolated LTR. Include synthetic-demo watermark and non-guaranteed transfer estimate. Hash output,upload private object and persist export record under lease token. Gateway returns signed URL only after authorization.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F61-enquiry-pdf.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install Jinja2==3.1.6 playwright==1.58.0 pypdf==6.8.0` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `services/worker/app/jobs/export_pdf.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/app/templates/enquiry_summary.html`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_17.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def export_enquiry(job: JobLease, request: ExportRequest) -> ExportResult; def render_export_html(snapshot: ApprovedSnapshot) -> str.

Implementation sequence:

1. Load approved immutable revision and decision; deny live draft export.
2. Use trusted HTML template with local brand/fonts and Chromium.
3. Disable external browser requests,JavaScript and arbitrary file access.
4. Render Arabic with RTL and identifiers isolated LTR.
5. Include synthetic-demo watermark and non-guaranteed transfer estimate.

6. Hash output,upload private object and persist export record under lease token.

7. Gateway returns signed URL only after authorization.

Required feature behaviours:

• F61 Enquiry PDF: Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download. Failure cases: Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title.

Exact package markers and bundled assets created by this step:

• `services/worker/app/static/brand/fonts/NotoNaskhArabic-OFL.txt`

• `services/worker/app/static/brand/fonts/NotoNaskhArabic.ttf`

• `services/worker/app/static/brand/fonts/NotoSans-OFL.txt`

• `services/worker/app/static/brand/fonts/NotoSans.ttf`

• `services/worker/app/static/brand/fonts/NotoSansArabic-OFL.txt`

• `services/worker/app/static/brand/fonts/NotoSansArabic.ttf`

• `services/worker/app/static/brand/logo-original.png`

FILES TO MODIFY

• `services/worker/app/dispatcher.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Worker; ARCHITECTURE.md section2. FLOW.md row `17` identifies the entry contract and preceding owners. Required predecessors: 16. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_17.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F61`: Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title. Assert export approved revision only.

`python tools/run_checks.py --prompt 17` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for approved enquiry pdf using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `17`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-17` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(worker): 17 approved enquiry pdf
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

066 prompts/18-frontend-shell-and-authentication.md

18 - Frontend shell and authentication

PHASE: Web DEPENDS ON: 17 RISK: high

GOAL

Create Next.js App Router shell from pinned package manifest. Public landing explains independent synthetic project; disable real-client signup. Supabase anonymous login supports demo; invited staff login supports operational roles. Keep all business API calls through gateway; only Auth and authorized private Storage transfer bypass it. Use bearer token in Authorization,not URL. Implement responsive skeleton,typed ApiError and request-ID display. Copy approved PNG icons and fonts from brand manifest.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F01-branded-landing.md
- features/F54-access-and-permissions.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `npm exec --yes --package=shadcn@4.20.1 -- shadcn --version`
- `npm exec --yes --package=pnpm@10.30.3 -- pnpm --version`
- `pnpm --dir apps/web add --save-exact next@16.2.12 react@19.2.6 react-dom@19.2.6 typescript@5.9.3 tailwindcss@4.1.18 @tailwindcss/postcss@4.1.18 postcss@8.5.6 @supabase/supabase-js@2.99.1 @supabase/ssr@0.8.0 next-intl@4.8.3 zod@4.6.5 vitest@4.1.11 @playwright/test@1.58.2 eslint@9.39.1 eslint-config-next@16.2.12 @testing-library/react@16.3.0 @testing-library/dom@10.4.1 @testing-library/jest-dom@6.9.1 @testing-library/user-event@14.6.1 jsdom@27.4.0 @vitejs/plugin-react@5.1.4 @types/react@19.2.14 @types/react-dom@19.2.3 @types/node@22.19.15 clsx@2.1.1 tailwind-merge@3.5.0 class-variance-authority@0.7.1 lucide-react@0.577.0 radix-ui@1.4.3`

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `apps/web/package.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/tsconfig.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/next.config.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/postcss.config.mjs`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/eslint.config.mjs`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/vitest.config.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/app/layout.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/app/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/app/login/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/styles/tokens.css`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/lib/api.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/lib/auth.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/components/BrandLogo.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/lib/contracts.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/pnpm-lock.yaml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/tests/setup.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/playwright.config.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/tests/features/f01.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `tests/foundations/test_18.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

export function BrandLogo(props: {compact?: boolean}): React.JSX.Element; export async function api<T>(path: string, options: RequestInit): Promise<T>; export async function startAnonymousDemo(): Promise<DemoSession>.

Implementation sequence:

1. Create Next.js App Router shell from pinned package manifest.

2. Public landing explains independent synthetic project; disable real-client signup.

3. Supabase anonymous login supports demo; invited staff login supports operational roles.

4. Keep all business API calls through gateway; only Auth and authorized private Storage transfer bypass it.

5. Use bearer token in Authorization,not URL.

6. Implement responsive skeleton,typed ApiError and request-ID display.

7. Copy approved PNG icons and fonts from brand manifest.

Required feature behaviours:

• F01 Branded landing: Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching. Failure cases: Missing repository link hides that action; logo has meaningful alternative text; no affiliation claim.

• F54 Access and permissions: Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup. Failure cases: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.

Exact package markers and bundled assets created by this step:

• `apps/web/public/brand/assets/PartsIQ-Asset-Pack/Preview.png`

• `apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-1024.png`

• `apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-128.png`

- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-144.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-152.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-16.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-180.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-192.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-24.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-256.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-32.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-384.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-48.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-512.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-64.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-transparent-96.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-1024.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-128.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-144.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-152.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-16.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-180.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-192.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-24.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-256.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-32.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-384.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-48.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-512.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-64.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/icons/partsiq-icon-white-96.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-1280.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-1740.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-320.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-transparent-640.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-1280.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-1740.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-320.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/logos/partsiq-logo-white-640.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-hdpi-72.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-mdpi-48.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xhdpi-96.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xxhdpi-144.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/android-launcher-xxxhdpi-192.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/google-play-512.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/mobile/ios-app-store-1024.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/social/partsiq-og-1200x630.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/social/partsiq-profile-1080.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/source/original-logo.png

- apps/web/public/brand/assets/PartsIQ-Asset-Pack/source/transparent-logo.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/android-chrome-192x192.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/android-chrome-512x512.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/apple-touch-icon.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-16x16.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-32x32.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-48x48.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon-96x96.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/favicon.ico
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/maskable-icon-192x192.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/maskable-icon-512x512.png
- apps/web/public/brand/assets/PartsIQ-Asset-Pack/web/site.webmanifest
- apps/web/public/brand/fonts/NotoNaskhArabic-OFL.txt
- apps/web/public/brand/fonts/NotoNaskhArabic.ttf
- apps/web/public/brand/fonts/NotoSans-OFL.txt
- apps/web/public/brand/fonts/NotoSans.ttf
- apps/web/public/brand/fonts/NotoSansArabic-OFL.txt
- apps/web/public/brand/fonts/NotoSansArabic.ttf
- apps/web/public/brand/logo-original.png

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Web; ARCHITECTURE.md section2. FLOW.md row 18 identifies the entry contract and preceding owners. Required predecessors: 17. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

- apps/web/tests/features/f01.test.tsx: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.
- tests/foundations/test_18.py: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.
- F01: Missing repository link hides that action; logo has meaningful alternative text; no affiliation claim. Assert explain the enquiry-to-approval journey.
- F54: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue. Assert verify supabase issuer,audience,signature,expiry.

python tools/run_checks.py --prompt 18 must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for frontend shell and authentication using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one

listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 18, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-18` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): 18 frontend shell and authentication
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

067 prompts/F01-branded-landing.md

F01 - Branded landing (covered by 18)

PHASE: Coverage verification DEPENDS ON: 18 RISK: low

GOAL

Verify F01 coverage without implementing it twice. Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching.

READ FIRST

- prompts/18-frontend-shell-and-authentication.md
- features/F01-branded-landing.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 18 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 18 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f01.test.tsx` is created by 18.

FILES TO MODIFY

`features/F01-branded-landing.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 18. Covered by 18. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Missing repository link hides that action; logo has meaningful alternative text; no affiliation claim.

ACCEPTANCE TESTS

`apps/web/tests/features/f01.test.tsx` must assert: Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching. Negative cases: Missing repository link hides that action; logo has meaningful alternative text; no affiliation claim.

Run `python tools/run_checks.py --feature F01`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise branded landing using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F01-branded-landing.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 18 with a separate trace.

COMMIT

```
docs(features): verify F01 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F01-branded-landing.md

F01 - Branded landing

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	18; foundations 18
Affected flows	FLOW.md row 18

Scope

Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Explain the enquiry-to-approval journey; show synthetic-demo notice; expose Start demo and architecture links; display the supplied bilingual logo without stretching.
- [ ] Missing repository link hides that action; logo has meaningful alternative text; no affiliation claim.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f01.test.tsx`; command `python tools/run_checks.py --feature F01`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

068 prompts/19-customer-chat-and-demo.md

19 - Customer chat and demo

PHASE: Web DEPENDS ON: 18 RISK: high

GOAL

Implement four scenario cards,chat,private upload progress,OCR confirmation and code-owned clarification display. Preserve unsent text when recoverable request fails. Poll enquiry with AbortController and bounded backoff; distinguish waiting customer/staff. Reset clears client cache and old generation. Only approved recommendation messages appear; customer messages and safe status/clarification messages remain visible without approval.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F02-demo-scenarios-and-assets.md
- features/F03-demo-role-switching.md
- features/F05-demo-reset.md
- features/F09-bilingual-chat.md
- features/F11-clarifying-questions.md
- features/F13-private-uploads.md
- features/F15-ocr-confirmation.md
- features/F18-progress-timeline.md
- features/F53-graceful-failures.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/app/chat/[id]/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/app/demo/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/chat/Chat.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/chat/UploadCard.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/chat/OcrConfirmation.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `apps/web/components/chat/Progress.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

• `tests/acceptance/test_f02.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/gateway/tests/features/test_f05.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/orchestrator/tests/features/test_f09.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/orchestrator/tests/features/test_f11.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `services/gateway/tests/features/test_f13.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `apps/web/tests/features/f15.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• `tests/foundations/test_19.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

export function Chat(props: {enquiryId: string}): React.JSX.Element; export function OcrConfirmation(props: {extraction: NameplateResult; onConfirm: (value: ConfirmExtraction) => Promise<void>}): React.JSX.Element.

Implementation sequence:

1. Implement four scenario cards,chat,private upload progress,OCR confirmation and code-owned clarification display.

2. Preserve unsent text when recoverable request fails.

3. Poll enquiry with AbortController and bounded backoff; distinguish waiting customer/staff.

4. Reset clears client cache and old generation.

5. Only approved recommendation messages appear; customer messages and safe status/clarification messages remain visible without approval.

Required feature behaviours:

• F02 Demo scenarios and assets: Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records. Failure cases: Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities.

• F03 Demo role switching: Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session. Failure cases: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.

• F05 Demo reset: Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection. Failure cases: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.

• F09 Bilingual chat: Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence. Failure cases: Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message.

• F11 Clarifying questions: Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only. Failure cases: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.

• F13 Private uploads: Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR. Failure cases: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.

• F15 OCR confirmation: Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text. Failure cases: Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409.

• F18 Progress timeline: Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff. Failure cases: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.

• F53 Graceful failures: Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input. Failure cases: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Web; ARCHITECTURE.md section2. FLOW.md row 19 identifies the entry contract and preceding owners. Required predecessors: 18. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f02.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/gateway/tests/features/test_f05.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f09.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f11.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/gateway/tests/features/test_f13.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `apps/web/tests/features/f15.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_19.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F02`: Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities. Assert seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage.

• `F03`: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404. Assert switch the presentation between customer and reviewer using the same anonymous identity and session.

• `F05`: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation. Assert lock session.

• `F09`: Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message. Assert accept arabic, english and mixed text.

• `F11`: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question. Assert use controller reason codes to choose code-owned bilingual questions.

• `F13`: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly. Assert create pending attachment and unique object path.

• `F15`: Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409. Assert show cropped or original nameplate beside extracted identifiers.

• `F18`: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling. Assert show persisted stages, wait reason and last update.

• F53: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported. Assert map stable error codes to bilingual recovery guidance.

`python tools/run_checks.py --prompt 19` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for customer chat and demo using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `19`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-19` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): 19 customer chat and demo
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

069 prompts/F02-demo-scenarios-and-assets.md

F02 - Demo scenarios and assets (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F02 coverage without implementing it twice. Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F02-demo-scenarios-and-assets.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 05, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f02.py` is created by 19.

FILES TO MODIFY

`features/F02-demo-scenarios-and-assets.md`: record actual verification results only.

HOW IT CONNECTS

Owner tools/catalog; FLOW rows 05, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities.

ACCEPTANCE TESTS

`tests/acceptance/test_f02.py` must assert: Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records. Negative cases: Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities.

Run `python tools/run_checks.py --feature F02`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise demo scenarios and assets using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F02-demo-scenarios-and-assets.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F02 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F02-demo-scenarios-and-assets.md

F02 - Demo scenarios and assets

Field	Value
Status	planned
Owner module(s)	tools/catalog
Implementing prompts	19; foundations 05, 19
Affected flows	FLOW.md row 19

Scope

Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Seed four deterministic scenarios: mixed-language photo, missing serial, incompatible part, local shortage; derive identifiers from seed 42; validate expected outcomes against generated records.
- [ ] Reject a scenario whose expected part lacks compatibility; reject stock claims not supported by fixture quantities.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

demo_sessions, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f02.py`; command `python tools/run_checks.py --feature F02`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

070 prompts/F05-demo-reset.md

F05 - Demo reset (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F05 coverage without implementing it twice. Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F05-demo-reset.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 07, 15, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/gateway/tests/features/test_f05.py` is created by 19.

FILES TO MODIFY

`features/F05-demo-reset.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/gateway; FLOW rows 07, 15, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.

ACCEPTANCE TESTS

`services/gateway/tests/features/test_f05.py` must assert: Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection. Negative cases: Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.

Run `python tools/run_checks.py --feature F05`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise demo reset using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F05-demo-reset.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F05 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F05-demo-reset.md

F05 - Demo reset

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	19; foundations 07, 15, 19
Affected flows	FLOW.md row 19

Scope

Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Lock session; increment generation; cancel unfinished jobs from prior generation; enqueue Storage cleanup; keep minimal audit; create fresh scenario enquiries on explicit selection.
- [ ] Late OCR from old generation cannot update current data; repeated reset idempotency key returns the same generation.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

demo_sessions, jobs. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f05.py`; command `python tools/run_checks.py --feature F05`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

071 prompts/F09-bilingual-chat.md

F09 - Bilingual chat (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F09 coverage without implementing it twice. Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F09-bilingual-chat.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 10, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f09.py` is created by 19.

FILES TO MODIFY

`features/F09-bilingual-chat.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 10, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f09.py` must assert: Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence. Negative cases: Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message.

Run `python tools/run_checks.py --feature F09`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise bilingual chat using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F09-bilingual-chat.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F09 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F09-bilingual-chat.md

F09 - Bilingual chat

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	19; foundations 10, 19
Affected flows	FLOW.md row 19

Scope

Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Accept Arabic, English and mixed text; preserve original message; detect preferred reply locale; render customer messages immediately after successful persistence.
- [ ] Empty and oversized input returns validation error; HTML is escaped; retry does not duplicate a message.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

messages, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f09.py`; command `python tools/run_checks.py --feature F09`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

072 prompts/F11-clarifying-questions.md

F11 - Clarifying questions (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F11 coverage without implementing it twice. Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F11-clarifying-questions.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 09, 11, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f11.py` is created by 19.

FILES TO MODIFY

`features/F11-clarifying-questions.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 09, 11, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f11.py` must assert: Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only. Negative cases: No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.

Run `python tools/run_checks.py --feature F11`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise clarifying questions using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F11-clarifying-questions.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F11 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F11-clarifying-questions.md

F11 - Clarifying questions

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	19; foundations 09, 11, 19
Affected flows	FLOW.md row 19

Scope

Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Use controller reason codes to choose code-owned bilingual questions; persist interrupt; associate answer with item; rerun affected resolution only.
- [ ] No final recommendation before approval; contradictory answer requires confirmation; repeat event creates one question.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

messages, enquiry_items. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f11.py`; command `python tools/run_checks.py --feature F11`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

073 prompts/F13-private-uploads.md

F13 - Private uploads (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F13 coverage without implementing it twice. Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F13-private-uploads.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 04, 08, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/gateway/tests/features/test_f13.py` is created by 19.

FILES TO MODIFY

`features/F13-private-uploads.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/gateway; FLOW rows 04, 08, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.

ACCEPTANCE TESTS

`services/gateway/tests/features/test_f13.py` must assert: Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR. Negative cases: Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.

Run `python tools/run_checks.py --feature F13`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise private uploads using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F13-private-uploads.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F13 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F13-private-uploads.md

F13 - Private uploads

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	19; foundations 04, 08, 19
Affected flows	FLOW.md row 19

Scope

Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Create pending attachment and unique object path; use signed upload; validate actual bytes and magic type after confirmation; strip EXIF and cap decoded pixels before OCR.
- [ ] Wrong MIME, oversized content, image decompression bomb, duplicate confirmation and nonexistent object handled explicitly.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_attachments. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f13.py`; command `python tools/run_checks.py --feature F13`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

074 prompts/F15-ocr-confirmation.md

F15 - OCR confirmation (covered by 19)

PHASE: Coverage verification DEPENDS ON: 19 RISK: low

GOAL

Verify F15 coverage without implementing it twice. Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text.

READ FIRST

- prompts/19-customer-chat-and-demo.md
- features/F15-ocr-confirmation.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 19 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 16, 19 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f15.test.tsx` is created by 19.

FILES TO MODIFY

`features/F15-ocr-confirmation.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 16, 19. Covered by 19. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409.

ACCEPTANCE TESTS

`apps/web/tests/features/f15.test.tsx` must assert: Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text. Negative cases: Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409.

Run `python tools/run_checks.py --feature F15`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise ocr confirmation using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F15-ocr-confirmation.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 19 with a separate trace.

COMMIT

```
docs(features): verify F15 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F15-ocr-confirmation.md

F15 - OCR confirmation

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	19; foundations 16, 19
Affected flows	FLOW.md row 19

Scope

Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Show cropped or original nameplate beside extracted identifiers; require user confirm or correction; record actor and time; preserve leading zeros in serial text.
- [ ] Unconfirmed OCR never authorizes matching; blank corrected serial triggers missing-serial flow; stale confirmation returns 409.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

extractions, enquiry_items. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f15.test.tsx`; command `python tools/run_checks.py --feature F15`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

075 prompts/20-staff-workspace-and-timeline.md

20 - Staff workspace and timeline

PHASE: Web DEPENDS ON: 19 RISK: high

GOAL

Render current review snapshot with revision badge,candidates,evidence references,allocation,draft and named verification results. Only current verified state enables approve. Post expected revision and idempotency key; on409 reload and require explicit new review. Edit prose uses structured fields. Reject requires reason. Real reviewers only see assignments; demo review remains ownsession. Add approved PDF action and display queue status.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F03-demo-role-switching.md
- features/F18-progress-timeline.md
- features/F27-match-explanations.md
- features/F32-warehouse-stock.md
- features/F37-review-workspace.md
- features/F39-review-actions.md
- features/F53-graceful-failures.md
- features/F61-enquiry-pdf.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/app/staff/inbox/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/app/staff/enquiries/[id]/page.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/review/Workspace.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/review/ApprovalControls.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/review/Timeline.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/tests/features/test_f03.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• apps/web/tests/features/f18.test.tsx: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• apps/web/tests/features/f27.test.tsx: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• services/search/tests/features/test_f32.py: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• apps/web/tests/features/f37.test.tsx: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• services/orchestrator/tests/features/test_f39.py: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• services/worker/tests/features/test_f61.py: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

• tests/foundations/test_20.py: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

export function Workspace(props: {enquiryId: string}): React.JSX.Element; export function ApprovalControls(props: {review: ReviewView; onChanged: () => Promise<void>}): React.JSX.Element.

Implementation sequence:

1. Render current review snapshot with revision badge,candidates,evidence references,allocation,draft and named verification results.

2. Only current verified state enables approve.

3. Post expected revision and idempotency key; on409 reload and require explicit new review.

4. Edit prose uses structured fields.

5. Reject requires reason.

6. Real reviewers only see assignments; demo review remains ownsession.

7. Add approved PDF action and display queue status.

Required feature behaviours:

• F03 Demo role switching: Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session. Failure cases: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.

• F18 Progress timeline: Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff. Failure cases: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.

• F27 Match explanations: Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose. Failure cases: Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability.

• F32 Warehouse stock: Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot. Failure cases: Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected.

• F37 Review workspace: Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages. Failure cases: Missing panels show explicit empty states; stale revision disables approval; unassigned reviewer receives 404.

• F39 Review actions: Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key. Failure cases: Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication.

• F53 Graceful failures: Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input. Failure cases: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.

• F61 Enquiry PDF: Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download. Failure cases: Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Web; ARCHITECTURE.md section2. FLOW.md row `20` identifies the entry contract and preceding owners. Required predecessors: 19. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f03.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `apps/web/tests/features/f18.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `apps/web/tests/features/f27.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/search/tests/features/test_f32.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `apps/web/tests/features/f37.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/orchestrator/tests/features/test_f39.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/worker/tests/features/test_f61.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_20.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F03`: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404. Assert switch the presentation between customer and reviewer using the same anonymous identity and session.

• `F18`: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling. Assert show persisted stages, wait reason and last update.

• `F27`: Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability. Assert show actual alias, machine configuration, supersession chain and fitment reference.

• `F32`: Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected. Assert show available quantity as on_hand minus reserved.

• `F37`: Missing panels show explicit empty states; stale revision disables approval; unassigned reviewer receives 404. Assert assemble current extraction, items, candidates, evidence, stock, draft and timeline.

• `F39`: Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication. Assert approve only verified current draft.

• `F53`: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported. Assert map stable error codes to bilingual recovery guidance.

• `F61`: Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title. Assert export approved revision only.

`python tools/run_checks.py --prompt 20` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for staff workspace and timeline using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `20`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-20` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): 20 staff workspace and timeline
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

076 prompts/F03-demo-role-switching.md

F03 - Demo role switching (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F03 coverage without implementing it twice. Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F03-demo-role-switching.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 07, 19, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/gateway/tests/features/test_f03.py` is created by 20.

FILES TO MODIFY

`features/F03-demo-role-switching.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/gateway; FLOW rows 07, 19, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.

ACCEPTANCE TESTS

`services/gateway/tests/features/test_f03.py` must assert: Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session. Negative cases: Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.

Run `python tools/run_checks.py --feature F03`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise demo role switching using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F03-demo-role-switching.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F03 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F03-demo-role-switching.md

F03 - Demo role switching

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	20; foundations 07, 19, 20
Affected flows	FLOW.md row 20

Scope

Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Switch the presentation between customer and reviewer using the same anonymous identity and session; authorize review only for enquiries belonging to that active session.
- [ ] Forged reviewer role never grants staff membership; another visitor's enquiry returns 404.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

demo_sessions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f03.py`; command `python tools/run_checks.py --feature F03`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

077 prompts/F18-progress-timeline.md

F18 - Progress timeline (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F18 coverage without implementing it twice. Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F18-progress-timeline.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 09, 19, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f18.test.tsx` is created by 20.

FILES TO MODIFY

`features/F18-progress-timeline.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 09, 19, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.

ACCEPTANCE TESTS

`apps/web/tests/features/f18.test.tsx` must assert: Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff. Negative cases: No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.

Run `python tools/run_checks.py --feature F18`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise progress timeline using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F18-progress-timeline.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F18 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F18-progress-timeline.md

F18 - Progress timeline

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	20; foundations 09, 19, 20
Affected flows	FLOW.md row 20

Scope

Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Show persisted stages, wait reason and last update; poll with bounded backoff; reconnect using server state; separate waiting for customer from waiting for staff.
- [ ] No fake completed progress; failed poll retains last state and offers retry; hidden tabs reduce polling.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

workflow_events, jobs. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f18.test.tsx`; command `python tools/run_checks.py --feature F18`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

078 prompts/F27-match-explanations.md

F27 - Match explanations (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F27 coverage without implementing it twice. Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F27-match-explanations.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f27.test.tsx` is created by 20.

FILES TO MODIFY

`features/F27-match-explanations.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 06, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability.

ACCEPTANCE TESTS

`apps/web/tests/features/f27.test.tsx` must assert: Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose. Negative cases: Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability.

Run `python tools/run_checks.py --feature F27`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise match explanations using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F27-match-explanations.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F27 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F27-match-explanations.md

F27 - Match explanations

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	20; foundations 06, 20
Affected flows	FLOW.md row 20

Scope

Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Show actual alias, machine configuration, supersession chain and fitment reference; derive explanations from search evidence rather than invented model prose.
- [ ] Absent evidence shown as unavailable; user HTML is escaped; scores never labelled calibrated probability.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_candidates. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f27.test.tsx`; command `python tools/run_checks.py --feature F27`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

079 prompts/F32-warehouse-stock.md

F32 - Warehouse stock (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F32 coverage without implementing it twice. Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F32-warehouse-stock.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 06, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/search/tests/features/test_f32.py` is created by 20.

FILES TO MODIFY

`features/F32-warehouse-stock.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/search; FLOW rows 06, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected.

ACCEPTANCE TESTS

`services/search/tests/features/test_f32.py` must assert: Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot. Negative cases: Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected.

Run `python tools/run_checks.py --feature F32`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise warehouse stock using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F32-warehouse-stock.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F32 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F32-warehouse-stock.md

F32 - Warehouse stock

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	20; foundations 06, 20
Affected flows	FLOW.md row 20

Scope

Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Show available quantity as on_hand minus reserved; include timestamp and warehouse name; keep display and planner using same snapshot.
- [ ] Negative stock blocked; stock unavailable never displayed as zero without explanation; unknown warehouse rejected.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

stock_levels, warehouses. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f32.py`; command `python tools/run_checks.py --feature F32`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

080 prompts/F37-review-workspace.md

F37 - Review workspace (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F37 coverage without implementing it twice. Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F37-review-workspace.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f37.test.tsx` is created by 20.

FILES TO MODIFY

`features/F37-review-workspace.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Missing panels show explicit empty states; stale revision disables approval; unassigned reviewer receives 404.

ACCEPTANCE TESTS

`apps/web/tests/features/f37.test.tsx` must assert: Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages. Negative cases: Missing panels show explicit empty states; stale revision disables approval; unassigned reviewer receives 404.

Run `python tools/run_checks.py --feature F37`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise review workspace using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F37-review-workspace.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F37 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F37-review-workspace.md

F37 - Review workspace

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	20; foundations 20
Affected flows	FLOW.md row 20

Scope

Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Assemble current extraction, items, candidates, evidence, stock, draft and timeline; show verification failures; separate staff-only notes from customer messages.
- [ ] Missing panels show explicit empty states; stale revision disables approval; unassigned reviewer receives 404.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

drafts, verification_results. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f37.test.tsx`; command `python tools/run_checks.py --feature F37`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

081 prompts/F39-review-actions.md

F39 - Review actions (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F39 coverage without implementing it twice. Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F39-review-actions.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 14, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/orchestrator/tests/features/test_f39.py` is created by 20.

FILES TO MODIFY

`features/F39-review-actions.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/orchestrator; FLOW rows 14, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication.

ACCEPTANCE TESTS

`services/orchestrator/tests/features/test_f39.py` must assert: Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key. Negative cases: Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication.

Run `python tools/run_checks.py --feature F39`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise review actions using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F39-review-actions.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F39 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F39-review-actions.md

F39 - Review actions

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	20; foundations 14, 20
Affected flows	FLOW.md row 20

Scope

Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Approve only verified current draft; edit prose through controlled schema then reverify; reject with reason; record actor and idempotency key.
- [ ] Double-click publishes once; concurrent edit defeats stale approval; failed verification blocks publication.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

review_decisions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f39.py`; command `python tools/run_checks.py --feature F39`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

082 prompts/F61-enquiry-pdf.md

F61 - Enquiry PDF (covered by 20)

PHASE: Coverage verification DEPENDS ON: 20 RISK: low

GOAL

Verify F61 coverage without implementing it twice. Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download.

READ FIRST

- prompts/20-staff-workspace-and-timeline.md
- features/F61-enquiry-pdf.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 20 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 17, 20 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/worker/tests/features/test_f61.py` is created by 20.

FILES TO MODIFY

`features/F61-enquiry-pdf.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/worker; FLOW rows 17, 20. Covered by 20. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title.

ACCEPTANCE TESTS

`services/worker/tests/features/test_f61.py` must assert: Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download. Negative cases: Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title.

Run `python tools/run_checks.py --feature F61`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise enquiry pdf using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F61-enquiry-pdf.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 20 with a separate trace.

COMMIT

```
docs(features): verify F61 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F61-enquiry-pdf.md

F61 - Enquiry PDF

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	20; foundations 17, 20
Affected flows	FLOW.md row 20

Scope

Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Export approved revision only; render logo,bilingual customer text,verified selections,estimates and references; embed Arabic fonts; store private object and signed download.
- [ ] Exclude notes and secrets; stale or unapproved revision rejected; long Arabic names wrap without clipping; no prices or quotation title.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

exports. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f61.py`; command `python tools/run_checks.py --feature F61`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

083 prompts/F16-photo-quality-feedback.md

F16 - Photo quality feedback

PHASE: worker DEPENDS ON: 16, 19 RISK: high

GOAL

Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning. Unsupported format rejected; dark or low-resolution sample gets guidance; no fabricated OCR confidence percentage.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F16-photo-quality-feedback.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/worker/tests/features/test_f16.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/worker/app/features/f16_photo_quality_feedback.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/PhotoQualityNotice.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f16_photo_quality_feedback.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

def assess_photo(request: F16Request) -> F16Result

`F16Request`: image_bytes: bytes; media_type: str.

`F16Result`: readable: bool; issues: list[str]; width: int; height: int.

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function PhotoQualityNotice(props: {value: F16Result | null; loading: boolean; error: ApiError | null; onAction: (request: F16Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning.

2. Unsupported format rejected; dark or low-resolution sample gets guidance; no fabricated OCR confidence percentage.

Required feature behaviours:

• F16 Photo quality feedback: Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning. Failure cases: Unsupported format rejected; dark or low-resolution sample gets guidance; no fabricated OCR confidence percentage.

Exact package markers and bundled assets created by this step:

• `services/worker/app/features/__init__.py`

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/components/chat/Chat.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/worker/app/dispatcher.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: worker; ARCHITECTURE.md section2. FLOW.md row `F16` identifies the entry contract and preceding owners. Required predecessors: 16, 19. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/worker/tests/features/test_f16.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F16`: Unsupported format rejected; dark or low-resolution sample gets guidance; no fabricated OCR confidence percentage. Assert check decodability, dimensions and blur heuristic before paid call.

`python tools/run_checks.py --prompt F16` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for photo quality feedback using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F16`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f16` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(worker): F16 photo quality feedback
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F16-photo-quality-feedback.md

F16 - Photo quality feedback

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	F16; foundations 16, 19
Affected flows	FLOW.md row F16

Scope

Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Check decodability, dimensions and blur heuristic before paid call; return actionable retake instructions; mark heuristic as advisory; allow a readable image despite noncritical blur warning.
- [ ] Unsupported format rejected; dark or low-resolution sample gets guidance; no fabricated OCR confidence percentage.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

extractions. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f16.py`; command `python tools/run_checks.py --feature F16`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

084 prompts/F17-multi-part-enquiries.md

F17 - Multi-part enquiries

PHASE: orchestrator DEPENDS ON: 09, 10, 11, 12, 13, 14, 19, 20 RISK: high

GOAL

Track up to 20 items with independent quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests. One unresolved item blocks final all-items approval; allocations across duplicate parts cannot double count available stock; zero quantity rejected.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F17-multi-part-enquiries.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/tests/features/test_f17.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/app/features/f17_multi_part_enquiries.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/ItemList.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f17_multi_part_enquiries.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def resolve_items(actor: ActorContext, request: F17Request) -> F17Result

`F17Request`: enquiry_id: UUID; expected_revision: int.

`F17Result`: items: list[ItemState]; unresolved_item_ids: list[UUID].

Endpoint: `POST /internal/items/resolve`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function ItemList(props: {value: F17Result | null; loading: boolean; error: ApiError | null; onAction: (request: F17Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Track up to 20 items with independent quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests.

2. One unresolved item blocks final all-items approval; allocations across duplicate parts cannot double count available stock; zero quantity rejected.

Required feature behaviours:

• F17 Multi-part enquiries: Track up to 20 items with independent quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests. Failure cases: One unresolved item blocks final all-items approval; allocations across duplicate parts cannot double count available stock; zero quantity rejected.

Exact package markers and bundled assets created by this step:

• `services/orchestrator/app/features/__init__.py`

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/components/chat/Chat.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Resolve each item independently but plan stock once across the combined requested quantities perpart. Allocate totals,then distribute to items in ordinal order so a repeated part cannot consume the same stock twice.

HOW IT CONNECTS

Owner: orchestrator; ARCHITECTURE.md section2. FLOW.md row `F17` identifies the entry contract and preceding owners. Required predecessors: 09, 10, 11, 12, 13, 14, 19, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f17.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F17`: One unresolved item blocks final all-items approval; allocations across duplicate parts cannot double count available stock; zero quantity rejected. Assert track up to 20 items with independent quantity, machine, candidate selection and status.

`python tools/run_checks.py --prompt F17` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for multi-part enquiries using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F17`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership

manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f17` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): F17 multi-part enquiries
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F17-multi-part-enquiries.md

F17 - Multi-part enquiries

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	F17; foundations 09, 10, 11, 12, 13, 14, 19, 20
Affected flows	FLOW.md row F17

Scope

Track up to 20 items with independent quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Track up to 20 items with independent quantity, machine, candidate selection and status; clarify individual items; verify aggregate allocation for repeated part requests.
- [ ] One unresolved item blocks final all-items approval; allocations across duplicate parts cannot double count available stock; zero quantity rejected.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_items, fulfilment_plans. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f17.py`; command `python tools/run_checks.py --feature F17`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

085 prompts/F19-enquiry-history.md

F19 - Enquiry history

PHASE: gateway DEPENDS ON: 07, 08, 19 RISK: high

GOAL

Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering. Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F19-enquiry-history.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f19.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f19_enquiry_history.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/HistoryList.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f19_enquiry_history.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def list_history(actor: ActorContext, request: F19Request) -> F19Result

`F19Request`: cursor: str | None = None; limit: int = 20; status: str | None = None.

`F19Result`: enquiries: list[EnquiryView]; next_cursor: str | None.

Endpoint: `GET /v1/enquiries`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function HistoryList(props: {value: F19Result | null; loading: boolean; error: ApiError | null; onAction: (request: F19Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.

2. Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

Required feature behaviours:

• F19 Enquiry history: Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering. Failure cases: Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.

Exact package markers and bundled assets created by this step:

• `services/gateway/app/features/__init__.py`

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Create apps/web/app/history/page.tsx and link from customer navigation. Request list_history via gateway; foreign or expired enquiry links produce scoped404.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F19` identifies the entry contract and preceding owners. Required predecessors: 07, 08, 19. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f19.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F19`: Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records. Assert return cursor-paginated owned history with status and created time.

`python tools/run_checks.py --prompt F19` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for enquiry history using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F19`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f19` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F19 enquiry history
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F19-enquiry-history.md

F19 - Enquiry history

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F19; foundations 07, 08, 19
Affected flows	FLOW.md row F19

Scope

Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Return cursor-paginated owned history with status and created time; filter active demo generation; link to full conversation; use stable created_at/id ordering.
- [ ] Foreign owner ID ignored; invalid cursor returns 422; equal timestamps do not duplicate or skip records.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f19.py`; command `python tools/run_checks.py --feature F19`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

086 prompts/F55-saved-machines.md

F55 - Saved machines

PHASE: gateway DEPENDS ON: 07, 19 RISK: high

GOAL

Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue. Duplicate label permitted with distinct ID; foreign ID denied; stale configuration never reused as authoritative.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F55-saved-machines.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f55.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f55_saved_machines.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/SavedMachines.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f55_saved_machines.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def save_machine(actor: ActorContext, request: F55Request) -> F55Result

`F55Request`: label: str; model_id: int; serial_text: str; idempotency_key: UUID.

`F55Result`: id: UUID; label: str; model_id: int; serial_text: str.

Endpoint: `POST /v1/machines`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function SavedMachines(props: {value: F55Result | null; loading: boolean; error: ApiError | null; onAction: (request: F55Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue.

2. Duplicate label permitted with distinct ID; foreign ID denied; stale configuration never reused as authoritative.

Required feature behaviours:

• F55 Saved machines: Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue. Failure cases: Duplicate label permitted with distinct ID; foreign ID denied; stale configuration never reused as authoritative.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Additional routes: GET /v1/machines returns own nonarchived saved machines; PATCH /v1/machines/{id} accepts label only; DELETE /v1/machines/{id} sets archived_at without deleting referenced data. Each checks active session generation and uses the common idempotency store.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F55` identifies the entry contract and preceding owners. Required predecessors: 07, 19. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f55.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F55`: Duplicate label permitted with distinct ID; foreign ID denied; stale configuration never reused as authoritative. Assert save confirmed model and serial with user label.

`python tools/run_checks.py --prompt F55` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for saved machines using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F55`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f55` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F55 saved machines
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F55-saved-machines.md

F55 - Saved machines

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F55; foundations 07, 19
Affected flows	FLOW.md row F55

Scope

Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Save confirmed model and serial with user label; list own active entries; archive entry; prefill a new enquiry but re-identify against current catalogue.
- [ ] Duplicate label permitted with distinct ID; foreign ID denied; stale configuration never reused as authoritative.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

saved_machines. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f55.py`; command `python tools/run_checks.py --feature F55`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

087 prompts/F56-repeat-enquiry.md

F56 - Repeat enquiry

PHASE: gateway DEPENDS ON: 09, 19, F19, F55 RISK: high

GOAL

Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again. Never copy approval,allocation,draft or signed file URL; source ownership required; repeated submission deduplicated.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F56-repeat-enquiry.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f56.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f56_repeat_enquiry.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/RepeatEnquiry.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f56_repeat_enquiry.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def repeat_enquiry(actor: ActorContext, request: F56Request) -> F56Result

`F56Request`: source_enquiry_id: UUID; idempotency_key: UUID.

`F56Result`: enquiry: EnquiryView.

Endpoint: `POST /v1/enquiries/{id}/repeat`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function RepeatEnquiry(props: {value: F56Result | null; loading: boolean; error: ApiError | null; onAction: (request: F56Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again.

2. Never copy approval,allocation,draft or signed file URL; source ownership required; repeated submission deduplicated.

Required feature behaviours:

• F56 Repeat enquiry: Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again. Failure cases: Never copy approval,allocation,draft or signed file URL; source ownership required; repeated submission deduplicated.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F56` identifies the entry contract and preceding owners. Required predecessors: 09, 19, F19, F55. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f56.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F56`: Never copy approval,allocation,draft or signed file URL; source ownership required; repeated submission deduplicated. Assert copy customer request and confirmed hints into a new enquiry.

`python tools/run_checks.py --prompt F56` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for repeat enquiry using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F56`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f56` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F56 repeat enquiry
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F56-repeat-enquiry.md

F56 - Repeat enquiry

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F56; foundations 09, 19
Affected flows	FLOW.md row F56

Scope

Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Copy customer request and confirmed hints into a new enquiry; issue new ID/revision; link origin in audit; run matching,stock and review again.
- [ ] Never copy approval,allocation,draft or signed file URL; source ownership required; repeated submission deduplicated.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries, enquiry_items. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f56.py`; command `python tools/run_checks.py --feature F56`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

088 prompts/F59-catalogue-versioning.md

F59 - Catalogue versioning

PHASE: catalog DEPENDS ON: 02, 05, 06 RISK: high

GOAL

Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action. Two publishers cannot leave two active versions; failed publish rolls back; old approval retains immutable evidence snapshot.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F59-catalogue-versioning.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `tests/acceptance/test_f59.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tools/catalog/catalog/versioning.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/CatalogueVersions.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f59_catalogue_versioning.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def publish_version(actor: ActorContext, request: F59Request) -> F59Result

`F59Request`: version_id: UUID; expected_active_version_id: UUID; idempotency_key: UUID.

`F59Result`: version_id: UUID; published_at: datetime; record_count: int.

Endpoint: `POST /v1/staff/catalogue/publish`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function CatalogueVersions(props: {value: F59Result | null; loading: boolean; error: ApiError | null; onAction: (request: F59Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action.

2. Two publishers cannot leave two active versions; failed publish rolls back; old approval retains immutable evidence snapshot.

Required feature behaviours:

• F59 Catalogue versioning: Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action. Failure cases: Two publishers cannot leave two active versions; failed publish rolls back; old approval retains immutable evidence snapshot.

• `services/worker/app/jobs/catalog_publish.py`: `async def publish_catalogue_job(lease: JobLease) -> JobResult: re-read initiating user catalogue_admin/admin membership; use catalogue role connection; invoke publish_version under publication lock; fence final job result.`

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `tools/catalog/catalog/__main__.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Gateway adapter services/gateway/app/features/f59_publish_routes.py validates catalogue_admin/admin,records idempotency and invokes restricted publication application; register in api_v1.py. Single transaction advisory lock710070 protects current-version switch. Publication uses catalogue DB credential held only by worker publication command,not gateway. Gateway queues publish request through worker internal route after signed actor verification.

• `services/gateway/app/api_v1.py`: register this prompt's route or handler.

• `services/worker/app/dispatcher.py`: register this prompt's route or handler.

HOW IT CONNECTS

Owner: catalog; ARCHITECTURE.md section2. FLOW.md row `F59` identifies the entry contract and preceding owners. Required predecessors: 02, 05, 06. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f59.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F59`: Two publishers cannot leave two active versions; failed publish rolls back; old approval retains immutable evidence snapshot. Assert snapshot validated catalogue canonically.

`python tools/run_checks.py --prompt F59` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for catalogue versioning using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F59`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership

manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f59` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(catalog): F59 catalogue versioning
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F59-catalogue-versioning.md

F59 - Catalogue versioning

Field	Value
Status	planned
Owner module(s)	tools/catalog
Implementing prompts	F59; foundations 02, 05, 06
Affected flows	FLOW.md row F59

Scope

Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Snapshot validated catalogue canonically; hash records; serialize publication; preserve numeric IDs; atomically retire old version and activate new; invalidate pending enquiries on next action.
- [ ] Two publishers cannot leave two active versions; failed publish rolls back; old approval retains immutable evidence snapshot.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

catalogue_versions, catalogue_records. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f59.py`; command `python tools/run_checks.py --feature F59`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

089 prompts/F57-catalogue-browser.md

F57 - Catalogue browser

PHASE: gateway DEPENDS ON: 02, 06, 18, F59 RISK: high

GOAL

Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect fitment and sources. No unpublished import rows exposed; missing number returns 404; literal query cannot become SQL.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F57-catalogue-browser.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f57.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f57_catalogue_browser.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/CatalogueBrowser.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f57_catalogue_browser.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def browse_catalogue(actor: ActorContext, request: F57Request) -> F57Result

`F57Request`: query: str | None = None; model_id: int | None = None; category: str | None = None; cursor: str | None = None; limit: int = 20.

`F57Result`: parts: list[PartView]; next_cursor: str | None; version_id: UUID.

Endpoint: `GET /v1/catalogue/parts`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function CatalogueBrowser(props: {value: F57Result | null; loading: boolean; error: ApiError | null; onAction: (request: F57Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect fitment and sources.

2. No unpublished import rows exposed; missing number returns 404; literal query cannot become SQL.

Required feature behaviours:

• F57 Catalogue browser: Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect fitment and sources. Failure cases: No unpublished import rows exposed; missing number returns 404; literal query cannot become SQL.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Create apps/web/app/staff/catalogue/page.tsx and route to CatalogueBrowser. Add GET /v1/catalogue/parts/{id} through existing search part_detail; do not return import staging rows.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F57` identifies the entry contract and preceding owners. Required predecessors: 02, 06, 18, F59. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f57.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F57`: No unpublished import rows exposed; missing number returns 404; literal query cannot become SQL. Assert read published catalogue with model/category/text filters.

`python tools/run_checks.py --prompt F57` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for catalogue browser using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F57`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f57` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F57 catalogue browser
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F57-catalogue-browser.md

F57 - Catalogue browser

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F57; foundations 02, 06, 18
Affected flows	FLOW.md row F57

Scope

Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect fitment and sources.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Read published catalogue with model/category/text filters; page results; show part detail and supersession; staff can inspect fitment and sources.
- [ ] No unpublished import rows exposed; missing number returns 404; literal query cannot become SQL.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

parts, equipment_models. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f57.py`; command `python tools/run_checks.py --feature F57`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

090 prompts/F58-catalogue-imports.md

F58 - Catalogue imports

PHASE: worker DEPENDS ON: 02, 05, 15, F59, F57 RISK: high

GOAL

Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish. Invalid batch changes no published data; file with macros/formulas rejected; 50000-row and 10MiB limits enforced; public demo cannot publish.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F58-catalogue-imports.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install openpyxl==3.1.5` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `services/worker/tests/features/test_f58.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/worker/app/features/f58_catalogue_imports.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/ImportWizard.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f58_catalogue_imports.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def validate_import(actor: ActorContext, request: F58Request) -> F58Result

`F58Request`: batch_id: UUID; lease: JobLease.

`F58Result`: valid: bool; row_count: int; errors: list[ImportErrorView].

Endpoint: `POST /v1/staff/imports`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function ImportWizard(props: {value: F58Result | null; loading: boolean; error: ApiError | null; onAction: (request: F58Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish.

2. Invalid batch changes no published data; file with macros/formulas rejected; 50000-row and 10MiB limits enforced; public demo cannot publish.

Required feature behaviours:

• F58 Catalogue imports: Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish. Failure cases: Invalid batch changes no published data; file with macros/formulas rejected; 50000-row and 10MiB limits enforced; public demo cannot publish.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/worker/app/dispatcher.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Gateway adapter must add POST /v1/staff/imports to create import_batches,GET /v1/staff/imports/{id} for validation report,and POST /v1/staff/imports/{id}/validate to enqueue through worker internal control. Public demo can inspect supplied sample validation reports but cannot alter shared catalogue. Explicit admin publication calls F59 only after valid status. Create services/gateway/app/features/f58_import_routes.py as the adapter; register in api_v1.py.

• `services/gateway/app/api_v1.py`: register this prompt's route or handler.

HOW IT CONNECTS

Owner: worker; ARCHITECTURE.md section2. FLOW.md row `F58` identifies the entry contract and preceding owners. Required predecessors: 02, 05, 15, F59, F57. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/worker/tests/features/test_f58.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F58`: Invalid batch changes no published data; file with macros/formulas rejected; 50000-row and 10MiB limits enforced; public demo cannot publish. Assert read csv utf-8 or xlsx without evaluating formulas.

`python tools/run_checks.py --prompt F58` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for catalogue imports using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F58`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing

reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f58` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(worker): F58 catalogue imports
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F58-catalogue-imports.md

F58 - Catalogue imports

Field	Value
Status	planned
Owner module(s)	services/worker
Implementing prompts	F58; foundations 02, 05, 15
Affected flows	FLOW.md row F58

Scope

Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Read CSV UTF-8 or XLSX without evaluating formulas; validate row schema,references,serial overlap,supersession cycles and source evidence; show dry-run report; explicit admin publish.
- [ ] Invalid batch changes no published data; file with macros/formulas rejected; 50000-row and 10MiB limits enforced; public demo cannot publish.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

import_batches, import_rows. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/worker/tests/features/test_f58.py`; command `python tools/run_checks.py --feature F58`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not

Date	Prompt	What changed	Commit
			executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

091 prompts/F60-missing-parts-queue.md

F60 - Missing-parts queue

PHASE: gateway DEPENDS ON: 03, 20 RISK: high

GOAL

Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation. Resolved status requires staff explanation; foreign item denied; empty search result is not fabricated stock.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F60-missing-parts-queue.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f60.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f60_missing_parts_queue.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/MissingPartsQueue.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f60_missing_parts_queue.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def update_missing_part(actor: ActorContext, request: F60Request) -> F60Result

`F60Request`: item_id: UUID; status: str; staff_note: str; expected_revision: int.

`F60Result`: id: UUID; status: str; updated_at: datetime.

Endpoint: `PATCH /v1/staff/missing-parts/{id}`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function MissingPartsQueue(props: {value: F60Result | null; loading: boolean; error: ApiError | null; onAction: (request: F60Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation.

2. Resolved status requires staff explanation; foreign item denied; empty search result is not fabricated stock.

Required feature behaviours:

• F60 Missing-parts queue: Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation. Failure cases: Resolved status requires staff explanation; foreign item denied; empty search result is not fabricated stock.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F60` identifies the entry contract and preceding owners. Required predecessors: 03, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f60.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F60`: Resolved status requires staff explanation; foreign item denied; empty search result is not fabricated stock. Assert record unresolved item once.

`python tools/run_checks.py --prompt F60` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for missing-parts queue using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F60`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f60` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F60 missing-parts queue
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F60-missing-parts-queue.md

F60 - Missing-parts queue

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F60; foundations 03, 20
Affected flows	FLOW.md row F60

Scope

Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Record unresolved item once; allow assigned staff to mark investigating,resolved or unavailable with note; link back to enquiry; no automatic catalogue creation.
- [ ] Resolved status requires staff explanation; foreign item denied; empty search result is not fabricated stock.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

missing_part_requests. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f60.py`; command `python tools/run_checks.py --feature F60`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

092 prompts/F35-stock-freshness.md

F35 - Stock freshness

PHASE: search DEPENDS ON: 06, 12, 14, 20 RISK: high

GOAL

Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible. Future timestamp flagged; changed stock returns 409 and reruns planning; stale stock never becomes a delivery guarantee.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F35-stock-freshness.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/search/tests/features/test_f35.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/search/app/features/f35_stock_freshness.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/StockFreshness.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f35_stock_freshness.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def check_stock_freshness(actor: ActorContext, request: F35Request) -> F35Result

`F35Request`: facts: VerifiedFacts; max_age_seconds: int.

`F35Result`: fresh: bool; changed_item_ids: list[UUID]; checked_at: datetime.

Endpoint: `POST /internal/stock/check`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function StockFreshness(props: {value: F35Result | null; loading: boolean; error: ApiError | null; onAction: (request: F35Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible.

2. Future timestamp flagged; changed stock returns 409 and reruns planning; stale stock never becomes a delivery guarantee.

Required feature behaviours:

• F35 Stock freshness: Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible. Failure cases: Future timestamp flagged; changed stock returns 409 and reruns planning; stale stock never becomes a delivery guarantee.

Exact package markers and bundled assets created by this step:

• `services/search/app/features/__init__.py`

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/approval.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: search; ARCHITECTURE.md section2. FLOW.md row `F35` identifies the entry contract and preceding owners. Required predecessors: 06, 12, 14, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/search/tests/features/test_f35.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F35`: Future timestamp flagged; changed stock returns 409 and reruns planning; stale stock never becomes a delivery guarantee. Assert compare stock timestamp with configurable freshness limit.

`python tools/run_checks.py --prompt F35` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for stock freshness using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F35`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f35` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(search): F35 stock freshness
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F35-stock-freshness.md

F35 - Stock freshness

Field	Value
Status	planned
Owner module(s)	services/search
Implementing prompts	F35; foundations 06, 12, 14, 20
Affected flows	FLOW.md row F35

Scope

Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Compare stock timestamp with configurable freshness limit; mark stale data in review; re-read current stock during approval; invalidate plan when allocation no longer possible.
- [ ] Future timestamp flagged; changed stock returns 409 and reruns planning; stale stock never becomes a delivery guarantee.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

stock_levels, fulfilment_plans. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/search/tests/features/test_f35.py`; command `python tools/run_checks.py --feature F35`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

093 prompts/F29-evidence-page-viewer.md

F29 - Evidence page viewer

PHASE: gateway DEPENDS ON: 04, 06, 20 RISK: high

GOAL

Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document. Customer receives reference only; arbitrary Storage key rejected; expired link can be renewed after authorization.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F29-evidence-page-viewer.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f29.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f29_evidence_page_viewer.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/EvidenceViewer.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f29_evidence_page_viewer.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def open_evidence(actor: ActorContext, request: F29Request) -> F29Result

`F29Request`: enquiry_id: UUID; evidence_id: UUID; expected_revision: int.

`F29Result`: url: str; page_number: int; expires_at: datetime; synthetic: bool.

Endpoint: `POST /v1/staff/evidence/open`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function EvidenceViewer(props: {value: F29Result | null; loading: boolean; error: ApiError | null; onAction: (request: F29Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document.

2. Customer receives reference only; arbitrary Storage key rejected; expired link can be renewed after authorization.

Required feature behaviours:

• F29 Evidence page viewer: Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document. Failure cases: Customer receives reference only; arbitrary Storage key rejected; expired link can be renewed after authorization.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F29` identifies the entry contract and preceding owners. Required predecessors: 04, 06, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f29.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F29`: Customer receives reference only; arbitrary Storage key rejected; expired link can be renewed after authorization. Assert authorize assigned reviewer or own demo reviewer.

`python tools/run_checks.py --prompt F29` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for evidence page viewer using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F29`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f29` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F29 evidence page viewer
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F29-evidence-page-viewer.md

F29 - Evidence page viewer

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F29; foundations 04, 06, 20
Affected flows	FLOW.md row F29

Scope

Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Authorize assigned reviewer or own demo reviewer; request short-lived worker-signed PDF URL; open exact cited page; label synthetic document.
- [ ] Customer receives reference only; arbitrary Storage key rejected; expired link can be renewed after authorization.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

catalogue_evidence. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f29.py`; command `python tools/run_checks.py --feature F29`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

094 prompts/F31-candidate-comparison.md

F31 - Candidate comparison

PHASE: web DEPENDS ON: 06, 20 RISK: high

GOAL

Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts. Foreign candidate or stale revision rejected; missing attribute shown as unavailable; keyboard navigation works.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F31-candidate-comparison.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/tests/features/f31.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `apps/web/lib/features/f31_candidate_comparison.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/CandidateComparison.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

export function compare_candidates(request: F31Request): F31Result

`F31Request`: candidates: CandidateView[]; selectedIds: number[].

`F31Result`: rows: ComparisonRow[].

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function CandidateComparison(props: {value: F31Result | null; loading: boolean; error: ApiError | null; onAction: (request: F31Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts.

2. Foreign candidate or stale revision rejected; missing attribute shown as unavailable; keyboard navigation works.

Required feature behaviours:

• F31 Candidate comparison: Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts. Failure cases: Foreign candidate or stale revision rejected; missing attribute shown as unavailable; keyboard navigation works.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: web; ARCHITECTURE.md section2. FLOW.md row `F31` identifies the entry contract and preceding owners. Required predecessors: 06, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `apps/web/tests/features/f31.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F31`: Foreign candidate or stale revision rejected; missing attribute shown as unavailable; keyboard navigation works. Assert compare two or three candidates from the same item/revision.

`python tools/run_checks.py --prompt F31` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for candidate comparison using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F31`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f31` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): F31 candidate comparison
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

• [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F31-candidate-comparison.md

F31 - Candidate comparison

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	F31; foundations 06, 20
Affected flows	FLOW.md row F31

Scope
Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts.

Out of scope
Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria
• [ ] Compare two or three candidates from the same item/revision; show number, names, evidence and warehouse availability; selecting a candidate sends its ID, not browser-authored facts.

• [ ] Foreign candidate or stale revision rejected; missing attribute shown as unavailable; keyboard navigation works.

• [ ] Test output and implementation commit recorded before status changes to done.

Data touched
enquiry_candidates. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests
`apps/web/tests/features/f31.test.tsx`; command `python tools/run_checks.py --feature F31`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence
Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

095 prompts/F38-correct-extracted-details.md

F38 - Correct extracted details

PHASE: orchestrator DEPENDS ON: 09, 14, 20 RISK: high

GOAL

Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after. Unknown field rejected; old revision returns 409; correction cannot directly set approved status or invent candidate facts.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F38-correct-extracted-details.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/tests/features/test_f38.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/app/features/f38_correct_extracted_details.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/CorrectionEditor.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f38_correct_extracted_details.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def correct_extraction(actor: ActorContext, request: F38Request) -> F38Result

`F38Request`: change: RevisionRequest.

`F38Result`: enquiry: EnquiryView.

Endpoint: `POST /internal/enquiries/correct`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function CorrectionEditor(props: {value: F38Result | null; loading: boolean; error: ApiError | null; onAction: (request: F38Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after.

2. Unknown field rejected; old revision returns 409; correction cannot directly set approved status or invent candidate facts.

Required feature behaviours:

• F38 Correct extracted details: Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after. Failure cases: Unknown field rejected; old revision returns 409; correction cannot directly set approved status or invent candidate facts.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Gateway adapter services/gateway/app/features/f38_correction_routes.py exposes POST /v1/staff/enquiries/{id}/corrections and calls internal orchestrator correction endpoint after require_access(review). Register in api_v1.py.

• `services/gateway/app/api_v1.py`: register this prompt's route or handler.

HOW IT CONNECTS

Owner: orchestrator; ARCHITECTURE.md section2. FLOW.md row `F38` identifies the entry contract and preceding owners. Required predecessors: 09, 14, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f38.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F38`: Unknown field rejected; old revision returns 409; correction cannot directly set approved status or invent candidate facts. Assert lock enquiry.

`python tools/run_checks.py --prompt F38` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for correct extracted details using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F38`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f38` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): F38 correct extracted details
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F38-correct-extracted-details.md

F38 - Correct extracted details

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	F38; foundations 09, 14, 20
Affected flows	FLOW.md row F38

Scope

Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Lock enquiry; compare expected revision; apply typed allowed corrections; increment revision; invalidate derived selections/draft/approval; enqueue recomputation; record before and after.
- [ ] Unknown field rejected; old revision returns 409; correction cannot directly set approved status or invent candidate facts.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_revisions, correction_records. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f38.py`; command `python tools/run_checks.py --feature F38`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

096 prompts/F42-assignments.md

F42 - Assignments

PHASE: gateway DEPENDS ON: 03, 07, 20 RISK: high

GOAL

Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped. Non-admin assignment denied; disabled/nonreviewer assignee rejected; previous reviewer loses access on next request.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F42-assignments.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f42.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f42_assignments.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/AssignmentControl.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f42_assignments.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def assign_enquiry(actor: ActorContext, request: F42Request) -> F42Result

`F42Request`: enquiry_id: UUID; assigned_to: UUID | None; expected_revision: int; idempotency_key: UUID.

`F42Result`: assigned_to: UUID | None; assigned_at: datetime.

Endpoint: `POST /v1/staff/enquiries/{id}/assignment`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function AssignmentControl(props: {value: F42Result | null; loading: boolean; error: ApiError | null; onAction: (request: F42Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped.

2. Non-admin assignment denied; disabled/nonreviewer assignee rejected; previous reviewer loses access on next request.

Required feature behaviours:

• F42 Assignments: Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped. Failure cases: Non-admin assignment denied; disabled/nonreviewer assignee rejected; previous reviewer loses access on next request.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Lock enquiry and compare expected revision for assignment; assignment itself does not change compatibility revision. Use assignment_events for history and revoke prior reviewer access on next request.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F42` identifies the entry contract and preceding owners. Required predecessors: 03, 07, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f42.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F42`: Non-admin assignment denied; disabled/nonreviewer assignee rejected; previous reviewer loses access on next request. Assert admin assigns active reviewer or unassigns.

`python tools/run_checks.py --prompt F42` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for assignments using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F42`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f42` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F42 assignments
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F42-assignments.md

F42 - Assignments

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F42; foundations 03, 07, 20
Affected flows	FLOW.md row F42

Scope

Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Admin assigns active reviewer or unassigns; persist assignment history atomically; access changes immediately; demo reviewer access remains session-scoped.
- [ ] Non-admin assignment denied; disabled/nonreviewer assignee rejected; previous reviewer loses access on next request.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries, assignment_events. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f42.py`; command `python tools/run_checks.py --feature F42`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

097 prompts/F41-internal-notes.md

F41 - Internal notes

PHASE: gateway DEPENDS ON: 03, 07, 20 RISK: high

GOAL

Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content. Empty or over-4000-character note rejected; customer route never includes note text; unassigned staff denied.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F41-internal-notes.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f41.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f41_internal_notes.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/InternalNotes.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f41_internal_notes.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def append_note(actor: ActorContext, request: F41Request) -> F41Result

`F41Request`: enquiry_id: UUID; body: str; idempotency_key: UUID.

`F41Result`: id: UUID; author_id: UUID; body: str; created_at: datetime.

Endpoint: `POST /v1/staff/enquiries/{id}/notes`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function InternalNotes(props: {value: F41Result | null; loading: boolean; error: ApiError | null; onAction: (request: F41Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content.

2. Empty or over-4000-character note rejected; customer route never includes note text; unassigned staff denied.

Required feature behaviours:

• F41 Internal notes: Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content. Failure cases: Empty or over-4000-character note rejected; customer route never includes note text; unassigned staff denied.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Use idempotency_records for note writes; notes are append-only in the application. GET /v1/staff/enquiries/{id}/notes lists in created_at/id order.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F41` identifies the entry contract and preceding owners. Required predecessors: 03, 07, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f41.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F41`: Empty or over-4000-character note rejected; customer route never includes note text; unassigned staff denied. Assert append reviewer notes with author and timestamp.

`python tools/run_checks.py --prompt F41` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for internal notes using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F41`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f41` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F41 internal notes
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F41-internal-notes.md

F41 - Internal notes

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F41; foundations 03, 07, 20
Affected flows	FLOW.md row F41

Scope

Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Append reviewer notes with author and timestamp; show only in staff workspace; prohibit customer/PDF exposure; escape rendered content.
- [ ] Empty or over-4000-character note rejected; customer route never includes note text; unassigned staff denied.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

internal_notes. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f41.py`; command `python tools/run_checks.py --feature F41`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

098 prompts/F43-overdue-indicators.md

F43 - Overdue indicators

PHASE: gateway DEPENDS ON: 20 RISK: high

GOAL

Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC. Approved/rejected items never overdue; daylight-saving display does not change deadline; thresholds not described as contractual SLA.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F43-overdue-indicators.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f43.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f43_overdue_indicators.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/OverdueBadge.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f43_overdue_indicators.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def classify_deadline(actor: ActorContext, request: F43Request) -> F43Result

`F43Request`: enquiry_id: UUID; now: datetime.

`F43Result`: due_at: datetime; overdue: bool; age_seconds: int.

Endpoint: `GET /v1/staff/enquiries/{id}/deadline`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function OverdueBadge(props: {value: F43Result | null; loading: boolean; error: ApiError | null; onAction: (request: F43Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC.

2. Approved/rejected items never overdue; daylight-saving display does not change deadline; thresholds not described as contractual SLA.

Required feature behaviours:

• F43 Overdue indicators: Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC. Failure cases: Approved/rejected items never overdue; daylight-saving display does not change deadline; thresholds not described as contractual SLA.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F43` identifies the entry contract and preceding owners. Required predecessors: 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f43.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F43`: Approved/rejected items never overdue; daylight-saving display does not change deadline; thresholds not described as contractual SLA. Assert set due_at using configurable urgent/normal thresholds.

`python tools/run_checks.py --prompt F43` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for overdue indicators using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F43`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f43` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F43 overdue indicators
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F43-overdue-indicators.md

F43 - Overdue indicators

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F43; foundations 20
Affected flows	FLOW.md row F43

Scope

Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Set due_at using configurable urgent/normal thresholds; display overdue only for unresolved states; sort overdue first when requested; store all times UTC.
- [ ] Approved/rejected items never overdue; daylight-saving display does not change deadline; thresholds not described as contractual SLA.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f43.py`; command `python tools/run_checks.py --feature F43`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

099 prompts/F44-conversation-summary.md

F44 - Conversation summary

PHASE: orchestrator DEPENDS ON: 09, 13, 20 RISK: high

GOAL

Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change. Outdated summary labelled stale; instructions inside messages ignored; exhausted budget leaves full conversation available.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F44-conversation-summary.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/tests/features/test_f44.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/app/features/f44_conversation_summary.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/ConversationSummary.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f44_conversation_summary.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def summarize_enquiry(actor: ActorContext, request: F44Request) -> F44Result

`F44Request`: enquiry_id: UUID; expected_revision: int.

`F44Result`: summary: str; source_message_ids: list[UUID]; revision: int.

Endpoint: `POST /internal/enquiries/summary`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function ConversationSummary(props: {value: F44Result | null; loading: boolean; error: ApiError | null; onAction: (request: F44Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change.

2. Outdated summary labelled stale; instructions inside messages ignored; exhausted budget leaves full conversation available.

Required feature behaviours:

• F44 Conversation summary: Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change. Failure cases: Outdated summary labelled stale; instructions inside messages ignored; exhausted budget leaves full conversation available.

FILES TO MODIFY

• `apps/web/app/staff/enquiries/[id]/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Gateway adapter services/gateway/app/features/f44_summary_routes.py exposes POST /v1/staff/enquiries/{id}/summary and calls internal summary endpoint. Return202 while processing,then GET review includes only current-revision summary. Register in api_v1.py.

• `services/gateway/app/api_v1.py`: register this prompt's route or handler.

HOW IT CONNECTS

Owner: orchestrator; ARCHITECTURE.md section2. FLOW.md row `F44` identifies the entry contract and preceding owners. Required predecessors: 09, 13, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f44.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F44`: Outdated summary labelled stale; instructions inside messages ignored; exhausted budget leaves full conversation available. Assert produce staff-only concise summary from authorized messages and confirmed fields.

`python tools/run_checks.py --prompt F44` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for conversation summary using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F44`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f44` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): F44 conversation summary
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F44-conversation-summary.md

F44 - Conversation summary

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	F44; foundations 09, 13, 20
Affected flows	FLOW.md row F44

Scope

Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Produce staff-only concise summary from authorized messages and confirmed fields; store message IDs and revision; use shared budget wrapper; regenerate on relevant change.
- [ ] Outdated summary labelled stale; instructions inside messages ignored; exhausted budget leaves full conversation available.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiry_summaries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f44.py`; command `python tools/run_checks.py --feature F44`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

100 prompts/F36-staff-inbox.md

F36 - Staff inbox

PHASE: gateway DEPENDS ON: 07, 20, F42, F43 RISK: high

GOAL

List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope. Unassigned real enquiries hidden from reviewer; admin can see all; filters do not leak global counts.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F36-staff-inbox.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f36.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f36_staff_inbox.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/InboxFilters.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f36_staff_inbox.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def list_inbox(actor: ActorContext, request: F36Request) -> F36Result

`F36Request`: cursor: str | None = None; limit: int = 20; status: str | None = None; priority: str | None = None; overdue: bool = False; query: str | None = None.

`F36Result`: enquiries: list[EnquiryView]; next_cursor: str | None; scoped_total: int.

Endpoint: `GET /v1/staff/enquiries`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function InboxFilters(props: {value: F36Result | null; loading: boolean; error: ApiError | null; onAction: (request: F36Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope.

2. Unassigned real enquiries hidden from reviewer; admin can see all; filters do not leak global counts.

Required feature behaviours:

• F36 Staff inbox: List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope. Failure cases: Unassigned real enquiries hidden from reviewer; admin can see all; filters do not leak global counts.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/app/staff/inbox/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F36` identifies the entry contract and preceding owners. Required predecessors: 07, 20, F42, F43. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f36.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F36`: Unassigned real enquiries hidden from reviewer; admin can see all; filters do not leak global counts. Assert list assigned operational enquiries and own active demo enquiries.

`python tools/run_checks.py --prompt F36` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for staff inbox using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F36`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f36` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production.

Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F36 staff inbox
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F36-staff-inbox.md

F36 - Staff inbox

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F36; foundations 07, 20
Affected flows	FLOW.md row F36

Scope

List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] List assigned operational enquiries and own active demo enquiries; filter status,priority,overdue and text; paginate stably; show counts in same access scope.
- [ ] Unassigned real enquiries hidden from reviewer; admin can see all; filters do not leak global counts.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f36.py`; command `python tools/run_checks.py --feature F36`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

101 prompts/F07-arabic-and-english-interface.md

F07 - Arabic and English interface

PHASE: web DEPENDS ON: 18, 19, 20 RISK: high

GOAL

Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi. Missing translation fails tests; language switch preserves enquiry ID and unsent draft; mixed Arabic identifiers remain readable.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F07-arabic-and-english-interface.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/tests/features/f07.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `apps/web/lib/features/f07_arabic_and_english_interface.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/LocaleSwitcher.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/messages/en.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/messages/ar.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/i18n/request.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

export function resolve_locale(request: F07Request): F07Result

`F07Request`: pathname: string; preference: "en" | "ar" | null.

`F07Result`: locale: "en" | "ar"; direction: "ltr" | "rtl".

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function LocaleSwitcher(props: {value: F07Result | null; loading: boolean; error: ApiError | null; onAction: (request: F07Request) => Promise<void>}): React.JSX.Element.` Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi.

2. Missing translation fails tests; language switch preserves enquiry ID and unsent draft; mixed Arabic identifiers remain readable.

Required feature behaviours:

• F07 Arabic and English interface: Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi. Failure cases: Missing translation fails tests; language switch preserves enquiry ID and unsent draft; mixed Arabic identifiers remain readable.

FILES TO MODIFY

• `apps/web/app/layout.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/next.config.ts`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Use locale cookie plus next-intl request configuration without renaming existing route files; locale switching preserves routes. The earlier locale-route-segment wording is replaced by cookie-based locale to avoid conflicting file paths.

HOW IT CONNECTS

Owner: web; ARCHITECTURE.md section2. FLOW.md row `F07` identifies the entry contract and preceding owners. Required predecessors: 18, 19, 20. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `apps/web/tests/features/f07.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F07`: Missing translation fails tests; language switch preserves enquiry ID and unsent draft; mixed Arabic identifiers remain readable. Assert add locale route segments and complete en/ar message catalogs.

`python tools/run_checks.py --prompt F07` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for arabic and english interface using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F07`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not

exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f07` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): F07 arabic and english interface
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F07-arabic-and-english-interface.md

F07 - Arabic and English interface

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	F07; foundations 18, 19, 20
Affected flows	FLOW.md row F07

Scope

Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Use a locale cookie and complete en/ar message catalogs; persist preference; set document lang and dir; isolate serials, numbers and part identifiers with bdi.
- [ ] Missing translation fails tests; language switch preserves enquiry ID and unsent draft; mixed Arabic identifiers remain readable.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

profiles. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f07.test.tsx`; command `python tools/run_checks.py --feature F07`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

102 prompts/F08-light-and-dark-themes.md

F08 - Light and dark themes

PHASE: web DEPENDS ON: 18 RISK: high

GOAL

Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme. No hydration mismatch; chart colors and focus rings meet contrast checks; storage-disabled browsers still work.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F08-light-and-dark-themes.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/tests/features/f08.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `apps/web/lib/features/f08_light_and_dark_themes.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/ThemeToggle.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

export function resolve_theme(request: F08Request): F08Result

`F08Request`: saved: "light" | "dark" | null; prefersDark: boolean.

`F08Result`: theme: "light" | "dark".

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function ThemeToggle(props: {value: F08Result | null; loading: boolean; error: ApiError | null; onAction: (request: F08Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme.

2. No hydration mismatch; chart colors and focus rings meet contrast checks; storage-disabled browsers still work.

Required feature behaviours:

• F08 Light and dark themes: Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme. Failure cases: No hydration mismatch; chart colors and focus rings meet contrast checks; storage-disabled browsers still work.

FILES TO MODIFY

• `apps/web/app/layout.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/styles/tokens.css`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: web; ARCHITECTURE.md section2. FLOW.md row `F08` identifies the entry contract and preceding owners. Required predecessors: 18. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `apps/web/tests/features/f08.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F08`: No hydration mismatch; chart colors and focus rings meet contrast checks; storage-disabled browsers still work. Assert use css semantic tokens.

`python tools/run_checks.py --prompt F08` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for light and dark themes using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F08`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f08` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): F08 light and dark themes
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F08-light-and-dark-themes.md

F08 - Light and dark themes

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	F08; foundations 18
Affected flows	FLOW.md row F08

Scope

Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Use CSS semantic tokens; initialize from saved preference then system setting; expose accessible toggle; honor reduced motion and browser color-scheme.
- [ ] No hydration mismatch; chart colors and focus rings meet contrast checks; storage-disabled browsers still work.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f08.test.tsx`; command `python tools/run_checks.py --feature F08`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

103 prompts/F06-responsive-layouts.md

F06 - Responsive layouts

PHASE: web DEPENDS ON: 18, 19, 20, F07, F08 RISK: high

GOAL

Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content. No page-wide horizontal overflow; zoom 200 percent remains usable; mobile dialogs retain visible close action.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F06-responsive-layouts.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `apps/web/tests/features/f06.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `apps/web/lib/features/f06_responsive_layouts.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/ResponsiveShell.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

export function get_layout(request: F06Request): F06Result

`F06Request`: width: number; reducedMotion: boolean.

`F06Result`: columns: 1 | 2 | 3; compact: boolean.

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function ResponsiveShell(props: {value: F06Result | null; loading: boolean; error: ApiError | null; onAction: (request: F06Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content.

2. No page-wide horizontal overflow; zoom 200 percent remains usable; mobile dialogs retain visible close action.

Required feature behaviours:

• F06 Responsive layouts: Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content. Failure cases: No page-wide horizontal overflow; zoom 200 percent remains usable; mobile dialogs retain visible close action.

FILES TO MODIFY

• `apps/web/app/layout.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `apps/web/styles/tokens.css`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: web; ARCHITECTURE.md section2. FLOW.md row `F06` identifies the entry contract and preceding owners. Required predecessors: 18, 19, 20, F07, F08. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `apps/web/tests/features/f06.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F06`: No page-wide horizontal overflow; zoom 200 percent remains usable; mobile dialogs retain visible close action. Assert support 360,768,1280 css-pixel widths.

`python tools/run_checks.py --prompt F06` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for responsive layouts using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F06`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f06` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(web): F06 responsive layouts
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F06-responsive-layouts.md

F06 - Responsive layouts

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	F06; foundations 18, 19, 20
Affected flows	FLOW.md row F06

Scope

Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Support 360,768,1280 CSS-pixel widths; stack review panels; horizontally scroll comparison tables within their region; preserve sticky primary review actions without covering content.
- [ ] No page-wide horizontal overflow; zoom 200 percent remains usable; mobile dialogs retain visible close action.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f06.test.tsx`; command `python tools/run_checks.py --feature F06`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

104 prompts/F48-input-isolation.md

F48 - Input isolation

PHASE: orchestrator DEPENDS ON: 10, 11, 12, 13, 16 RISK: high

GOAL

Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates. Prompt injection cannot call write tools,change roles or reveal secrets; tool result with rogue instructions remains inert.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F48-input-isolation.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/orchestrator/tests/features/test_f48.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/orchestrator/app/features/f48_input_isolation.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

def validate_tool_call(request: F48Request) -> F48Result

`F48Request`: tool_name: str; arguments_json: str; context: ToolContext.

`F48Result`: allowed: bool; reason_code: str | None.

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

Implementation sequence:

1. Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates.

2. Prompt injection cannot call write tools,change roles or reveal secrets; tool result with rogue instructions remains inert.

Required feature behaviours:

- F48 Input isolation: Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates. Failure cases: Prompt injection cannot call write tools,change roles or reveal secrets; tool result with rogue instructions remains inert.

FILES TO MODIFY

• `services/orchestrator/app/handlers.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/orchestrator/app/routes.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: orchestrator; ARCHITECTURE.md section2. FLOW.md row `F48` identifies the entry contract and preceding owners. Required predecessors: 10, 11, 12, 13, 16. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/orchestrator/tests/features/test_f48.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F48`: Prompt injection cannot call write tools,change roles or reveal secrets; tool result with rogue instructions remains inert. Assert treat customer text,ocr and imported evidence as data.

`python tools/run_checks.py --prompt F48` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for input isolation using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F48`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f48` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(orchestrator): F48 input isolation
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

• [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F48-input-isolation.md

F48 - Input isolation

Field	Value
Status	planned
Owner module(s)	services/orchestrator
Implementing prompts	F48; foundations 10, 11, 12, 13, 16
Affected flows	FLOW.md row F48

Scope

Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

• [ ] Treat customer text,OCR and imported evidence as data; separate trusted instructions; bound tool names,arguments,steps and output size; never execute model-produced templates.

• [ ] Prompt injection cannot call write tools,change roles or reveal secrets; tool result with rogue instructions remains inert.

• [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/orchestrator/tests/features/test_f48.py`; command `python tools/run_checks.py --feature F48`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

105 prompts/21-labelled-evaluations.md

21 - Labelled evaluations

PHASE: Quality DEPENDS ON: F48 RISK: high

GOAL

Build labelled expected outcomes from deterministic catalogue fixtures. Include all four demo scenarios,serial endpoints,Arabic normalization,typos,successor fitment,close scores,multiitem stock and malicious instructions. Separate deterministic checks from optional live-model evaluation. Deterministic suite must not require OpenAI credit. Record dataset hash,git SHA and model ID for live runs. Accuracy denominator excludes unreadable/unlabelled cases and reports abstention separately.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F62-labelled-evaluations.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `tools/evals/run.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/evals/cases.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/evals/test_scenarios.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/evals/__init__.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/acceptance/test_f62.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_21.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def evaluate_case(case: EvalCase, actual: EvalOutput) -> EvalResult; async def run_suite(cases: list[EvalCase], live: bool=False) -> EvalRun.

Implementation sequence:

1. Build labelled expected outcomes from deterministic catalogue fixtures.

2. Include all four demo scenarios,serial endpoints,Arabic normalization,typos,successor fitment,close scores,multiitem stock and malicious instructions.

3. Separate deterministic checks from optional live-model evaluation.

4. Deterministic suite must not require OpenAI credit.

5. Record dataset hash,git SHA and model ID for live runs.

6. Accuracy denominator excludes unreadable/unlabelled cases and reports abstention separately.

Required feature behaviours:

• F62 Labelled evaluations: Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls. Failure cases: No self-reported model confidence as accuracy; evaluation fixture disjoint from examples used in prompts; failures keep actual output.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Quality; ARCHITECTURE.md section2. FLOW.md row `21` identifies the entry contract and preceding owners. Required predecessors: F48. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f62.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_21.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F62`: No self-reported model confidence as accuracy; evaluation fixture disjoint from examples used in prompts; failures keep actual output. Assert include four demos plus arabic,typo,serial-boundary,ambiguity,stock and injection cases.

`python tools/run_checks.py --prompt 21` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for labelled evaluations using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `21`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-21` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated

later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(quality): 21 labelled evaluations
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

106 prompts/F62-labelled-evaluations.md

F62 - Labelled evaluations (covered by 21)

PHASE: Coverage verification DEPENDS ON: 21 RISK: low

GOAL

Verify F62 coverage without implementing it twice. Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls.

READ FIRST

- prompts/21-labelled-evaluations.md
- features/F62-labelled-evaluations.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 21 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 21 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f62.py` is created by 21.

FILES TO MODIFY

`features/F62-labelled-evaluations.md`: record actual verification results only.

HOW IT CONNECTS

Owner tools/evals; FLOW rows 21. Covered by 21. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No self-reported model confidence as accuracy; evaluation fixture disjoint from examples used in prompts; failures keep actual output.

ACCEPTANCE TESTS

`tests/acceptance/test_f62.py` must assert: Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls. Negative cases: No self-reported model confidence as accuracy; evaluation fixture disjoint from examples used in prompts; failures keep actual output.

Run `python tools/run_checks.py --feature F62`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise labelled evaluations using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F62-labelled-evaluations.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 21 with a separate trace.

COMMIT

```
docs(features): verify F62 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F62-labelled-evaluations.md

F62 - Labelled evaluations

Field	Value
Status	planned
Owner module(s)	tools/evals
Implementing prompts	21; foundations 21
Affected flows	FLOW.md row 21

Scope

Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Include four demos plus Arabic,typo,serial-boundary,ambiguity,stock and injection cases; record expected compatible set and clarification outcome; run deterministic suite without API calls.
- [ ] No self-reported model confidence as accuracy; evaluation fixture disjoint from examples used in prompts; failures keep actual output.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

eval_cases, eval_runs, eval_results. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f62.py`; command `python tools/run_checks.py --feature F62`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

107 prompts/23-observability-and-credit-limits.md

23 - Observability and credit limits

PHASE: Operations DEPENDS ON: 21 RISK: high

GOAL

Use common telemetry and AI wrapper from01; add optional Langfuse export with redacted payload and strict timeouts. Record latency,cost,token and stage outcomes locally. Allowance is manually confirmed free credit,not inferred from key. Require amount,expiry and acknowledged no-paid-fallback; default disabled. Reserve worst-case token/image usage atomically for each attempt; uncertain requests retain reservation. Throttle per session and daily quota. Trace exporter failure cannot bypass local accounting.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F64-latency-and-ai-cost.md
- features/F67-traces.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install langfuse==3.14.5` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `services/gateway/app/metrics.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/credit_allowance.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/check_readiness.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/acceptance/test_f64.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/acceptance/test_f67.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_23.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def get_metrics(actor: ActorContext, query: MetricsQuery) -> MetricsView; def validate_allowance(amount: Decimal, expires_at: datetime, paid_budget: Decimal) -> CreditAllowance; async def readiness_report() -> ReadinessReport.

Implementation sequence:

1. Use common telemetry and AI wrapper from01; add optional Langfuse export with redacted payload and strict timeouts.

2. Record latency,cost,token and stage outcomes locally.

3. Allowance is manually confirmed free credit,not inferred from key.

4. Require amount,expiry and acknowledged no-paid-fallback; default disabled.

5. Reserve worst-case token/image usage atomically for each attempt; uncertain requests retain reservation.

6. Throttle per session and daily quota.

7. Trace exporter failure cannot bypass local accounting.

Required feature behaviours:

• F64 Latency and AI cost: Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance. Failure cases: Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit.

• F67 Traces: Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable. Failure cases: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.

FILES TO MODIFY

• `libs/partsiq_common/partsiq_common/ai/budget.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `libs/partsiq_common/partsiq_common/ai/client.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `libs/partsiq_common/partsiq_common/telemetry.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Operations; ARCHITECTURE.md section2. FLOW.md row 23 identifies the entry contract and preceding owners. Required predecessors: 21. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f64.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/acceptance/test_f67.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_23.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F64`: Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit. Assert measure stages with monotonic clock.

• `F67`: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative. Assert propagate request/enquiry/job ids.

`python tools/run_checks.py --prompt 23` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for observability and credit limits using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 23, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-23` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(operations): 23 observability and credit limits
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

108 prompts/F64-latency-and-ai-cost.md

F64 - Latency and AI cost (covered by 23)

PHASE: Coverage verification DEPENDS ON: 23 RISK: low

GOAL

Verify F64 coverage without implementing it twice. Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance.

READ FIRST

- prompts/23-observability-and-credit-limits.md
- features/F64-latency-and-ai-cost.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 23 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 01, 23 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f64.py` is created by 23.

FILES TO MODIFY

`features/F64-latency-and-ai-cost.md`: record actual verification results only.

HOW IT CONNECTS

Owner libs/partsiq_common; FLOW rows 01, 23. Covered by 23. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit.

ACCEPTANCE TESTS

`tests/acceptance/test_f64.py` must assert: Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance. Negative cases: Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit.

Run `python tools/run_checks.py --feature F64`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise latency and ai cost using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F64-latency-and-ai-cost.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 23 with a separate trace.

COMMIT

```
docs(features): verify F64 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F64-latency-and-ai-cost.md

F64 - Latency and AI cost

Field	Value
Status	planned
Owner module(s)	libs/partsiq_common
Implementing prompts	23; foundations 01, 23
Affected flows	FLOW.md row 23

Scope

Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Measure stages with monotonic clock; log model/token usage and estimated cost; reserve confirmed free credit before each attempt; expose consumed/reserved allowance.
- [ ] Unknown price or expired credit blocks calls; concurrent reservations cannot exceed cap; API key validity never implies free credit.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

ai_usage, ai_budget_accounts. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f64.py`; command `python tools/run_checks.py --feature F64`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

109 prompts/F67-traces.md

F67 - Traces (covered by 23)

PHASE: Coverage verification DEPENDS ON: 23 RISK: low

GOAL

Verify F67 coverage without implementing it twice. Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable.

READ FIRST

- prompts/23-observability-and-credit-limits.md
- features/F67-traces.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 23 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 01, 09, 23 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f67.py` is created by 23.

FILES TO MODIFY

`features/F67-traces.md`: record actual verification results only.

HOW IT CONNECTS

Owner libs/partsiq_common; FLOW rows 01, 09, 23. Covered by 23. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.

ACCEPTANCE TESTS

`tests/acceptance/test_f67.py` must assert: Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable. Negative cases: No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.

Run `python tools/run_checks.py --feature F67`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise traces using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F67-traces.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 23 with a separate trace.

COMMIT

```
docs(features): verify F67 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F67-traces.md

F67 - Traces

Field	Value
Status	planned
Owner module(s)	libs/partsiq_common
Implementing prompts	23; foundations 01, 09, 23
Affected flows	FLOW.md row 23

Scope

Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Propagate request/enquiry/job IDs; capture stage timing and safe status; export optional Langfuse traces with payload redaction; operate normally when tracing unavailable.
- [ ] No raw key or full photo logged; exporter outage never breaks approval; budget accounting remains local and authoritative.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

ai_usage, audit_log. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f67.py`; command `python tools/run_checks.py --feature F67`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

110 prompts/F63-metrics-dashboard.md

F63 - Metrics dashboard

PHASE: gateway DEPENDS ON: 21, 23 RISK: high

GOAL

Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments. Zero denominator displays unavailable; no fabricated percentage; demo traffic never silently improves operational metrics.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F63-metrics-dashboard.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f63.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f63_metrics_dashboard.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/MetricsDashboard.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f63_metrics_dashboard.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def metrics_dashboard(actor: ActorContext, request: F63Request) -> F63Result

`F63Request`: query: MetricsQuery.

`F63Result`: metrics: MetricsView.

Endpoint: `GET /v1/staff/analytics`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function MetricsDashboard(props: {value: F63Result | null; loading: boolean; error: ApiError | null; onAction: (request: F63Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments.

2. Zero denominator displays unavailable; no fabricated percentage; demo traffic never silently improves operational metrics.

Required feature behaviours:

• F63 Metrics dashboard: Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments. Failure cases: Zero denominator displays unavailable; no fabricated percentage; demo traffic never silently improves operational metrics.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

Create apps/web/app/staff/analytics/page.tsx. Administrative evaluation metrics require admin; reviewers receive assignment-scoped operational counts only. Use duration aggregates from ai_usage and decision timestamps with explicit denominator.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F63` identifies the entry contract and preceding owners. Required predecessors: 21, 23. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f63.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F63`: Zero denominator displays unavailable; no fabricated percentage; demo traffic never silently improves operational metrics. Assert show volume,status,review time,clarification and labelled accuracy with denominators.

`python tools/run_checks.py --prompt F63` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for metrics dashboard using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F63`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f63` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated

later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F63 metrics dashboard
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F63-metrics-dashboard.md

F63 - Metrics dashboard

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F63; foundations 21, 23
Affected flows	FLOW.md row F63

Scope

Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Show volume,status,review time,clarification and labelled accuracy with denominators; separate demo traffic; filter time window; scope reviewer data to assignments.
- [ ] Zero denominator displays unavailable; no fabricated percentage; demo traffic never silently improves operational metrics.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

eval_runs, eval_results, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f63.py`; command `python tools/run_checks.py --feature F63`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

111 prompts/F65-correction-feedback.md

F65 - Correction feedback

PHASE: gateway DEPENDS ON: 14, 21, 23, F63 RISK: high

GOAL

List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits. Private text excluded from exported fixtures; duplicate case source detected; unreviewed correction cannot become ground truth.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F65-correction-feedback.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `services/gateway/tests/features/test_f65.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/app/features/f65_correction_feedback.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/components/features/CorrectionFeedback.tsx`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `apps/web/lib/features/f65_correction_feedback.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

async def curate_correction(actor: ActorContext, request: F65Request) -> F65Result

`F65Request`: correction_id: UUID; case_id: str; approved_expected: str.

`F65Result`: case_id: str; source_revision: int; approved_by: UUID.

Endpoint: `POST /v1/staff/corrections/{id}/eval-case`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

`export function CorrectionFeedback(props: {value: F65Result | null; loading: boolean; error: ApiError | null; onAction: (request: F65Request) => Promise<void>}): React.JSX.Element`. Display loading,empty,error and success states; submit only typed identifiers,never browser-authored compatibility facts.

Implementation sequence:

1. List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits.

2. Private text excluded from exported fixtures; duplicate case source detected; unreviewed correction cannot become ground truth.

Required feature behaviours:

• F65 Correction feedback: List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits. Failure cases: Private text excluded from exported fixtures; duplicate case source detected; unreviewed correction cannot become ground truth.

FILES TO MODIFY

• `apps/web/app/page.tsx`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/gateway/app/api_v1.py`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: gateway; ARCHITECTURE.md section2. FLOW.md row `F65` identifies the entry contract and preceding owners. Required predecessors: 14, 21, 23, F63. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f65.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F65`: Private text excluded from exported fixtures; duplicate case source detected; unreviewed correction cannot become ground truth. Assert list authorized before/after corrections.

`python tools/run_checks.py --prompt F65` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for correction feedback using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F65`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f65` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(gateway): F65 correction feedback
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F65-correction-feedback.md

F65 - Correction feedback

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	F65; foundations 14, 21, 23
Affected flows	FLOW.md row F65

Scope

List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] List authorized before/after corrections; let admin curate anonymized labelled eval cases; record reviewer and source revision; never auto-train on unreviewed edits.
- [ ] Private text excluded from exported fixtures; duplicate case source detected; unreviewed correction cannot become ground truth.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

correction_records, eval_cases. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f65.py`; command `python tools/run_checks.py --feature F65`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

112 prompts/22-security-and-failure-recovery-gate.md

22 - Security and failure recovery gate

PHASE: Quality DEPENDS ON: F65 RISK: high

GOAL

Run against disposable local Supabase including auth and storage. Exercise foreign sessions,assigned-only reviewer,reset races,expired JWT,upload bombs,prompt injection,Redis down,worker crash,stale lease,API timeout and simultaneous approvals. Assert no secret in browser bundle or logs. Block tests when DB hostname/project matches production. Retain actual command output and failed cases; never mark blocked tests passed.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F04-demo-isolation.md
- features/F53-graceful-failures.md
- features/F54-access-and-permissions.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `tests/integration/test_security_recovery.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/integration/test_rls.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/integration/test_budget_races.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/e2e/security.spec.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/gateway/tests/features/test_f04.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `apps/web/tests/features/f53.test.tsx`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `services/gateway/tests/features/test_f54.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_22.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def assert_disposable_environment(config: TestConfig) -> None; test suites instantiate existing service entry points without new business logic.

Implementation sequence:

1. Run against disposable local Supabase including auth and storage.

2. Exercise foreign sessions,assigned-only reviewer,reset races,expired JWT,upload bombs,prompt injection,Redis down,worker crash,stale lease,API timeout and simultaneous approvals.

3. Assert no secret in browser bundle or logs.

4. Block tests when DB hostname/project matches production.

5. Retain actual command output and failed cases; never mark blocked tests passed.

Required feature behaviours:

• F04 Demo isolation: Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default. Failure cases: Expired, reset and foreign sessions fail; missing identity context returns no rows.

• F53 Graceful failures: Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input. Failure cases: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.

• F54 Access and permissions: Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup. Failure cases: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Quality; ARCHITECTURE.md section2. FLOW.md row 22 identifies the entry contract and preceding owners. Required predecessors: F65. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `services/gateway/tests/features/test_f04.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `apps/web/tests/features/f53.test.tsx`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `services/gateway/tests/features/test_f54.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_22.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F04`: Expired, reset and foreign sessions fail; missing identity context returns no rows. Assert bind each session to verified anonymous supabase identity.

• `F53`: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported. Assert map stable error codes to bilingual recovery guidance.

• `F54`: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue. Assert verify supabase issuer,audience,signature,expiry.

`python tools/run_checks.py --prompt 22` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner

exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for security and failure recovery gate using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `22`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-22` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(quality): 22 security and failure recovery gate
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

113 prompts/F04-demo-isolation.md

F04 - Demo isolation (covered by 22)

PHASE: Coverage verification DEPENDS ON: 22 RISK: low

GOAL

Verify F04 coverage without implementing it twice. Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default.

READ FIRST

- prompts/22-security-and-failure-recovery-gate.md
- features/F04-demo-isolation.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 22 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 03, 07, 22 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/gateway/tests/features/test_f04.py` is created by 22.

FILES TO MODIFY

`features/F04-demo-isolation.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/gateway; FLOW rows 03, 07, 22. Covered by 22. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Expired, reset and foreign sessions fail; missing identity context returns no rows.

ACCEPTANCE TESTS

`services/gateway/tests/features/test_f04.py` must assert: Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default. Negative cases: Expired, reset and foreign sessions fail; missing identity context returns no rows.

Run `python tools/run_checks.py --feature F04`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise demo isolation using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F04-demo-isolation.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 22 with a separate trace.

COMMIT

```
docs(features): verify F04 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F04-demo-isolation.md

F04 - Demo isolation

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	22; foundations 03, 07, 22
Affected flows	FLOW.md row 22

Scope

Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Bind each session to verified anonymous Supabase identity; check generation and expiry on every read, write and worker completion; exclude demos from operational metrics by default.
- [ ] Expired, reset and foreign sessions fail; missing identity context returns no rows.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

demo_sessions, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f04.py`; command `python tools/run_checks.py --feature F04`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

114 prompts/F53-graceful-failures.md

F53 - Graceful failures (covered by 22)

PHASE: Coverage verification DEPENDS ON: 22 RISK: low

GOAL

Verify F53 coverage without implementing it twice. Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input.

READ FIRST

- prompts/22-security-and-failure-recovery-gate.md
- features/F53-graceful-failures.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 22 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 19, 20, 22 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `apps/web/tests/features/f53.test.tsx` is created by 22.

FILES TO MODIFY

`features/F53-graceful-failures.md`: record actual verification results only.

HOW IT CONNECTS

Owner apps/web; FLOW rows 19, 20, 22. Covered by 22. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.

ACCEPTANCE TESTS

`apps/web/tests/features/f53.test.tsx` must assert: Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input. Negative cases: No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.

Run `python tools/run_checks.py --feature F53`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise graceful failures using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F53-graceful-failures.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 22 with a separate trace.

COMMIT

```
docs(features): verify F53 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F53-graceful-failures.md

F53 - Graceful failures

Field	Value
Status	planned
Owner module(s)	apps/web
Implementing prompts	22; foundations 19, 20, 22
Affected flows	FLOW.md row 22

Scope

Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Map stable error codes to bilingual recovery guidance; distinguish validation,quota,service outage and staff-required states; preserve user input.
- [ ] No raw stack traces or keys; retry only safe idempotent action; search lexical fallback clearly reported.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

jobs, enquiries. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`apps/web/tests/features/f53.test.tsx`; command `python tools/run_checks.py --feature F53`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

115 prompts/F54-access-and-permissions.md

F54 - Access and permissions (covered by 22)

PHASE: Coverage verification DEPENDS ON: 22 RISK: low

GOAL

Verify F54 coverage without implementing it twice. Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup.

READ FIRST

- prompts/22-security-and-failure-recovery-gate.md
- features/F54-access-and-permissions.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 22 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 07, 18, 22 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `services/gateway/tests/features/test_f54.py` is created by 22.

FILES TO MODIFY

`features/F54-access-and-permissions.md`: record actual verification results only.

HOW IT CONNECTS

Owner services/gateway; FLOW rows 07, 18, 22. Covered by 22. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.

ACCEPTANCE TESTS

`services/gateway/tests/features/test_f54.py` must assert: Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup. Negative cases: Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.

Run `python tools/run_checks.py --feature F54`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise access and permissions using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F54-access-and-permissions.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 22 with a separate trace.

COMMIT

```
docs(features): verify F54 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F54-access-and-permissions.md

F54 - Access and permissions

Field	Value
Status	planned
Owner module(s)	services/gateway
Implementing prompts	22; foundations 07, 18, 22
Affected flows	FLOW.md row 22

Scope

Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Verify Supabase issuer,audience,signature,expiry; derive identity server-side; enforce assigned-only review; keep credentials server-side; disable public real-client signup.
- [ ] Forged role claim ignored; missing context denies rows; preview cannot reach prod database; demo cannot publish shared catalogue.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

profiles, staff_roles. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`services/gateway/tests/features/test_f54.py`; command `python tools/run_checks.py --feature F54`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

116 prompts/24-production-containers-and-caddy.md

24 - Production containers and Caddy

PHASE: Infrastructure DEPENDS ON: 22 RISK: high

GOAL

Build immutable service images; pin base image digests from verified registry manifest. Preserve gateway/search Dockerfile lineage and modify them explicitly. One search process avoids duplicate model memory. Separate worker,sweeper and orchestrator event-consumer commands; private Docker network for internal ports; publish only Caddy80/443. Add read-only rootfs where possible,tmpfs for PDF,bounded logs and resource limits measured during smoke. Use no Kubernetes or public Redis.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `docker-compose.prod.yml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `Caddyfile`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/orchestrator/Dockerfile`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `services/worker/Dockerfile`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `infra/verify_images.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `infra/images.lock.json`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_24.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def verify_images(compose_file: Path) -> list[str]: reject mutable application tags,missing healthchecks,public internal ports and embedded credentials.

Implementation sequence:

1. Build immutable service images; pin base image digests from verified registry manifest.
2. Preserve gateway/search Dockerfile lineage and modify them explicitly.

3. One search process avoids duplicate model memory.

4. Separate worker,sweeper and orchestrator event-consumer commands; private Docker network for internal ports; publish only Caddy80/443.

5. Add read-only rootfs where possible,tmpfs for PDF,bounded logs and resource limits measured during smoke.

6. Use no Kubernetes or public Redis.

FILES TO MODIFY

• `services/gateway/Dockerfile`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

• `services/search/Dockerfile`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Infrastructure; ARCHITECTURE.md section2. FLOW.md row `24` identifies the entry contract and preceding owners. Required predecessors: 22. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_24.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

`python tools/run_checks.py --prompt 24` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for production containers and caddy using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `24`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-24` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(infrastructure): 24 production containers and caddy
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

117 prompts/25-ci-and-controlled-deployment.md

25 - CI and controlled deployment

PHASE: Infrastructure DEPENDS ON: 24 RISK: high

GOAL

Pin GitHub actions by verified commit SHA. Run Python service tests in separate jobs or processes,frontend type/lint/test,local Supabase RLS integration and evals. Build/push GHCR images only on trusted branch. Deploy uses environment approval and OIDC-to-SSM with exact repository/environment subject. Keep deployment disabled until26 marks infrastructure ready. Never expose secrets to fork pull requests. Run backend migration separately before image activation.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F66-tests-and-ci.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `.github/workflows/ci.yml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `.github/workflows/deploy-backend.yml`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/ci_checks.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/acceptance/test_f66.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `tests/foundations/test_25.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def check_ci_configuration(root: Path) -> list[str]: inspect permissions,action pinning,prod-test protection and deployment guards.

Implementation sequence:

1. Pin GitHub actions by verified commit SHA.
2. Run Python service tests in separate jobs or processes,frontend type/lint/test,local Supabase RLS integration and evals.
3. Build/push GHCR images only on trusted branch.
4. Deploy uses environment approval and OIDC-to-SSM with exact repository/environment subject.

5. Keep deployment disabled until26 marks infrastructure ready.

6. Never expose secrets to fork pull requests.

7. Run backend migration separately before image activation.

Required feature behaviours:

• F66 Tests and CI: Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs. Failure cases: No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Infrastructure; ARCHITECTURE.md section2. FLOW.md row `25` identifies the entry contract and preceding owners. Required predecessors: 24. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f66.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `tests/foundations/test_25.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F66`: No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment. Assert run service-isolated python tests,frontend checks,rls integration and deterministic evals.

`python tools/run_checks.py --prompt 25` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for ci and controlled deployment using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `25`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-25` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(infrastructure): 25 ci and controlled deployment
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

118 prompts/F66-tests-and-ci.md

F66 - Tests and CI (covered by 25)

PHASE: Coverage verification DEPENDS ON: 25 RISK: low

GOAL

Verify F66 coverage without implementing it twice. Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs.

READ FIRST

- prompts/25-ci-and-controlled-deployment.md
- features/F66-tests-and-ci.md
- docs/CONSTRAINTS.md
- docs/CONTRACTS.md

PLAN CHECK

Confirm foundation 25 has executed and its acceptance evidence exists. If coverage is incomplete,complete that foundation's named implementation rather than creating a second owner.

PACKAGES

No installation here. Packages are installed by foundations 00, 25 and recorded in spec/dependency-receipts.json.

FILES TO CREATE

None. `tests/acceptance/test_f66.py` is created by 25.

FILES TO MODIFY

`features/F66-tests-and-ci.md`: record actual verification results only.

HOW IT CONNECTS

Owner infra; FLOW rows 00, 25. Covered by 25. Follow existing entry contracts in the foundation prompt.

CONSTRAINTS FOR THIS STEP

No duplicate domain function,route or state. No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment.

ACCEPTANCE TESTS

`tests/acceptance/test_f66.py` must assert: Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs. Negative cases: No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment.

Run `python tools/run_checks.py --feature F66`. Expected: exit0,all feature assertions pass,no required skips. Record actual test count.

MANUAL VERIFICATION

Exercise tests and ci using synthetic fixtures. Compare persisted and visible outcomes; inspect its failure cases. Do not mark done from source inspection alone.

DOC UPDATES

Update features/F66-tests-and-ci.md,docs/HANDOVER.md and docs/SESSION_LOG.md with foundation owner and verification output.

ROLLBACK

This coverage check changes documentation only. Revert its documentation commit if wrong; any code defect is fixed under foundation 25 with a separate trace.

COMMIT

```
docs(features): verify F66 foundation coverage
```

DONE WHEN

- [ ] Foundation evidence exists.
- [ ] Feature tests pass.
- [ ] Trace links exact command and commit.
- [ ] No duplicate implementation introduced.

Feature trace: features/F66-tests-and-ci.md

F66 - Tests and CI

Field	Value
Status	planned
Owner module(s)	infra
Implementing prompts	25; foundations 00, 25
Affected flows	FLOW.md row 25

Scope

Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

- [ ] Run service-isolated Python tests,frontend checks,RLS integration and deterministic evals; build images; require explicit environment approval for deployment; pin action SHAs.
- [ ] No prod DB in tests; no secret available to untrusted fork jobs; failed gate blocks deployment.
- [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f66.py`; command `python tools/run_checks.py --feature F66`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

119 prompts/26-aws-free-plan-deployment.md

26 - AWS free-plan deployment

PHASE: Infrastructure DEPENDS ON: 25 RISK: high

GOAL

Operator verifies account free-plan eligibility,remaining credits,expiry and region service availability before provisioning. No automatic paid upgrade. Default region ap-south-1 is an operational proposal selected for user proximity; verify account availability before use. Choose instance only after measured CPU/RAM of BGE-M3 and Chromium; free micro capacity is not assumed adequate. Use SSM without public SSH,encrypted volume and minimal IAM. Domain is operator input or no-cost DNS name; HTTPS required. Deploy by immutable release SHA with lock,healthcheck and rollback to previous images.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `infra/ec2/bootstrap.sh`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `infra/ec2/deploy.sh`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `infra/ec2/rollback.sh`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `infra/ec2/preflight.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_26.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

def preflight(account: AccountReceipt, sizing: SizingReceipt) -> DeploymentDecision; bootstrap.sh accepts a validated environment file path; deploy.sh accepts one hexadecimal release SHA.

Implementation sequence:

1. Operator verifies account free-plan eligibility,remaining credits,expiry and region service availability before provisioning.

2. No automatic paid upgrade.

3. Default region ap-south-1 is an operational proposal selected for user proximity; verify account availability before use.

4. Choose instance only after measured CPU/RAM of BGE-M3 and Chromium; free micro capacity is not assumed adequate.

5. Use SSM without public SSH,encrypted volume and minimal IAM.

6. Domain is operator input or no-cost DNS name; HTTPS required.

7. Deploy by immutable release SHA with lock,healthcheck and rollback to previous images.

FILES TO MODIFY

• `.github/workflows/deploy-backend.yml`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: Infrastructure; ARCHITECTURE.md section2. FLOW.md row `26` identifies the entry contract and preceding owners. Required predecessors: 25. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_26.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

`python tools/run_checks.py --prompt 26` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for aws free-plan deployment using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `26`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-26` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(infrastructure): 26 aws free-plan deployment
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

- [ ] Failure and permission cases pass.
- [ ] No forward runtime dependency introduced.
- [ ] Documentation and feature traces updated.
- [ ] Diff reviewed; no unrelated change.
- [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

120 prompts/27-vercel-and-hosted-acceptance.md

27 - Vercel and hosted acceptance

PHASE: Infrastructure DEPENDS ON: 26 RISK: high

GOAL

Create separate Vercel preview/dev and production environment bindings. Only publishable Supabase key and gateway URL are NEXT_PUBLIC. Verify CORS matches exact deployment origin; do not add wildcard credentials. Run four scenarios on hosted demo,check approval and PDF,measure actual timings and inspect mobile/Arabic. Supabase inactivity pausing and AWS credit expiry are documented limits. Deployed application is not claimed available unless real hosted checks pass.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `infra/vercel/README.md`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/e2e/hosted.spec.ts`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tools/hosted_acceptance.py`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `tests/foundations/test_27.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.

Typed entry contracts:

async def run_hosted_acceptance(base_url: str, api_url: str) -> HostedReceipt; reject non-HTTPS URLs outside localhost.

Implementation sequence:

1. Create separate Vercel preview/dev and production environment bindings.
2. Only publishable Supabase key and gateway URL are NEXT_PUBLIC.
3. Verify CORS matches exact deployment origin; do not add wildcard credentials.
4. Run four scenarios on hosted demo,check approval and PDF,measure actual timings and inspect mobile/Arabic.
5. Supabase inactivity pausing and AWS credit expiry are documented limits.
6. Deployed application is not claimed available unless real hosted checks pass.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: Infrastructure; ARCHITECTURE.md section2. FLOW.md row 27 identifies the entry contract and preceding owners. Required predecessors: 26. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/foundations/test_27.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

`python tools/run_checks.py --prompt 27` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for vercel and hosted acceptance using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step 27, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-27` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(infrastructure): 27 vercel and hosted acceptance
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

• [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

121 prompts/F68-recruiter-documentation.md

F68 - Recruiter documentation

PHASE: docs DEPENDS ON: 21, 23, 22, 24, 25, 26, 27 RISK: high

GOAL

Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools. No invented accuracy,uptime,client relationship or production adoption; distinguish implementation status from specification.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F68-recruiter-documentation.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

- `uv pip install python-docx==1.2.0` in the task-specific development environment. Add the identical pins to the owning workspace project; synchronize from the shared workspace lock for application execution.

Release dates and registry receipts are in spec/dependency-receipts.json. Use spec/python-requirements.lock and spec/pnpm-lock.yaml as resolution references; do not claim installing these packages proves application compatibility.

FILES TO CREATE

- `tests/acceptance/test_f68.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `docs/RECRUITER_GUIDE.md`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

Documentation contract: README.md and docs/RECRUITER_GUIDE.md

`F68Request`: source: validated hosted and test receipts.

`F68Result`: README.md and docs/RECRUITER_GUIDE.md.

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

Implementation sequence:

1. Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools.

2. No invented accuracy,uptime,client relationship or production adoption; distinguish implementation status from specification.

Required feature behaviours:

• F68 Recruiter documentation: Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools. Failure cases: No invented accuracy,uptime,client relationship or production adoption; distinguish implementation status from specification.

FILES TO MODIFY

• `README.md`: connect the new implementation through the existing owner; preserve callers and add regression coverage.

HOW IT CONNECTS

Owner: docs; ARCHITECTURE.md section2. FLOW.md row `F68` identifies the entry contract and preceding owners. Required predecessors: 21, 23, 22, 24, 25, 26, 27. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

• `tests/acceptance/test_f68.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.

• `F68`: No invented accuracy,uptime,client relationship or production adoption; distinguish implementation status from specification. Assert write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations.

`python tools/run_checks.py --prompt F68` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for recruiter documentation using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F68`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f68` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(docs): F68 recruiter documentation
```

DONE WHEN

• [ ] Exact paths and contracts implemented.

• [ ] Listed tests executed with actual evidence.

• [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F68-recruiter-documentation.md

F68 - Recruiter documentation

Field	Value
Status	planned
Owner module(s)	docs
Implementing prompts	F68; foundations 21, 22, 23, 24, 25, 26, 27
Affected flows	FLOW.md row F68

Scope
Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools.

Out of scope
Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria
• [ ] Write concise overview,real architecture,quick start,demo paths,tests,trade-offs and limitations; link measured evidence; explain deterministic decisions and specialist tools.

• [ ] No invented accuracy,uptime,client relationship or production adoption; distinguish implementation status from specification.

• [ ] Test output and implementation commit recorded before status changes to done.

Data touched
none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests
`tests/acceptance/test_f68.py`; command `python tools/run_checks.py --feature F68`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence
Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

122 prompts/F69-demo-walkthrough.md

F69 - Demo walkthrough

PHASE: docs DEPENDS ON: 27, F68 RISK: high

GOAL

Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions. Do not claim video exists until recorded; use synthetic data; show failure handling and reset; verify links before sending recruiters.

READ FIRST

- AGENTS.md
- docs/HANDOVER.md
- docs/CONSTRAINTS.md
- docs/ARCHITECTURE.md sections 2 and 4
- docs/TECH_SPEC.md
- docs/CONTRACTS.md
- docs/FLOW.md
- spec/file-ownership.json
- features/F69-demo-walkthrough.md

PLAN CHECK

State the exact files to create/modify, packages, validation commands and risk before implementation. The approved build sequence authorizes these listed changes. Ask only if a new decision changes locked scope, introduces a destructive action or crosses a boundary. Do not repeatedly ask for approval already granted. Verify prerequisite prompts are complete, not merely present as documents.

PACKAGES

No new package. Use the existing exact pins and committed locks; do not install a second library for this feature.

FILES TO CREATE

- `tests/acceptance/test_f69.py`: acceptance source for the cases below; arrange isolated fixtures, invoke the production entry point, assert persisted and visible outcomes.
- `docs/DEMO_SCRIPT.md`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.
- `docs/RECORDING_CHECKLIST.md`: implement the responsibility specified in the ordered logic and typed contracts below; no unregistered duplicate implementation.

Typed entry contracts:

Documentation contract: docs/DEMO_SCRIPT.md and docs/RECORDING_CHECKLIST.md

`F69Request`: source: four verified demo scenarios.

`F69Result`: docs/DEMO_SCRIPT.md and docs/RECORDING_CHECKLIST.md.

Endpoint: `none`. For a public route whose owner is worker/catalogue, gateway validates role and records a durable job/event; it never executes the job inline. Internal routes use the internal token and propagated actor context.

Implementation sequence:

1. Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions.

2. Do not claim video exists until recorded; use synthetic data; show failure handling and reset; verify links before sending recruiters.

Required feature behaviours:

- F69 Demo walkthrough: Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions. Failure cases: Do

not claim video exists until recorded; use synthetic data; show failure handling and reset; verify links before sending recruiters.

FILES TO MODIFY

No existing runtime file requires modification beyond the documentation updates and package manifests named here.

HOW IT CONNECTS

Owner: docs; ARCHITECTURE.md section2. FLOW.md row `F69` identifies the entry contract and preceding owners. Required predecessors: 27, F68. HTTP routes call application functions,which invoke domain rules and infrastructure adapters. Specialist tools are read-only. Gateway passes verified identity and request ID; worker events carry revision and lease generation. Register new routes in the named existing entry point; never create a parallel public backend.

CONSTRAINTS FOR THIS STEP

Preserve legacy domain names and compatible-set filtering. OpenAI only through common client. BGE-M3 only in search. Assigned-only real review; own-session demo review. No final recommendation without current-revision approval. No paid fallback,external message,quotation,voice,reservation or order creation. Document assumptions as measured facts or explicit configuration,never as guarantees.

ACCEPTANCE TESTS

- `tests/acceptance/test_f69.py`: run success,invalid-input,unauthorized,stale-revision and dependency-failure cases applicable to its feature; use the feature-specific assertions below.
- `F69`: Do not claim video exists until recorded; use synthetic data; show failure handling and reset; verify links before sending recruiters. Assert create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review.

`python tools/run_checks.py --prompt F69` must exit0 and report all named tests passed, with no required test skipped. It dispatches each service test in its own process and runs UI tests through pnpm. Before that runner exists, prompt00 creates and tests it using the standard library. Do not fabricate a passing count. Record actual output in the feature trace. Test databases must be disposable local Supabase; no production reset.

MANUAL VERIFICATION

Run the relevant demo path for demo walkthrough using synthetic fixtures. Inspect customer view and assigned-reviewer view where applicable. Confirm the success outcome,then exercise at least one listed failure. Inspect persisted revision,audit event and request ID without exposing secrets. For UI,check Arabic,keyboard operation and 360px width. For infrastructure,save the real provider/preflight receipt; missing account access is blocked,not passed.

DOC UPDATES

Update docs/HANDOVER.md with completed step `F69`, actual commands, blockers and next ID from prompts/README.md. Append docs/SESSION_LOG.md with actual model identifier if exposed,otherwise 'not exposed'. Record decision changes in docs/DECISIONS.md; update matching FLOW row and file-ownership manifest for approved additions. Update docs/TECH_SPEC.md schema/package receipts only after comparing reality. Set each affected feature trace to done only when its tests pass; retain original failures and corrective evidence.

ROLLBACK

Before edits create `git tag pre-f69` if absent. Commit this task independently. To undo,run `git log -1 --format=%H`,verify the displayed commit is this task,then `git revert --no-edit HEAD`. Do not revert unrelated later work. Applied SQL requires a new compensating migration and backup review; never reset production. Publication history and audit are retained. Deployment rollback selects previous immutable images and a schema-compatible release as described in docs/ROLLBACK.md.

COMMIT

```
feat(docs): F69 demo walkthrough
```

DONE WHEN

- [ ] Exact paths and contracts implemented.
- [ ] Listed tests executed with actual evidence.
- [ ] Failure and permission cases pass.

• [ ] No forward runtime dependency introduced.

• [ ] Documentation and feature traces updated.

• [ ] Diff reviewed; no unrelated change.

• [ ] Commit recorded; unexecuted deployment checks remain explicitly blocked.

Feature trace: features/F69-demo-walkthrough.md

F69 - Demo walkthrough

Field	Value
Status	planned
Owner module(s)	docs
Implementing prompts	F69; foundations 27
Affected flows	FLOW.md row F69

Scope

Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions.

Out of scope

Voice,quotations,external messaging,payments,actual stock reservations and automatic approval are excluded.

Acceptance criteria

• [ ] Create 3-minute and 8-minute scripts covering happy path,missing serial,incompatibility,stock split and human review; include recording checklist and captions.

• [ ] Do not claim video exists until recorded; use synthetic data; show failure handling and reset; verify links before sending recruiters.

• [ ] Test output and implementation commit recorded before status changes to done.

Data touched

none. Exact columns,keys,indexes and policies are in TECH_SPEC.md section4 and spec/sql. No feature may silently introduce a column absent from that schema.

Tests

`tests/acceptance/test_f69.py`; command `python tools/run_checks.py --feature F69`. Live provider checks require confirmed credit and are separate from deterministic acceptance.

Trace log

Date	Prompt	What changed	Commit
2026-10-11	Build kit	Scope,owner and required tests specified	Specification artifact; application implementation not executed

Verification evidence

Build-kit coverage verified by tools/kit/verify_kit.py. Application tests have not run because this deliverable is an implementation kit. Do not report this feature as implemented until its owning prompt executes.

123 features/README.md

Feature trace index

All69 feature traces are planned until application implementation executes. A kit coverage pass is not application acceptance. Read spec/features.json for complete mapping and prompts/README.md for execution order.

Typed common contracts

```
"""Executable contract specification copied by foundation 01; no provider or database side effects."""
from datetime import datetime
from decimal import Decimal
from enum import StrEnum
from typing import Literal
from uuid import UUID
from pydantic import BaseModel, ConfigDict, Field

class Contract(BaseModel):
    model_config = ConfigDict(extra='forbid', strict=True)

Locale = Literal['en', 'ar']
class AccessAction(StrEnum):
    READ='read'
    REVIEW='review'
    ASSIGN='assign'
    IMPORT='import'
class ActorContext(Contract):
    user_id: UUID
    request_id: UUID
    roles: list[Literal['reviewer','catalogue_admin','admin']]
    session_id: UUID | None = None
    session_generation: int | None = None
class DemoSession(Contract):
    id: UUID
    generation: int = Field(ge=1)
    expires_at: datetime
class ItemInput(Contract):
    description: str = Field(min_length=1,max_length=1000)
    quantity: Decimal | None = Field(default=None,gt=0,le=100000)
    model_text: str | None = None
    serial_text: str | None = None
    source_message_id: UUID
    source_start: int = Field(ge=0)
    source_end: int = Field(ge=0)
class IntakeResult(Contract):
    language: Literal['en','ar','mixed']
    preferred_locale: Locale
    urgency: Literal['normal','urgent']
    items: list[ItemInput] = Field(min_length=1,max_length=20)
class EnquiryCreate(Contract):
    text: str = Field(min_length=1,max_length=12000)
    locale: Locale
    destination_warehouse_id: int = Field(gt=0)
    session_id: UUID | None = None
class MessageCreate(Contract):
    text: str = Field(min_length=1,max_length=12000)
    expected_revision: int = Field(ge=1)
    item_id: UUID | None = None
class MessageView(Contract):
    id: UUID
    enquiry_id: UUID
    revision: int
    kind: Literal['customer','clarification','status','recommendation']
    body: str
    created_at: datetime
class EnquiryView(Contract):
    id: UUID
    revision: int
    status:
Literal['received','processing','needs_information','awaiting_review','approved','rejected','failed']
    messages: list[MessageView]
    item_ids: list[UUID]
    catalogue_version_id: UUID | None
    updated_at: datetime
class UploadRequest(Contract):
    enquiry_id: UUID
    expected_revision: int = Field(ge=1)
    media_type: Literal['image/jpeg','image/png','image/webp']
    size_bytes: int = Field(gt=0,le=10485760)
class UploadTicket(Contract):
    attachment_id: UUID
    upload_url: str
    expires_at: datetime
class AttachmentView(Contract):
    id: UUID
    status: Literal['pending','validated','rejected','deleted']
    error_code: str | None = None
class DownloadRequest(Contract):
    actor: ActorContext
    resource_type: Literal['evidence','export','attachment']
    resource_id: UUID
class DownloadTicket(Contract):
    url: str
    expires_at: datetime
class ImageInfo(Contract):
    width: int = Field(gt=0,le=12000)
    height: int = Field(gt=0,le=12000)
```

```
    media_type: str
    sha256: str = Field(pattern='^[a-f0-9]{64}$')
class NameplateResult(Contract):
    attachment_id: UUID
    model_text: str | None
    serial_text: str | None
    issues: list[Literal['unreadable','blur','low_resolution','glare','partial_plate']]
    confirmation_required: Literal[True] = True
class ConfirmExtraction(Contract):
    extraction_id: UUID
    expected_revision: int = Field(ge=1)
    model_text: str = Field(min_length=1,max_length=120)
    serial_text: str = Field(min_length=1,max_length=120)
class Allocation(Contract):
    warehouse_id: int = Field(gt=0)
    quantity: Decimal = Field(gt=0)
    transfer_days: int = Field(ge=0,le=60)
    stock_updated_at: datetime
class VerifiedItem(Contract):
    item_id: UUID
    configuration_id: int
    part_id: int
    part_number: str
    name_en: str
    name_ar: str
    requested_quantity: Decimal = Field(gt=0)
    allocations: list[Allocation]
    shortage: Decimal = Field(ge=0)
    source_reference: str
    evidence_id: UUID
class VerifiedFacts(Contract):
    enquiry_id: UUID
    revision: int
    catalogue_version_id: UUID
    items: list[VerifiedItem] = Field(min_length=1,max_length=20)
class CandidateEvidence(Contract):
    item_id: UUID
    part_id: int
    configuration_id: int
    evidence_id: UUID
    source_reference: str
class StockEvidence(Contract):
    part_id: int
    warehouse_id: int
    available: Decimal = Field(ge=0)
    updated_at: datetime
class VerificationEvidence(Contract):
    current_revision: int
    active_catalogue_version_id: UUID
    candidates: list[CandidateEvidence]
    stock: list[StockEvidence]
class VerificationFailure(Contract):
    code: Literal['STALE_REVISION','STALE_CATALOGUE','INCOMPATIBLE','MISSING_EVIDENCE','INVALID_QUANTITY
','OVERALLOCATION','STALE_STOCK','UNRESOLVED_ITEM','UNSAFE_PROSE']
    item_id: UUID | None
    detail: str
class VerificationResult(Contract):
    passed: bool
    failures: list[VerificationFailure]
    facts_sha256: str
class ReplySlots(Contract):
    greeting: str = Field(max_length=200)
    explanation: str = Field(max_length=1500)
    closing: str = Field(max_length=300)
class ApprovalRequest(Contract):
    enquiry_id: UUID
    expected_revision: int = Field(ge=1)
    draft_id: UUID
    edited_slots: ReplySlots | None = None
class RejectionRequest(Contract):
    enquiry_id: UUID
    expected_revision: int = Field(ge=1)
    reason: str = Field(min_length=1,max_length=1000)
class ApprovalResult(Contract):
    decision_id: UUID
    message_id: UUID
    revision: int
    already_applied: bool
class ReviewResult(Contract):
    decision_id: UUID
    revision: int
    status: Literal['approved','rejected']
class ItemCorrection(Contract):
    item_id: UUID
    description: str | None = None
    quantity: Decimal | None = Field(default=None,gt=0,le=100000)
    model_text: str | None = None
    serial_text: str | None = None
class RevisionRequest(Contract):
    enquiry_id: UUID
    expected_revision: int
```

```
    corrections: list[ItemCorrection] = Field(min_length=1,max_length=20)
    reason: str = Field(min_length=1,max_length=1000)
class JobLease(Contract):
    id: UUID
    kind: Literal['nameplate','export_pdf','catalog_import','catalog_publish','storage_cleanup']
    enquiry_id: UUID | None
    revision: int | None
    session_generation: int | None
    lease_token: UUID
    expires_at: datetime
    attempt: int
    payload: dict[str,str | int | bool | None]
class JobResult(Contract):
    status: Literal['succeeded','failed','cancelled']
    result_id: UUID | None
    error_code: str | None
class ExportRequest(Contract):
    enquiry_id: UUID
    revision: int
    decision_id: UUID
class ExportResult(Contract):
    export_id: UUID
    storage_key: str
    sha256: str
class ApprovedSnapshot(Contract):
    decision_id: UUID
    approved_at: datetime
    locale: Locale
    facts: VerifiedFacts
    reply: str
    synthetic: bool
class AuditEvent(Contract):
    actor_type: Literal['system','ai','user']
    actor_id: str
    action: str
    entity_type: str
    entity_id: str
    request_id: UUID
    enquiry_id: UUID | None
    revision: int | None
    payload: dict[str,str | int | bool | None]
class AIRequest(Contract):
    request_key: UUID
    enquiry_id: UUID | None
    stage: Literal['intake','nameplate','parts','reply','summary']
    model: str
    trusted_instructions: str
    untrusted_text: str
    image_base64: str | None = None
    output_schema_name:
Literal['IntakeResult','NameplateResult','PartsProposal','ReplySlots','SummaryResult']
    max_output_tokens: int = Field(ge=1,le=2048)
class AIResult(Contract):
    output_json: str
    input_tokens: int
    output_tokens: int
    provider_request_id: str
class BudgetReservation(Contract):
    id: UUID
    account_id: str
    reserved_usd: Decimal
class CreditAllowance(Contract):
    amount_usd: Decimal = Field(ge=0)
    expires_at: datetime
    paid_budget_usd: Decimal = Field(ge=0,le=0)
class ItemState(Contract):
    id: UUID
    request: ItemInput
    configuration_id: int | None
    candidate_part_ids: list[int]
    selected_part_id: int | None
    status: Literal['unresolved','needs_information','ready','unavailable']
class EnquiryState(Contract):
    enquiry_id: UUID
    revision: int
    session_generation: int | None
    catalogue_version_id: UUID | None
    items: list[ItemState]
    step_count: int = Field(ge=0)
    pending_question: str | None
    draft_id: UUID | None
    verification_id: UUID | None
    phase: str
class ToolContext(Contract):
    enquiry_id: UUID
    revision: int
    catalogue_version_id: UUID
    item_id: UUID
    max_calls: int = Field(ge=0,le=4)
class PartsProposal(Contract):
    explanation: str
```

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    clarification_code: Literal['needs_serial','needs_model','ambiguous_part','mismatch','none']
    candidate_part_ids: list[int]
class SummaryResult(Contract):
    summary: str = Field(max_length=2000)
    source_message_ids: list[UUID]
class MetricsQuery(Contract):
    start: datetime
    end: datetime
    include_demo: bool = False
class MetricsView(Contract):
    enquiry_count: int
    approved_count: int
    clarification_count: int
    evaluated_count: int
    correct_count: int
    median_review_seconds: float | None
    p95_latency_ms: float | None
    estimated_cost_usd: Decimal
class ReviewView(Contract):
    enquiry: EnquiryView
    facts: VerifiedFacts | None
    draft: ReplySlots | None
    verification: VerificationResult | None
    assigned_to: UUID | None
class DemoScenario(Contract):
    id: Literal['mixed_photo','missing_serial','incompatible_number','local_shortage']
    text: str
    attachment_path: str | None
    expected_status: str
    expected_configuration_id: int | None
    expected_part_ids: list[int]
class WorkflowStep(StrEnum):
    INTAKE='intake'
    AWAIT_OCR='await_ocr'
    CONFIRM='confirm'
    IDENTIFY='identify'
    CLARIFY='clarify'
    SEARCH='search'
    CHOOSE='choose'
    PLAN='plan'
    DRAFT='draft'
    VERIFY='verify'
    REVIEW='review'
    COMPLETE='complete'
    NEEDS_STAFF='needs_staff'
class EmbeddingReceipt(Contract):
    version_id: UUID
    embedded_count: int
    model_revision: str
class EventOutcome(Contract):
    event_id: UUID
    state: Literal['processed','obsolete','retry','failed']
class EvalCase(Contract):
    id: str
    scenario: str
    input_text: str
    expected_status: str
    expected_part_ids: list[int]
class EvalOutput(Contract):
    status: str
    part_ids: list[int]
    clarification_code: str | None
class EvalResult(Contract):
    case_id: str
    passed: bool
    actual: EvalOutput
class EvalRun(Contract):
    id: UUID
    dataset_sha256: str
    results: list[EvalResult]
class SchemaReceipt(Contract):
    postgres_version: str
    extensions: dict[str,str]
    columns: dict[str,list[str]]
    indexes: dict[str,list[str]]
    policies: dict[str,list[str]]
class TestConfig(Contract):
    app_env: Literal['local','test']
    database_url: str
    production_project_ref: str
class ReadinessReport(Contract):
    database: bool
    redis: bool
    search_lexical: bool
    search_semantic: bool
    ai_credit_enabled: bool
    storage: bool
class AccountReceipt(Contract):
    account_id: str
    plan: Literal['free']
    region: str
```

```
    credit_usd: Decimal
    expires_at: datetime
class SizingReceipt(Contract):
    instance_type: str
    peak_memory_mib: int
    instance_memory_mib: int
    measured_at: datetime
class DeploymentDecision(Contract):
    allowed: bool
    reasons: list[str]
class HostedReceipt(Contract):
    base_url: str
    api_url: str
    scenarios_passed: list[str]
    scenarios_failed: list[str]
    checked_at: datetime
```